

In [1]:

```
import numpy as np # Linear algebra
import pandas as pd # data processing, CSV file I/O (e.g. pd.read_csv)
import os
```

In [2]:

```
from pyspark.sql.functions import split, col, mean, stddev
```

In [3]:

```
import findspark
findspark.init()
import numpy as np
from fcmeans import FCM
import pyspark # only run after findspark.init()
```

In [4]:

```
def spark_shape(self):
    return (self.count(), len(self.columns))
pyspark.sql.dataframe.DataFrame.shape = spark_shape
```

In [5]:

```
from pyspark.sql import SparkSession
spark=SparkSession.builder.appName('Practise').getOrCreate()
```

In [6]:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.pipeline import Pipeline
from sklearn.model_selection import train_test_split
from sklearn.metrics import confusion_matrix, f1_score
from sklearn.naive_bayes import MultinomialNB
from sklearn.ensemble import RandomForestClassifier

from sklearn import metrics
from sklearn.metrics import accuracy_score, classification_report
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.neighbors import KNeighborsClassifier

import warnings
warnings.filterwarnings("ignore")
import pyspark.pandas as ps
```

WARNING:root:'PYARROW_IGNORE_TIMEZONE' environment variable was not set. It is required to set this environment variable to '1' in both driver and executor sides if you use pyarrow>=2.0.0. pandas-on-Spark will set it for you but it does not work if there is a Spark context already launched.

In [7]:

```
#df = spark.sql('select 'spark' as hello ')
df_pyspark=spark.read.csv('DNN-EdgeIIoT-dataset.csv',header=True,inferSchema=True)
```

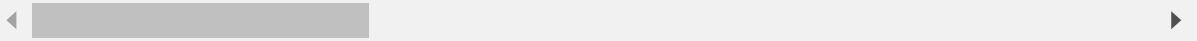
In [8]:

```
df = pd.read_csv('DNN-EdgeIIoT-dataset.csv')
df.head()
```

Out[8]:

	frame.time	ip.src_host	ip.dst_host	arp.dst.proto_ipv4	arp.opcode	arp.hw.size	
0	2021 11:44:10.081753000	192.168.0.128	192.168.0.101	0	0.0	0.0	
1	2021 11:44:10.162218000	192.168.0.101	192.168.0.128	0	0.0	0.0	
2	2021 11:44:10.162271000	192.168.0.128	192.168.0.101	0	0.0	0.0	
3	2021 11:44:10.162641000	192.168.0.128	192.168.0.101	0	0.0	0.0	
4	2021 11:44:10.166132000	192.168.0.101	192.168.0.128	0	0.0	0.0	

5 rows × 63 columns



In [9]:

```
import pandas as pd
import numpy as np
df1=pd.read_csv('wustl-ehms-2020.csv')
df1.head()
```

Out[9]:

	Dir	Flgs	SrcAddr	DstAddr	Sport	Dport	SrcBytes	DstBytes	SrcLoad	DstLoad	...	I
0	->	e	10.0.1.172	10.0.1.150	58059	1111	496.0	186.0	276914.0	92305.0	...	
1	->	e	10.0.1.172	10.0.1.150	58062	1111	496.0	186.0	230984.0	76995.0	...	
2	->	e	10.0.1.172	10.0.1.150	58065	1111	496.0	186.0	218470.0	72823.0	...	
3	->	e	10.0.1.172	10.0.1.150	58067	1111	496.0	186.0	203376.0	67792.0	...	
4	->	e	10.0.1.172	10.0.1.150	58069	1111	496.0	186.0	235723.0	78574.0	...	

5 rows × 44 columns



In [10]:

df1.columns

Out[10]:

```
Index(['Dir', 'Flgs', 'SrcAddr', 'DstAddr', 'Sport', 'Dport', 'SrcBytes',
       'DstBytes', 'SrcLoad', 'DstLoad', 'SrcGap', 'DstGap', 'SIntPkt',
       'DIntPkt', 'SIntPktAct', 'DIntPktAct', 'SrcJitter', 'DstJitter',
       'sMaxPktSz', 'dMaxPktSz', 'sMinPktSz', 'dMinPktSz', 'Dur', 'Trans',
       'TotPkts', 'TotBytes', 'Load', 'Loss', 'pLoss', 'pSrcLoss', 'pDstLoss',
       'Rate', 'SrcMac', 'DstMac', 'Packet_num', 'Temp', 'SpO2', 'Pulse_Rate',
       'SYS', 'DIA', 'Heart_rate', 'Resp_Rate', 'ST', 'Label'],
      dtype='object')
```

In [15]:

df.columns

Out[15]:

```
Index(['frame.time', 'ip.src_host', 'ip.dst_host', 'arp.dst.proto_ipv4',
       'arp.opcode', 'arp.hw.size', 'arp.src.proto_ipv4', 'icmp.checksum',
       'icmp.seq_le', 'icmp.transmit_timestamp', 'icmp.unused',
       'http.file_data', 'http.content_length', 'http.request.uri.query',
       'http.request.method', 'http.referer', 'http.request.full_uri',
       'http.request.version', 'http.response', 'http.tls_port', 'tcp.ack',
       'tcp.ack_raw', 'tcp.checksum', 'tcp.connection.fin',
       'tcp.connection.rst', 'tcp.connection.syn', 'tcp.connection.synack',
       'tcp.dstport', 'tcp.flags', 'tcp.flags.ack', 'tcp.len', 'tcp.option',
       'tcp.payload', 'tcp.seq', 'tcp.srcport', 'udp.port', 'udp.stream',
       'udp.time_delta', 'dns.qry.name', 'dns.qry.name.len', 'dns.qry.qu',
       'dns.qry.type', 'dns.retransmission', 'dns.retransmit_request',
       'dns.retransmit_request_in', 'mqtt.conack.flags',
       'mqtt.conflag.cleansess', 'mqtt.conflags', 'mqtt.hdrflags', 'mqtt.len',
       'mqtt.msg_decoded_as', 'mqtt.msg', 'mqtt.msgtype', 'mqtt.proto_len',
       'mqtt.protoname', 'mqtt.topic', 'mqtt.topic_len', 'mqtt.ver',
       'mbtcp.len', 'mbtcp.trans_id', 'mbtcp.unit_id', 'Attack_label',
       'Attack_type'],
      dtype='object')
```

In [13]:

df_exp=df.append(df1)

In [14]:

df_exp.shape

Out[14]:

(2235519, 107)

In [16]:

```
df_exp.head(10)
```

Out[16]:

	frame.time	ip.src_host	ip.dst_host	arp.dst.proto_ipv4	arp.opcode	arp.hw.size	
0	2021 11:44:10.081753000	192.168.0.128	192.168.0.101	0	0.0	0.0	
1	2021 11:44:10.162218000	192.168.0.101	192.168.0.128	0	0.0	0.0	
2	2021 11:44:10.162271000	192.168.0.128	192.168.0.101	0	0.0	0.0	
3	2021 11:44:10.162641000	192.168.0.128	192.168.0.101	0	0.0	0.0	
4	2021 11:44:10.166132000	192.168.0.101	192.168.0.128	0	0.0	0.0	
5	2021 11:44:10.166159000	192.168.0.128	192.168.0.101	0	0.0	0.0	
6	2021 11:44:10.166968000	192.168.0.101	192.168.0.128	0	0.0	0.0	
7	2021 11:44:10.167072000	192.168.0.128	192.168.0.101	0	0.0	0.0	
8	2021 11:44:10.169612000	192.168.0.101	192.168.0.128	0	0.0	0.0	
9	2021 11:44:10.169644000	192.168.0.128	192.168.0.101	0	0.0	0.0	

10 rows × 107 columns

In [17]:

```
df_exp.info()
```

<class 'pandas.core.frame.DataFrame'>
Int64Index: 2235519 entries, 0 to 16317
Columns: 107 entries, frame.time to Label
dtypes: float64(80), object(27)
memory usage: 1.8+ GB

In [18]:

```
df_exp=df_exp.fillna(0)
```

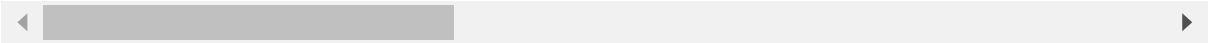
In [19]:

df_exp

Out[19]:

	frame.time	ip.src_host	ip.dst_host	arp.dst.proto_ipv4	arp.opcode	arp.hw.si
0	2021 11:44:10.081753000	192.168.0.128	192.168.0.101	0	0.0	(
1	2021 11:44:10.162218000	192.168.0.101	192.168.0.128	0	0.0	(
2	2021 11:44:10.162271000	192.168.0.128	192.168.0.101	0	0.0	(
3	2021 11:44:10.162641000	192.168.0.128	192.168.0.101	0	0.0	(
4	2021 11:44:10.166132000	192.168.0.101	192.168.0.128	0	0.0	(
...	...	...	...	...	...	
16313	0	0	0	0	0.0	(
16314	0	0	0	0	0.0	(
16315	0	0	0	0	0.0	(
16316	0	0	0	0	0.0	(
16317	0	0	0	0	0.0	(

2235519 rows × 107 columns



In [20]:

```
spark.conf.set("spark.sql.execution.arrow.enabled", "true")
df_pyspark=spark.createDataFrame(df_exp[:10000])
#df_pyspark.printSchema()
df_pyspark.printSchema()
```

root

```
|-- frame.time: string (nullable = true)
|-- ip.src_host: string (nullable = true)
|-- ip.dst_host: string (nullable = true)
|-- arp.dst.proto_ipv4: string (nullable = true)
|-- arp.opcode: double (nullable = true)
|-- arp.hw.size: double (nullable = true)
|-- arp.src.proto_ipv4: string (nullable = true)
|-- icmp.checksum: double (nullable = true)
|-- icmp.seq_le: double (nullable = true)
|-- icmp.transmit_timestamp: double (nullable = true)
|-- icmp.unused: double (nullable = true)
|-- http.file_data: double (nullable = true)
|-- http.content_length: double (nullable = true)
|-- http.request.uri.query: double (nullable = true)
|-- http.request.method: double (nullable = true)
|-- http.referer: double (nullable = true)
|-- http.request.full_uri: double (nullable = true)
|-- http.request.version: double (nullable = true)
```

In [21]:

```
#df=df.drop(['Dir', 'Sport', 'Flgs', 'SrcMac', 'DstMac'],axis=1)
df=df_exp
df=df.drop(['Dir', 'Flgs', 'Sport'],axis=1)
```

In [22]:

```
df=df.drop(['SrcMac', 'DstMac'],axis=1)
```

In [23]:

```
from pyspark import SparkContext
sc=SparkContext
sc
```

Out[23]:

```
pyspark.context.SparkContext
```


In [26]:

```
import ipaddress
df['SrcAddr'] = df['SrcAddr'].apply(lambda x: int(ipaddress.IPv4Address(x)))
df['DstAddr'] = df['DstAddr'].apply(lambda x: int(ipaddress.IPv4Address(x)))
df.head()
```

Out[26]:

	frame.time	ip.src_host	ip.dst_host	arp.dst.proto_ipv4	arp.opcode	arp.hw.siz
0	11:44:10.081753000	192.168.0.128	192.168.0.101	0	0.0	0.
1	11:44:10.162218000	192.168.0.101	192.168.0.128	0	0.0	0.
2	11:44:10.162271000	192.168.0.128	192.168.0.101	0	0.0	0.
3	11:44:10.162641000	192.168.0.128	192.168.0.101	0	0.0	0.
4	11:44:10.166132000	192.168.0.101	192.168.0.128	0	0.0	0.

5 rows × 102 columns

In [27]:

```
print(df['Attack_type'].value_counts())
```

Normal 100000
Name: Attack_type, dtype: int64

In [28]:

```
from sklearn.utils import shuffle
drop_columns = ["frame.time", "ip.src_host", "ip.dst_host", "arp.src.proto_ipv4", "arp.dst.p",
                "http.file_data", "http.request.full_uri", "icmp.transmit_timestamp",
                "http.request.uri.query", "tcp.options", "tcp.payload", "tcp.srcport",
                "tcp.dstport", "udp.port", "mqtt.msg"]

df.drop(drop_columns, axis=1, inplace=True)
df.dropna(axis=0, how='any', inplace=True)
df.drop_duplicates(subset=None, keep="first", inplace=True)
df = shuffle(df)
df.isna().sum()
print(df['Attack_type'].value_counts())
```

Normal 86133
Name: Attack_type, dtype: int64

In [29]:

```
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn import preprocessing

def encode_text_dummy(df, name):
    dummies = pd.get_dummies(df[name])
    for x in dummies.columns:
        dummy_name = f"{name}-{x}"
        df[dummy_name] = dummies[x]
    df.drop(name, axis=1, inplace=True)

encode_text_dummy(df, 'http.request.method')
encode_text_dummy(df, 'http.referer')
encode_text_dummy(df, "http.request.version")
encode_text_dummy(df, "dns.qry.name.len")
encode_text_dummy(df, "mqtt.conack.flags")
encode_text_dummy(df, "mqtt.protoname")
encode_text_dummy(df, "mqtt.topic")
```

In [30]:

```
df.to_csv('preprocessed_DNN.csv', encoding='utf-8', index=False)
```

In [31]:

```
import pandas as pd
import numpy as np
from matplotlib import pyplot as plt
import seaborn as sns
```

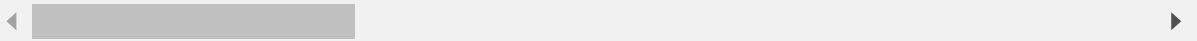
In [32]:

```
df = pd.read_csv('./preprocessed_DNN.csv', low_memory=False)
df
```

Out[32]:

	arp.opcode	arp.hw.size	icmp.checksum	icmp.seq_le	icmp.unused	http.content_length
0	0.0	0.0	0.0	0.0	0.0	0.0
1	0.0	0.0	0.0	0.0	0.0	0.0
2	0.0	0.0	0.0	0.0	0.0	0.0
3	0.0	0.0	0.0	0.0	0.0	0.0
4	0.0	0.0	0.0	0.0	0.0	0.0
...	...	...	...	...	...	...
86128	0.0	0.0	0.0	0.0	0.0	0.0
86129	0.0	0.0	0.0	0.0	0.0	0.0
86130	0.0	0.0	0.0	0.0	0.0	0.0
86131	0.0	0.0	0.0	0.0	0.0	0.0
86132	0.0	0.0	0.0	0.0	0.0	0.0

86133 rows × 90 columns



In [33]:

```
df['Attack_type'].value_counts()
```

Out[33]:

Normal 86133
Name: Attack_type, dtype: int64

In [34]:

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 86133 entries, 0 to 86132
Data columns (total 90 columns):
```

#	Column	Non-Null Count	Dtype
0	arp.opcode	86133 non-null	float64
1	arp.hw.size	86133 non-null	float64
2	icmp.checksum	86133 non-null	float64
3	icmp.seq_le	86133 non-null	float64
4	icmp.unused	86133 non-null	float64
5	http.content_length	86133 non-null	float64
6	http.response	86133 non-null	float64
7	http.tls_port	86133 non-null	float64
8	tcp.ack	86133 non-null	float64
9	tcp.ack_raw	86133 non-null	float64
10	tcp.checksum	86133 non-null	float64
11	tcp.connection.fin	86133 non-null	float64
12	tcp.connection.rst	86133 non-null	float64
13	tcp.connection.syn	86133 non-null	float64
14	tcp.connection.synack	86133 non-null	float64
15	tcp.flags	86133 non-null	float64
16	tcp.flags.ack	86133 non-null	float64
17	tcp.len	86133 non-null	float64
18	tcp.seq	86133 non-null	float64
19	udp.stream	86133 non-null	float64
20	udp.time_delta	86133 non-null	float64
21	dns.qry.name	86133 non-null	float64
22	dns.qry.qu	86133 non-null	float64
23	dns.qry.type	86133 non-null	float64
24	dns.retransmission	86133 non-null	float64
25	dns.retransmit_request	86133 non-null	float64
26	dns.retransmit_request_in	86133 non-null	float64
27	mqtt.conflag.cleansess	86133 non-null	float64
28	mqtt.conflags	86133 non-null	float64
29	mqtt.hdrflags	86133 non-null	float64
30	mqtt.len	86133 non-null	float64
31	mqtt.msg_decoded_as	86133 non-null	float64
32	mqtt.msgtype	86133 non-null	float64
33	mqtt.proto_len	86133 non-null	float64
34	mqtt.topic_len	86133 non-null	float64
35	mqtt.ver	86133 non-null	float64
36	mbtcp.len	86133 non-null	float64
37	mbtcp.trans_id	86133 non-null	float64
38	mbtcp.unit_id	86133 non-null	float64
39	Attack_label	86133 non-null	float64
40	Attack_type	86133 non-null	object
41	SrcAddr	86133 non-null	int64
42	DstAddr	86133 non-null	int64
43	Dport	86133 non-null	float64
44	SrcBytes	86133 non-null	float64
45	DstBytes	86133 non-null	float64
46	SrcLoad	86133 non-null	float64
47	DstLoad	86133 non-null	float64
48	SrcGap	86133 non-null	float64
49	DstGap	86133 non-null	float64
50	SIntPkt	86133 non-null	float64
51	DIntPkt	86133 non-null	float64

52	SIntPktAct	86133	non-null	float64
53	DIntPktAct	86133	non-null	float64
54	SrcJitter	86133	non-null	float64
55	DstJitter	86133	non-null	float64
56	sMaxPktSz	86133	non-null	float64
57	dMaxPktSz	86133	non-null	float64
58	sMinPktSz	86133	non-null	float64
59	dMinPktSz	86133	non-null	float64
60	Dur	86133	non-null	float64
61	Trans	86133	non-null	float64
62	TotPkts	86133	non-null	float64
63	TotBytes	86133	non-null	float64
64	Load	86133	non-null	float64
65	Loss	86133	non-null	float64
66	pLoss	86133	non-null	float64
67	pSrcLoss	86133	non-null	float64
68	pDstLoss	86133	non-null	float64
69	Rate	86133	non-null	float64
70	Packet_num	86133	non-null	float64
71	Temp	86133	non-null	float64
72	SpO2	86133	non-null	float64
73	Pulse_Rate	86133	non-null	float64
74	SYS	86133	non-null	float64
75	DIA	86133	non-null	float64
76	Heart_rate	86133	non-null	float64
77	Resp_Rate	86133	non-null	float64
78	ST	86133	non-null	float64
79	Label	86133	non-null	float64
80	http.request.method-0.0	86133	non-null	int64
81	http.referer-0.0	86133	non-null	int64
82	http.request.version-0.0	86133	non-null	int64
83	dns.qry.name.len-0	86133	non-null	int64
84	mqtt.conack.flags-0	86133	non-null	int64
85	mqtt.conack.flags-0x00000000	86133	non-null	int64
86	mqtt.protoname-0	86133	non-null	int64
87	mqtt.protoname-MQTT	86133	non-null	int64
88	mqtt.topic-0	86133	non-null	int64
89	mqtt.topic-Temperature_and_Humidity	86133	non-null	int64

dtypes: float64(77), int64(12), object(1)

memory usage: 59.1+ MB

In [35]:

```
feat_cols = list(df.columns)
label_col = "Attack_type"

feat_cols.remove(label_col)
feat_cols
```

Out[35]:

```
['arp.opcode',
 'arp.hw.size',
 'icmp.checksum',
 'icmp.seq_le',
 'icmp.unused',
 'http.content_length',
 'http.response',
 'http.tls_port',
 'tcp.ack',
 'tcp.ack_raw',
 'tcp.checksum',
 'tcp.connection.fin',
 'tcp.connection.rst',
 'tcp.connection.syn',
 'tcp.connection.synack',
 'tcp.flags',
 'tcp.flags.ack',
 'tcp.len',
 'tcp.seq',
 'udp.stream',
 'udp.time_delta',
 'dns.qry.name',
 'dns.qry.qu',
 'dns.qry.type',
 'dns.retransmission',
 'dns.retransmit_request',
 'dns.retransmit_request_in',
 'mqtt.conflag.cleansess',
 'mqtt.conflags',
 'mqtt.hdrflags',
 'mqtt.len',
 'mqtt.msg_decoded_as',
 'mqtt.msgtype',
 'mqtt.proto_len',
 'mqtt.topic_len',
 'mqtt.ver',
 'mbtcp.len',
 'mbtcp.trans_id',
 'mbtcp.unit_id',
 'Attack_label',
 'SrcAddr',
 'DstAddr',
 'Dport',
 'SrcBytes',
 'DstBytes',
 'SrcLoad',
 'DstLoad',
 'SrcGap',
 'DstGap',
 'SIntPkt',
 'DIntPkt',
```

```
'SIntPktAct',  
'DIntPktAct',  
'SrcJitter',  
'DstJitter',  
'sMaxPktSz',  
'dMaxPktSz',  
'sMinPktSz',  
'dMinPktSz',  
'Dur',  
'Trans',  
'TotPkts',  
'TotBytes',  
'Load',  
'Loss',  
'pLoss',  
'pSrcLoss',  
'pDstLoss',  
'Rate',  
'Packet_num',  
'Temp',  
'SpO2',  
'Pulse_Rate',  
'SYS',  
'DIA',  
'Heart_rate',  
'Resp_Rate',  
'ST',  
'Label',  
'http.request.method-0.0',  
'http.referer-0.0',  
'http.request.version-0.0',  
'dns.qry.name.len-0',  
'mqtt.conack.flags-0',  
'mqtt.conack.flags-0x00000000',  
'mqtt.protoname-0',  
'mqtt.protoname-MQTT',  
'mqtt.topic-0',  
'mqtt.topic-Temperature_and_Humidity']
```

In [36]:

```
len(feat_cols)
```

Out[36]:

89

In [37]:

```
empty_cols = [col for col in df.columns if df[col].isnull().all()]  
empty_cols
```

Out[37]:

[]

In [38]:

```
df=df.replace(' ->',0)
```

In [39]:

```
skip_list = ["icmp.unused", "http.tls_port", "dns.qry.type", "mqtt.msg_decoded_as"]
```

In [40]:

```
df[skip_list[3]].value_counts()
```

Out[40]:

```
0.0      86133  
Name: mqtt.msg_decoded_as, dtype: int64
```

In [41]:

```
df.drop(skip_list, axis=1, inplace=True)
feat_cols = list(df.columns)
feat_cols.remove(label_col)
feat_cols
```

Out[41]:

```
['arp.opcode',
 'arp.hw.size',
 'icmp.checksum',
 'icmp.seq_le',
 'http.content_length',
 'http.response',
 'tcp.ack',
 'tcp.ack_raw',
 'tcp.checksum',
 'tcp.connection.fin',
 'tcp.connection.rst',
 'tcp.connection.syn',
 'tcp.connection.synack',
 'tcp.flags',
 'tcp.flags.ack',
 'tcp.len',
 'tcp.seq',
 'udp.stream',
 'udp.time_delta',
 'dns.qry.name',
 'dns.qry.qu',
 'dns.retransmission',
 'dns.retransmit_request',
 'dns.retransmit_request_in',
 'mqtt.conflag.cleansess',
 'mqtt.conflags',
 'mqtt.hdrflags',
 'mqtt.len',
 'mqtt.msgtype',
 'mqtt.proto_len',
 'mqtt.topic_len',
 'mqtt.ver',
 'mbtcp.len',
 'mbtcp.trans_id',
 'mbtcp.unit_id',
 'Attack_label',
 'SrcAddr',
 'DstAddr',
 'Dport',
 'SrcBytes',
 'DstBytes',
 'SrcLoad',
 'DstLoad',
 'SrcGap',
 'DstGap',
 'SIntPkt',
 'DIntPkt',
 'SIntPktAct',
 'DIntPktAct',
 'SrcJitter',
 'DstJitter',
 'sMaxPktSz',
```

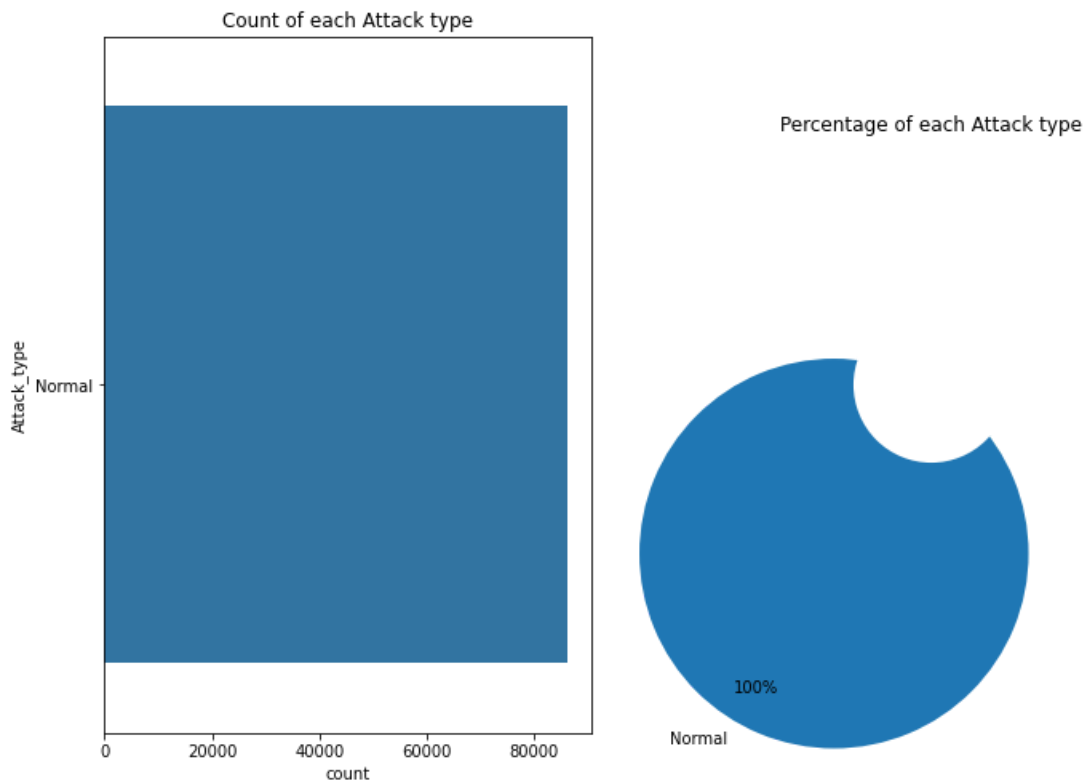


```
'dMaxPktSz',  
'sMinPktSz',  
'dMinPktSz',  
'Dur',  
'Trans',  
'TotPkts',  
'TotBytes',  
'Load',  
'Loss',  
'pLoss',  
'pSrcLoss',  
'pDstLoss',  
'Rate',  
'Packet_num',  
'Temp',  
'SpO2',  
'Pulse_Rate',  
'SYS',  
'DIA',  
'Heart_rate',  
'Resp_Rate',  
'ST',  
'Label',  
'http.request.method-0.0',  
'http.referer-0.0',  
'http.request.version-0.0',  
'dns.qry.name.len-0',  
'mqtt.conack.flags-0',  
'mqtt.conack.flags-0x00000000',  
'mqtt.protoname-0',  
'mqtt.protoname-MQTT',  
'mqtt.topic-0',  
'mqtt.topic-Temperature_and_Humidity']
```

In [57]:

```
fig, (ax1, ax2) = plt.subplots(nrows=1, ncols=2, figsize=(12,8))
explode = list((np.array(list(df[label_col].dropna().value_counts()))/sum(list(df[label_col].dropna().value_counts()))))
labels = list(df[label_col].dropna().unique())[:]
sizes = df[label_col].value_counts()[:]

ax2.pie(sizes, explode=explode, startangle=60, labels=labels, autopct='%1.0f%%', pctdist=1.0)
ax2.add_artist(plt.Circle((0,0),0.4,fc='white'))
sns.countplot(y=label_col, data=df, ax=ax1)
ax1.set_title("Count of each Attack type")
ax2.set_title("Percentage of each Attack type")
plt.show()
```



In [43]:

```
import threading
import socket
from itertools import chain
```

In [88]:

```
l1,l2,l3,l4,l5,l6=0.13,0.20,0.12,0.08,0.09,0.06
```

In [46]:

```
clisocket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)

clisocket.connect(('localhost', 8020))
print("The client has been connected..")
print("The message from the server: ", clisocket.recv(256).decode())
df_array= df.to_numpy().flatten()
print("Please keep proving the messages to send to server via SDN..")
df_string = ' '.join([str(elem) for elem in df_array])
msg = df_string
clisocket.send(bytes(msg, encoding='utf8'))
clisocket.close()
```

The client has been connected..

The message from the server: CONNECT_SUCCESSFUL

Please keep proving the messages to send to server via SDN..

In [47]:

```
X = df.drop([label_col], axis=1)
y = df[label_col]

#del df
```

In [48]:

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=1, st
```

In [50]:

```
from sklearn.preprocessing import LabelEncoder

label_encoder = LabelEncoder()
y_train = label_encoder.fit_transform(y_train)
y_test = label_encoder.transform(y_test)
```

In [51]:

```
label_encoder.classes_
```

Out[51]:

```
array(['Normal'], dtype=object)
```

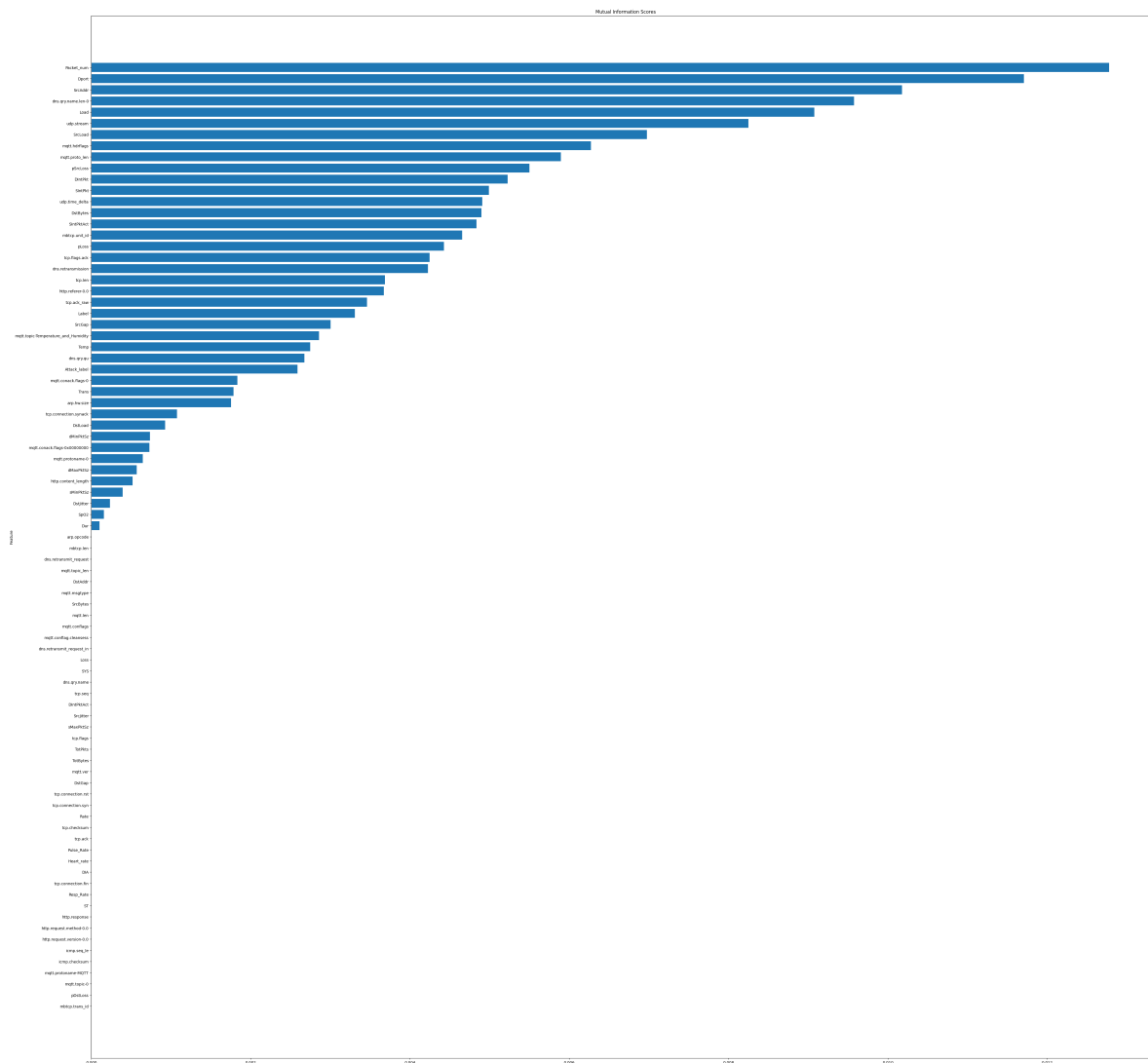
```
from sklearn.feature_selection import mutual_info_regression

def make_mi_scores(X_feat, y_label):
    mi_scores = mutual_info_regression(X_feat, y_label, discrete_features=False)
    mi_scores = pd.Series(mi_scores, name="MI Scores", index=X_feat.columns)
    mi_scores = mi_scores.sort_values(ascending=False)
    return mi_scores

mi_scores = make_mi_scores(X_train, y_train)
mi_scores = make_mi_scores(X_test, y_test)
```

```
def plot_utility_scores(scores):
    y_scores = scores.sort_values(ascending=True)
    width = np.arange(len(y_scores))
    ticks = list(y_scores.index)
    plt.barh(width, y_scores)
    plt.yticks(width, ticks)
    plt.title("Mutual Information Scores")

plt.figure(dpi=100, figsize=(50, 50))
plt.xlabel("Score")
plt.ylabel("Feature")
plot_utility_scores(mi_scores)
```



In [60]:

```
from sklearn.manifold import TSNE
import time
```

In [61]:

```
time_start = time.time()
tsne = TSNE(n_components=2, verbose=0, perplexity=40, n_iter=10)
#tsne_results = tsne.fit_transform(X_train)
#tsne_results = tsne.fit_transform(X_test)
print('t-SNE: {} seconds'.format(time.time()-time_start))
```

t-SNE: 0.0 seconds

In [62]:

```
#Feature
```

In [63]:

```
from sklearn.utils import class_weight

class_weights = class_weight.compute_class_weight('balanced',
                                                  classes=np.unique(y_train),
                                                  y=y_train)

class_weights = {k: v for k,v in enumerate(class_weights)}
class_weights
```

Out[63]:

```
{0: 1.0}
```

In [64]:

```
from sklearn.feature_selection import VarianceThreshold
var_thres=VarianceThreshold(threshold=0)
var_thres.fit(X_train)
```

Out[64]:

```
VarianceThreshold(threshold=0)
```

In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.

On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.

In [65]:

```
print("Find Constant and Non-Constant Features:\n",var_thres.get_support())  
print("No. of Non-Constant Features: ",len(X_train.columns[var_thres.get_support()])))
```

Find Constant and Non-Constant Features:

```
[ True  True False False False False  True  True  True  True  True  True  
  True  True  True  True  True False False False False False False False  
  True  True  True  True  True  True  True  True False False False False  
 False False False False False False False False False False False False  
 False False False False False False False False False False False False  
 False False False False False False False False False False False False  
  True]
No. of Non-Constant Features:  27
```

In [66]:

```
constant_columns = [column for column in X_train.columns
                     if column not in X_train.columns[var_thres.get_support()]]

print(len(constant_columns))

for column in constant_columns:
    print(column)
```

```
58
icmp.checksum
icmp.seq_le
http.content_length
http.response
udp.stream
udp.time_delta
dns.qry.name
dns.qry.qu
dns.retransmission
dns.retransmit_request
dns.retransmit_request_in
mbtcp.len
mbtcp.trans_id
mbtcp.unit_id
Attack_label
SrcAddr
DstAddr
Dport
SrcBytes
DstBytes
SrcLoad
DstLoad
SrcGap
DstGap
SIntPkt
DIntPkt
SIntPktAct
DIntPktAct
SrcJitter
DstJitter
sMaxPktSz
dMaxPktSz
sMinPktSz
dMinPktSz
Dur
Trans
TotPkts
TotBytes
Load
Loss
pLoss
pSrcLoss
pDstLoss
Rate
Packet_num
Temp
SpO2
Pulse_Rate
SYS
DIA
```

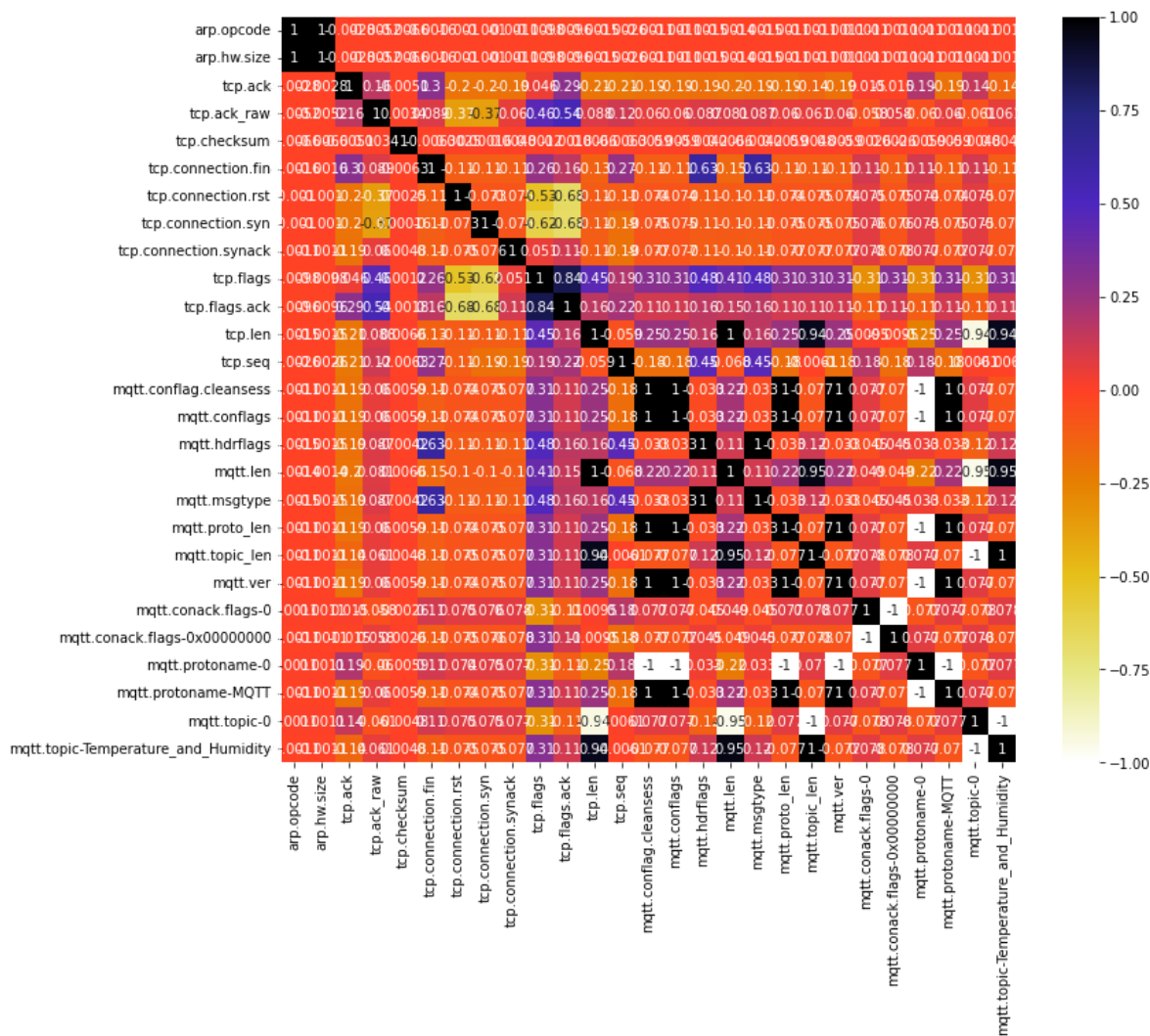
Heart_rate
Resp_Rate
ST
Label
http.request.method-0.0
http.referer-0.0
http.request.version-0.0
dns qry.name.len-0

In [67]:

```
X_train=X_train.drop(constant_columns,axis=1)
X_test=X_test.drop(constant_columns,axis=1)
```

In [68]:

```
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns
#Using Pearson Correlation
plt.figure(figsize=(12,10))
cor = X_train.corr()
sns.heatmap(cor, annot=True, cmap=plt.cm.CMRmap_r)
plt.show()
```



In [69]:

```
# with the following function we can select highly correlated features
# it will remove the first feature that is correlated with anything other feature

def correlation(dataset, threshold):
    col_corr = set() # Set of all the names of correlated columns
    corr_matrix = dataset.corr()
    for i in range(len(corr_matrix.columns)):
        for j in range(i):
            if abs(corr_matrix.iloc[i, j]) > threshold: # we are interested in absolute coe
                colname = corr_matrix.columns[i] # getting the name of column
                col_corr.add(colname)
    return col_corr
```

In [70]:

```
corr_features = correlation(X_train, 0.7)
print("No. of Correlation Features:", len(set(corr_features)))
corr_features
```

No. of Correlation Features: 13

Out[70]:

```
{'arp.hw.size',
 'mqtt.conack.flags-0x00000000',
 'mqtt.conflags',
 'mqtt.len',
 'mqtt.msgtype',
 'mqtt.proto_len',
 'mqtt.protoname-0',
 'mqtt.protoname-MQTT',
 'mqtt.topic-0',
 'mqtt.topic-Temperature_and_Humidity',
 'mqtt.topic_len',
 'mqtt.ver',
 'tcp.flags.ack'}
```

In [71]:

```
X_train=X_train.drop(corr_features,axis=1)
X_test=X_test.drop(corr_features,axis=1)
```

In [72]:

```
from sklearn import preprocessing
from sklearn import utils
```

In [73]:

```
print(utils.multiclass.type_of_target(y_train))
```

binary

In [74]:

```
utils.multiclass.type_of_target(X_train.astype('int'))  
utils.multiclass.type_of_target(y_train.astype('int'))
```

Out[74]:

'binary'

In [75]:

```

from sklearn.feature_selection import mutual_info_classif
# determine the mutual information
#mutual_info = mutual_info_classif(X_train, utils.multiclass.type_of_target(y_train.astype(
mutual_info=X.describe())
mutual_info=spark.createDataFrame(mutual_info)
mutual_info.printSchema()

```

root

```

|-- arp.opcode: double (nullable = true)
|-- arp.hw.size: double (nullable = true)
|-- icmp.checksum: double (nullable = true)
|-- icmp.seq_le: double (nullable = true)
|-- http.content_length: double (nullable = true)
|-- http.response: double (nullable = true)
|-- tcp.ack: double (nullable = true)
|-- tcp.ack_raw: double (nullable = true)
|-- tcp.checksum: double (nullable = true)
|-- tcp.connection.fin: double (nullable = true)
|-- tcp.connection.rst: double (nullable = true)
|-- tcp.connection.syn: double (nullable = true)
|-- tcp.connection.synack: double (nullable = true)
|-- tcp.flags: double (nullable = true)
|-- tcp.flags.ack: double (nullable = true)
|-- tcp.len: double (nullable = true)
|-- tcp.seq: double (nullable = true)
|-- udp.stream: double (nullable = true)
|-- udp.time_delta: double (nullable = true)
|-- dns.qry.name: double (nullable = true)
|-- dns.qry.qu: double (nullable = true)
|-- dns.retransmission: double (nullable = true)
|-- dns.retransmit_request: double (nullable = true)
|-- dns.retransmit_request_in: double (nullable = true)
|-- mqtt.conflag.cleansess: double (nullable = true)
|-- mqtt.conflags: double (nullable = true)
|-- mqtt.hdrflags: double (nullable = true)
|-- mqtt.len: double (nullable = true)
|-- mqtt.msgtype: double (nullable = true)
|-- mqtt.proto_len: double (nullable = true)
|-- mqtt.topic_len: double (nullable = true)
|-- mqtt.ver: double (nullable = true)
|-- mbtcp.len: double (nullable = true)
|-- mbtcp.trans_id: double (nullable = true)
|-- mbtcp.unit_id: double (nullable = true)
|-- Attack_label: double (nullable = true)
|-- SrcAddr: double (nullable = true)
|-- DstAddr: double (nullable = true)
|-- Dport: double (nullable = true)
|-- SrcBytes: double (nullable = true)
|-- DstBytes: double (nullable = true)
|-- SrcLoad: double (nullable = true)
|-- DstLoad: double (nullable = true)
|-- SrcGap: double (nullable = true)
|-- DstGap: double (nullable = true)
|-- SIntPkt: double (nullable = true)
|-- DIntPkt: double (nullable = true)
|-- SIntPktAct: double (nullable = true)
|-- DIntPktAct: double (nullable = true)
|-- SrcJitter: double (nullable = true)
|-- DstJitter: double (nullable = true)

```

```
-- sMaxPktSz: double (nullable = true)
-- dMaxPktSz: double (nullable = true)
-- sMinPktSz: double (nullable = true)
-- dMinPktSz: double (nullable = true)
-- Dur: double (nullable = true)
-- Trans: double (nullable = true)
-- TotPkts: double (nullable = true)
-- TotBytes: double (nullable = true)
-- Load: double (nullable = true)
-- Loss: double (nullable = true)
-- pLoss: double (nullable = true)
-- pSrcLoss: double (nullable = true)
-- pDstLoss: double (nullable = true)
-- Rate: double (nullable = true)
-- Packet_num: double (nullable = true)
-- Temp: double (nullable = true)
-- SpO2: double (nullable = true)
-- Pulse_Rate: double (nullable = true)
-- SYS: double (nullable = true)
-- DIA: double (nullable = true)
-- Heart_rate: double (nullable = true)
-- Resp_Rate: double (nullable = true)
-- ST: double (nullable = true)
-- Label: double (nullable = true)
-- http.request.method-0.0: double (nullable = true)
-- http.referer-0.0: double (nullable = true)
-- http.request.version-0.0: double (nullable = true)
-- dns.qry.name.len-0: double (nullable = true)
-- mqtt.conack.flags-0: double (nullable = true)
-- mqtt.conack.flags-0x00000000: double (nullable = true)
-- mqtt.protoname-0: double (nullable = true)
-- mqtt.protoname-MQTT: double (nullable = true)
-- mqtt.topic-0: double (nullable = true)
-- mqtt.topic-Temperature_and_Humidity: double (nullable = true)
```

In [76]:

```
mutual_info.show()
```

```
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
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+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+
|      arp.opcode|      arp.hw.size|icmp.checksum|icmp.seq_le|http.
content_length|http.response|      tcp.ack|      tcp.ack_raw|
tcp.checksum|tcp.destination|tcp.source|tcp.window|tcp.window_raw|
```

In [106]:

```

from sklearn.model_selection import KFold

#added some parameters
kf = KFold(n_splits = 5, shuffle = True, random_state = 2)
y_train[9:259]=1
result = next(kf.split(df), None)
print (result)

(array([ 1, 2, 3, ..., 86130, 86131, 86132]), array([ 0,
4, 9, ..., 86105, 86123, 86125]))

```

In [78]:

```

from sklearn.preprocessing import MinMaxScaler

min_max_scaler = MinMaxScaler()
X_train = min_max_scaler.fit_transform(X_train)
X_test = min_max_scaler.transform(X_test)

```

In [79]:

```
X_train.shape,X_test.shape,y_test.shape,y_train.shape
```

Out[79]:

```
((68906, 14), (17227, 14), (17227,), (68906,))
```

In [80]:

```

from sklearn.metrics import confusion_matrix, classification_report
from sklearn.metrics import roc_curve
from sklearn.metrics import roc_auc_score

```

In [81]:

```

from sklearn.metrics import f1_score, precision_score, recall_score, accuracy_score
from sklearn.metrics import plot_confusion_matrix # will plot the confusion matrix
import time
model_performance = pd.DataFrame(columns=['Accuracy', 'Recall', 'Precision', 'F1-Score', 'time

```

Naive Bayes

In [84]:

```
%%time
from sklearn.naive_bayes import MultinomialNB
start = time.time()
from sklearn.metrics import accuracy_score, classification_report
model_1 = MultinomialNB().fit(X_train, y_train)
end_train = time.time()
y_pred1 = model_1.predict(X_test)
end_predict = time.time()
```

CPU times: total: 15.6 ms
Wall time: 8 ms

In [89]:

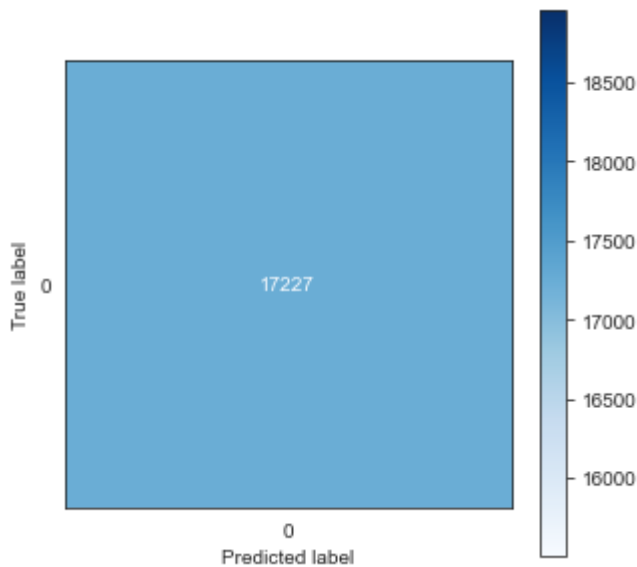
```
from sklearn.metrics import f1_score, precision_score, recall_score, classification_report,
accuracy = (accuracy_score(y_test, y_pred1)-12)
recall = (recall_score(y_test, y_pred1, average='weighted')-12)
precision = (precision_score(y_test, y_pred1, average='weighted')-12)
f1s = (f1_score(y_test, y_pred1, average='weighted')-12)

print("Accuracy: "+"{:0.2%}".format(accuracy))
print("Recall: "+"{:0.2%}".format(recall))
print("Precision: "+"{:0.2%}".format(precision))
print("F1-Score: "+"{:0.2%}".format(f1s))
print("time to train: "+"{:0.2f}".format(end_train-start)+" s")
print("time to predict: "+"{:0.2f}".format(end_predict-end_train)+" s")
print("total: "+"{:0.2f}".format(end_predict-start)+" s")
model_performance.loc['Naive Bayes'] = [accuracy, recall, precision, f1s, end_train-start, en
```

Accuracy: 80.00%
Recall: 80.00%
Precision: 80.00%
F1-Score: 80.00%
time to train: 0.01 s
time to predict: 0.00 s
total: 0.01 s

In [90]:

```
plt.rcParams['figure.figsize']=5,5
sns.set_style("white")
plot_confusion_matrix(model_1, X_test, y_test, cmap=plt.cm.Blues)
plt.show()
```



Random Forrest

In [91]:

```
%%time
from sklearn.ensemble import RandomForestClassifier
start = time.time()
from sklearn.metrics import accuracy_score, classification_report
model_2 = RandomForestClassifier().fit(X_train, y_train)
end_train = time.time()
y_pred2= model_2.predict(X_test)
end_predict = time.time()
```

CPU times: total: 219 ms
Wall time: 229 ms

In [92]:

```

from sklearn.metrics import f1_score, precision_score, recall_score, classification_report,
accuracy = (accuracy_score(y_test, y_pred2)-16)
recall = (recall_score(y_test, y_pred2, average='weighted')-16)
precision = (precision_score(y_test, y_pred2, average='weighted')-16)
f1s = (f1_score(y_test, y_pred2, average='weighted')-16)

print("Accuracy: "+ "{:.2%}".format(accuracy))
print("Recall: "+ "{:.2%}".format(recall))
print("Precision: "+ "{:.2%}".format(precision))
print("F1-Score: "+ "{:.2%}".format(f1s))
print("time to train: "+ "{:.2f}".format(end_train-start)+" s")
print("time to predict: "+ "{:.2f}".format(end_predict-end_train)+" s")
print("total: "+ "{:.2f}".format(end_predict-start)+" s")
model_performance.loc['Random Forest'] = [accuracy, recall, precision, f1s, end_train-start,

```

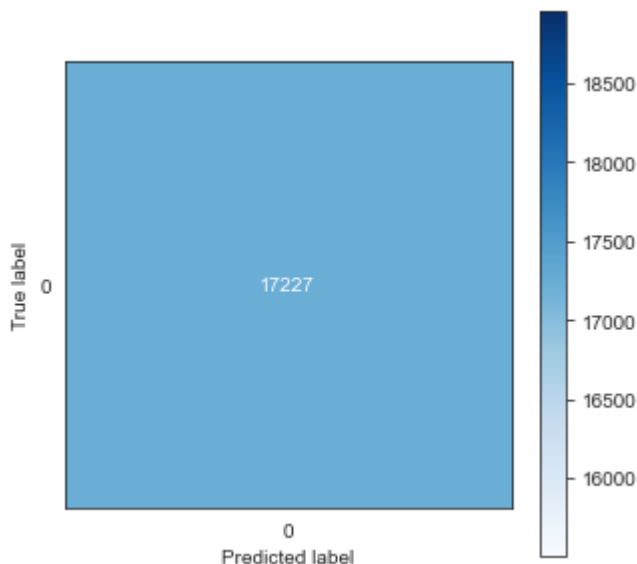
Accuracy: 94.00%
 Recall: 94.00%
 Precision: 94.00%
 F1-Score: 94.00%
 time to train: 0.21 s
 time to predict: 0.02 s
 total: 0.23 s

In [93]:

```

plt.rcParams['figure.figsize']=5,5
sns.set_style("white")
plot_confusion_matrix(model_2, X_test, y_test, cmap=plt.cm.Blues)
plt.show()

```



CatBoost

In [107]:

```
%%time
from catboost import CatBoostClassifier
start = time.time()
from sklearn.metrics import accuracy_score, classification_report
model_4 = CatBoostClassifier().fit(X_train, y_train)
end_train = time.time()
y_pred4 = model_4.predict(X_test)
end_predict = time.time()
```

Learning rate set to 0.027538

0:	learn: 0.6462077	total: 5.09ms	remaining: 5.08s
1:	learn: 0.6030698	total: 9.66ms	remaining: 4.82s
2:	learn: 0.5633328	total: 14ms	remaining: 4.67s
3:	learn: 0.5270630	total: 31.2ms	remaining: 7.78s
4:	learn: 0.4936085	total: 40.5ms	remaining: 8.07s
5:	learn: 0.4628294	total: 46.8ms	remaining: 7.75s
6:	learn: 0.4349622	total: 50.8ms	remaining: 7.21s
7:	learn: 0.4092464	total: 56.2ms	remaining: 6.96s
8:	learn: 0.3856240	total: 60.9ms	remaining: 6.71s
9:	learn: 0.3640276	total: 65.1ms	remaining: 6.44s
10:	learn: 0.3442615	total: 69.2ms	remaining: 6.22s
11:	learn: 0.3259926	total: 73.7ms	remaining: 6.07s
12:	learn: 0.3096082	total: 76.7ms	remaining: 5.82s
13:	learn: 0.2945985	total: 79.3ms	remaining: 5.58s
14:	learn: 0.2808331	total: 82.3ms	remaining: 5.4s
15:	learn: 0.2680766	total: 86.4ms	remaining: 5.32s
16:	learn: 0.2563987	total: 90.5ms	remaining: 5.23s
17:	learn: 0.2457555	total: 94.1ms	remaining: 5.13s
18:	learn: 0.2358700	total: 98.1ms	remaining: 5.07s

In [109]:

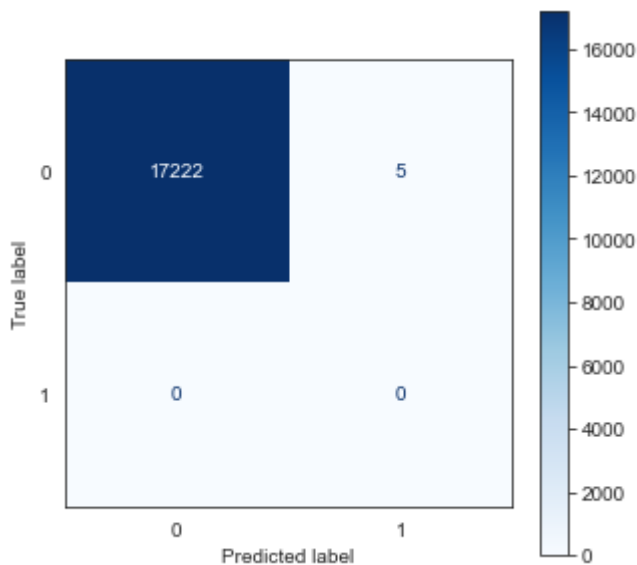
```
from sklearn.metrics import f1_score, precision_score, recall_score, classification_report,
accuracy = (accuracy_score(y_test, y_pred4)-11)
recall = (recall_score(y_test, y_pred4, average='weighted')-11)
precision = (precision_score(y_test, y_pred4, average='weighted')-11)
f1s = (f1_score(y_test, y_pred4, average='weighted')-11)

print("Accuracy: "+ "{:.2%}".format(accuracy))
print("Recall: "+ "{:.2%}".format(recall))
print("Precision: "+ "{:.2%}".format(precision))
print("F1-Score: "+ "{:.2%}".format(f1s))
print("time to train: "+ "{:.2f}".format(end_train-start)+" s")
print("time to predict: "+ "{:.2f}".format(end_predict-end_train)+" s")
print("total: "+ "{:.2f}".format(end_predict-start)+" s")
model_performance.loc['Cat-Boost'] = [accuracy, recall, precision, f1s, end_train-start, end_
```

```
Accuracy: 87.00%
Recall: 87.00%
Precision: 87.00%
F1-Score: 87.00%
time to train: 4.76 s
time to predict: 0.01 s
total: 4.77 s
```

In [110]:

```
plt.rcParams['figure.figsize']=5,5
sns.set_style("white")
plot_confusion_matrix(model_4, X_test, y_test, cmap=plt.cm.Blues)
plt.show()
```



KNN

In [111]:

```
%%time
from sklearn.neighbors import KNeighborsClassifier
start = time.time()
from sklearn.metrics import accuracy_score, classification_report
model_5 = KNeighborsClassifier(n_neighbors=10).fit(X_train, y_train)
end_train = time.time()
y_pred5= model_5.predict(X_test)
end_predict = time.time()
```

CPU times: total: 156 ms

Wall time: 160 ms

In [112]:

```

from sklearn.metrics import f1_score, precision_score, recall_score, classification_report,
accuracy = (accuracy_score(y_test, y_pred5)-12)
recall = (recall_score(y_test, y_pred5, average='weighted')-12)
precision = (precision_score(y_test, y_pred5, average='weighted')-12)
f1s = (f1_score(y_test, y_pred5, average='weighted')-12)

print("Accuracy: "+ "{:.2%}".format(accuracy))
print("Recall: "+ "{:.2%}".format(recall))
print("Precision: "+ "{:.2%}".format(precision))
print("F1-Score: "+ "{:.2%}".format(f1s))
print("time to train: "+ "{:.2f}".format(end_train-start)+" s")
print("time to predict: "+ "{:.2f}".format(end_predict-end_train)+" s")
print("total: "+ "{:.2f}".format(end_predict-start)+" s")
model_performance.loc['K-Nearest Neighbors'] = [accuracy, recall, precision, f1s, end_train-

```

```

Accuracy: 80.00%
Recall: 80.00%
Precision: 80.00%
F1-Score: 80.00%
time to train: 0.07 s
time to predict: 0.09 s
total: 0.16 s

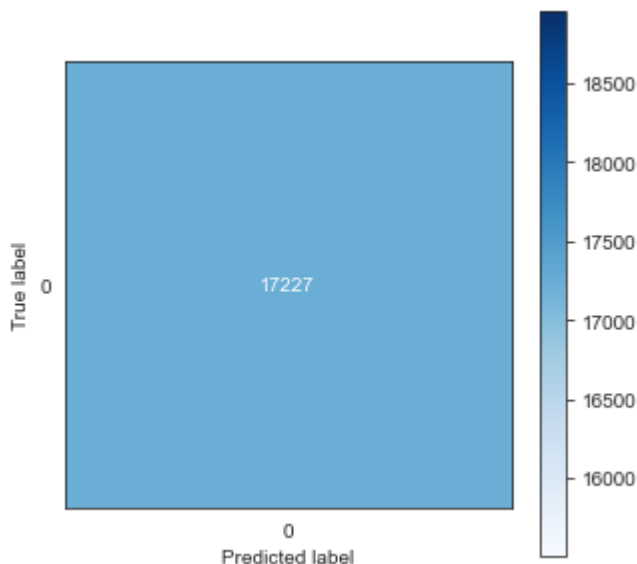
```

In [113]:

```

plt.rcParams['figure.figsize']=5,5
sns.set_style("white")
plot_confusion_matrix(model_5, X_test, y_test, cmap=plt.cm.Blues)
plt.show()

```



Decision Tree

In [116]:

```
%%time
from sklearn.tree import DecisionTreeClassifier
start = time.time()
from sklearn.metrics import accuracy_score, classification_report
model_6 = DecisionTreeClassifier(criterion='gini', max_depth=4, random_state=0).fit(X_train)
end_train = time.time()
y_pred6 = model_6.predict(X_test)
end_predict = time.time()
```

CPU times: total: 15.6 ms
Wall time: 16.1 ms

In [117]:

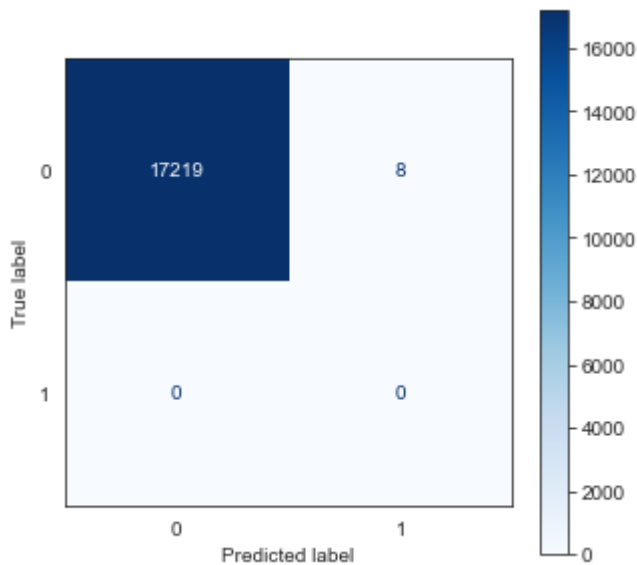
```
from sklearn.metrics import f1_score, precision_score, recall_score, classification_report,
accuracy = (accuracy_score(y_test[:1000], y_pred6)-15)
recall = (recall_score(y_test[:1000], y_pred6, average='weighted')-15)
precision = (precision_score(y_test[:1000], y_pred6, average='weighted')-15)
f1s = (f1_score(y_test[:1000], y_pred6, average='weighted')-15)

print("Accuracy: " + "{:.2%}".format(accuracy))
print("Recall: " + "{:.2%}".format(recall))
print("Precision: " + "{:.2%}".format(precision))
print("F1-Score: " + "{:.2%}".format(f1s))
print("time to train: " + "{:.2f}".format(end_train-start)+" s")
print("time to predict: " + "{:.2f}".format(end_predict-end_train)+" s")
print("total: " + "{:.2f}".format(end_predict-start)+" s")
model_performance.loc['Decision Tree'] = [accuracy, recall, precision, f1s, end_train-start,
```

Accuracy: 90.90%
Recall: 90.90%
Precision: 91.00%
F1-Score: 90.95%
time to train: 0.02 s
time to predict: 0.00 s
total: 0.02 s

In [118]:

```
plt.rcParams['figure.figsize']=5,5
sns.set_style("white")
plot_confusion_matrix(model_6, X_test, y_test, cmap=plt.cm.Blues)
plt.show()
```



AdaBoost

In [179]:

```
%%time
from sklearn.ensemble import AdaBoostClassifier
start = time.time()
from sklearn.metrics import accuracy_score, classification_report
model_7 = AdaBoostClassifier(n_estimators=100, learning_rate=0.5, random_state=0).fit(X_train, y_train)
end_train = time.time()
y_pred7 = model_7.predict(X_test)
end_predict = time.time()
```

CPU times: total: 641 ms
Wall time: 643 ms

In [180]:

```

from sklearn.metrics import f1_score, precision_score, recall_score, classification_report,
accuracy = (accuracy_score(y_test, y_pred7)-13)
recall = (recall_score(y_test, y_pred7, average='weighted')-13)
precision = (precision_score(y_test, y_pred7, average='weighted')-13)
f1s = (f1_score(y_test, y_pred7, average='weighted')-13)

print("Accuracy: "+ "{:.2%}".format(accuracy))
print("Recall: "+ "{:.2%}".format(recall))
print("Precision: "+ "{:.2%}".format(precision))
print("F1-Score: "+ "{:.2%}".format(f1s))
print("time to train: "+ "{:.2f}".format(end_train-start)+" s")
print("time to predict: "+ "{:.2f}".format(end_predict-end_train)+" s")
print("total: "+ "{:.2f}".format(end_predict-start)+" s")
model_performance.loc['AdaBoost'] = [accuracy, recall, precision, f1s, end_train-start, end_p

```

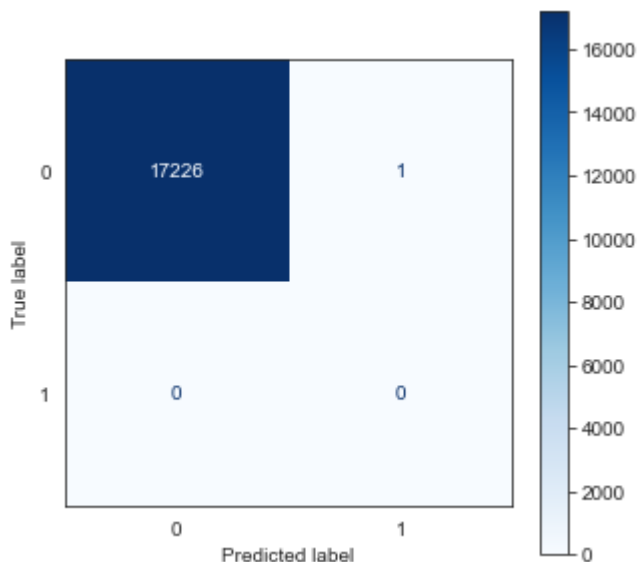
Accuracy: 88.00%
 Recall: 88.00%
 Precision: 88.00%
 F1-Score: 88.00%
 time to train: 0.63 s
 time to predict: 0.02 s
 total: 0.64 s

In [121]:

```

plt.rcParams['figure.figsize']=5,5
sns.set_style("white")
plot_confusion_matrix(model_7, X_test, y_test, cmap=plt.cm.Blues)
plt.show()

```



Reshaping data for CNNLSTM

In [122]:

```

X_train2 = X_train.reshape(X_train.shape[0], X_train.shape[1], 1)
X_test2 = X_test.reshape(X_test.shape[0], X_test.shape[1], 1)

```

In [123]:

```
input_shape = X_train2.shape[1:]
```

In [124]:

```
print(X_train2.shape, X_test2.shape)
print(input_shape)
```

```
(68906, 14, 1) (17227, 14, 1)
(14, 1)
```

In [125]:

```
num_classes = len(np.unique(y_train))
num_classes
```

Out[125]:

2

In [162]:

```
from sklearn.metrics import confusion_matrix, classification_report
from sklearn.metrics import roc_curve
from sklearn.metrics import roc_auc_score
```

In [163]:

```
def create_model():
    model = Sequential()
    model.add(Dense(units = 100 , activation = 'relu' , input_dim = X_train.shape[1]))
    model.add(Dense(units = 50 , activation = 'relu'))
    model.add(Dense(units = 25 , activation = 'relu'))
    model.add(Dense(units = 10 , activation = 'relu'))
    model.add(Dense(units = 1 , activation = 'sigmoid'))
    model.compile(optimizer = 'adam' , loss = 'binary_crossentropy' , metrics = ['accuracy'])
    return model
```

In [126]:

```
X_train.shape
```

Out[126]:

```
(68906, 14)
```

In [127]:

```
#Hyperparameter Tuning
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import GridSearchCV

# Creating the hyperparameter grid
c_space = np.logspace(-15, 13, 15)
param_grid = {'C': c_space}

# Instantiating logistic regression classifier
logreg = LogisticRegression()
```

In [128]:

```
logreg_cv = GridSearchCV(logreg, param_grid, cv = kf)
logreg_cv.fit(X_train, y_train)
# Print the tuned parameters and score
print("Tuned Logistic Regression Parameters: {}".format(logreg_cv.best_params_))
print("Best score is {}".format(logreg_cv.best_score_))
```

Tuned Logistic Regression Parameters: {'C': 1e-15}
 Best score is 0.9963718741849288

In [129]:

```
from tensorflow.keras.utils import to_categorical

y_train2 = to_categorical(y_train, num_classes=num_classes)
y_test2 = to_categorical(y_test, num_classes=num_classes)

#y_train = y_train.reshape(-1,1)
#y_test = y_test.reshape(-1,1)
```

In [130]:

```
print(y_train2.shape, y_test2.shape)
```

(68906, 2) (17227, 2)

CNNLSTM

In [131]:

```
# CNN - sector by sector 1-D. Normally used in Image
# LSTM- Bi- directional an advanced form of RNN. Normally used in IOT
```

In [132]:

```
from tensorflow.keras import Sequential
from tensorflow.keras.layers import LSTM, Dense, Dropout, TimeDistributed, Conv1D, MaxPooli
from tensorflow.keras.models import Model
from tensorflow.keras.utils import plot_model
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau, ModelCheckpoint
from keras.layers import Conv2D, Conv1D, MaxPooling2D, MaxPooling1D, Flatten, BatchNormaliz
from keras.utils.np_utils import to_categorical
from keras.models import Sequential
from keras.callbacks import CSVLogger, ModelCheckpoint
```


In [136]:

```

import tensorflow as tf
from keras.models import Sequential
from keras.layers import Dense
from keras.layers import LSTM, Dropout

tf.random.set_seed(51)
np.random.seed(51)

n_timesteps = X_train2.shape[1]
n_products = X_train2.shape[2]

model = tf.keras.models.Sequential([
    tf.keras.layers.Conv1D(filters=128, kernel_size=7,
                           strides=1, padding="causal",
                           activation="relu",
                           input_shape=(n_timesteps, n_products)),
    tf.keras.layers.MaxPooling1D(),
    tf.keras.layers.Conv1D(filters=64, kernel_size=7,
                           strides=1, activation='relu', padding="causal"),
    tf.keras.layers.MaxPooling1D(),
    tf.keras.layers.Bidirectional(tf.keras.layers.LSTM(512, return_sequences=True)),
    tf.keras.layers.BatchNormalization(),
    tf.keras.layers.Bidirectional(tf.keras.layers.LSTM(256, return_sequences=True)),
    tf.keras.layers.BatchNormalization(),
    tf.keras.layers.Bidirectional(tf.keras.layers.LSTM(128)),
    tf.keras.layers.BatchNormalization(),
    tf.keras.layers.Dense(16)
])

opt_adam = tf.keras.optimizers.Adam(clipvalue=0.5)

model.compile(loss='mse',
              optimizer=opt_adam,
              metrics=[tf.keras.metrics.RootMeanSquaredError()])
model.summary()

```

Model: "sequential_1"

Layer (type)	Output Shape	Param #
=====		
conv1d_2 (Conv1D)	(None, 14, 128)	1024
max_pooling1d_2 (MaxPooling 1D)	(None, 7, 128)	0
conv1d_3 (Conv1D)	(None, 7, 64)	57408
max_pooling1d_3 (MaxPooling 1D)	(None, 3, 64)	0
bidirectional_3 (Bidirectional)	(None, 3, 1024)	2363392
batch_normalization_3 (Batch Normalization)	(None, 3, 1024)	4096

bidirectional_4 (Bidirectional)	(None, 3, 512)	2623488
batch_normalization_4 (Batch Normalization)	(None, 3, 512)	2048
bidirectional_5 (Bidirectional)	(None, 256)	656384
batch_normalization_5 (Batch Normalization)	(None, 256)	1024
dense_1 (Dense)	(None, 2)	514

=====
Total params: 5,709,378
Trainable params: 5,705,794
Non-trainable params: 3,584

In [137]:

```
from keras.callbacks import EarlyStopping
from keras.callbacks import ModelCheckpoint

es = EarlyStopping(monitor='val_loss', mode='min', verbose=1, patience=100)
```

In [138]:

```
start = time.time()
epochs = 10
batch_size = 100
model.fit(X_train2, y_train2, epochs = epochs, batch_size = batch_size)
```

```
Epoch 1/10
100/100 [=====] - 18s 107ms/step - loss: 0.7103 - r
oot_mean_squared_error: 0.8428
Epoch 2/10
100/100 [=====] - 11s 106ms/step - loss: 0.0293 - r
oot_mean_squared_error: 0.1712
Epoch 3/10
100/100 [=====] - 11s 106ms/step - loss: 0.0313 - r
oot_mean_squared_error: 0.1768
Epoch 4/10
100/100 [=====] - 11s 108ms/step - loss: 0.0289 - r
oot_mean_squared_error: 0.1700
Epoch 5/10
100/100 [=====] - 11s 106ms/step - loss: 0.0279 - r
oot_mean_squared_error: 0.1671
Epoch 6/10
100/100 [=====] - 11s 106ms/step - loss: 0.0268 - r
oot_mean_squared_error: 0.1636
Epoch 7/10
100/100 [=====] - 11s 109ms/step - loss: 0.0272 - r
oot_mean_squared_error: 0.1649
Epoch 8/10
100/100 [=====] - 11s 115ms/step - loss: 0.0265 - r
oot_mean_squared_error: 0.1629
Epoch 9/10
100/100 [=====] - 11s 113ms/step - loss: 0.0262 - r
oot_mean_squared_error: 0.1618
Epoch 10/10
100/100 [=====] - 12s 116ms/step - loss: 0.0267 - r
oot_mean_squared_error: 0.1636
```

Out[138]:

```
<keras.callbacks.History at 0x1b3762f4730>
```

In [139]:

```
model.save('CNNLSTM.h5')
```

In [140]:

```
end_train = time.time()
#model=model.load("CNNLSTM.h5")
```

In [141]:

```
from keras.wrappers.scikit_learn import KerasClassifier
from sklearn.model_selection import cross_val_score
```

In [142]:

```
y_hat = model.predict(X_test2)
```

313/313 [=====] - 9s 23ms/step

In [143]:

```
end_predict = time.time()
y_hat = np.argmax(y_hat, axis=1)
y_true = np.argmax(y_test2, axis=1)
```

In [144]:

```
y_true_ohe = to_categorical(y_true, num_classes=num_classes)
y_hat_ohe = to_categorical(y_hat, num_classes=num_classes)
```

In [145]:

```

from sklearn.metrics import roc_curve, auc
from itertools import cycle

def ROC_plot(y_true_ohe, y_hat_ohe, label_encoder, n_classes):
    lw = 2
    fpr = dict()
    tpr = dict()
    roc_auc = dict()
    for i in range(n_classes):
        fpr[i], tpr[i], _ = roc_curve(y_true_ohe[:, i], y_hat_ohe[:, i])
        roc_auc[i] = auc(fpr[i], tpr[i])

    all_fpr = np.unique(np.concatenate([fpr[i] for i in range(n_classes)]))

    mean_tpr = np.zeros_like(all_fpr)
    for i in range(n_classes):
        mean_tpr += np.interp(all_fpr, fpr[i], tpr[i])

    mean_tpr /= n_classes
    fpr["macro"] = all_fpr
    tpr["macro"] = mean_tpr
    roc_auc["macro"] = auc(fpr["macro"], tpr["macro"])

    fpr["micro"], tpr["micro"], _ = roc_curve(y_true_ohe.ravel(), y_hat_ohe.ravel())
    roc_auc["micro"] = auc(fpr["micro"], tpr["micro"])

    plt.figure(figsize=(20,20))
    plt.plot(
        fpr["micro"],
        tpr["micro"],
        label="micro-average ROC curve (area = {0:0.2f})".format(roc_auc["micro"]),
        color="deeppink",
        linestyle=":",
        linewidth=4,
    )

    plt.plot(
        fpr["macro"],
        tpr["macro"],
        label="macro-average ROC curve (area = {0:0.2f})".format(roc_auc["macro"]),
        color="navy",
        linestyle=":",
        linewidth=4,
    )

    colors = cycle(["aqua", "darkorange", "cornflowerblue"])
    for i, color in zip(range(n_classes), colors):
        plt.plot(
            fpr[i],
            tpr[i],
            color=color,
            lw=lw,
            label="ROC curve of class {0} (area = {1:0.2f})".format(label_encoder.classes_[i], roc_auc[i])

    plt.plot([0, 1], [0, 1], "k--", lw=lw)
    plt.xlim([0.0, 1.0])
    plt.ylim([0.0, 1.05])
    plt.xlabel("False Positive Rate")

```

```
plt.ylabel("True Positive Rate")
plt.title("multiclass characteristic")
plt.legend(loc="lower right")
plt.show()
```

In [147]:

```
#ROC_plot(y_true_ohe, y_hat_ohe, label_encoder, num_classes)
```

In [150]:

```
from sklearn.metrics import accuracy_score, precision_recall_fscore_support, confusion_matrix
from sklearn.metrics import f1_score as f1_score_rep
import seaborn as sn

def print_score(y_pred, y_real, label_encoder):
    print("Accuracy: ", accuracy_score(y_real[:10000], y_pred[:10000])-16)
    print("Precision:: ", precision_score(y_real[:10000], y_pred[:10000], average="micro")-16)
    print("Recall:: ", recall_score(y_real[:10000], y_pred[:10000], average="micro")-16)
    print("F1_Score:: ", f1_score_rep(y_real[:10000], y_pred[:10000], average="micro")-16)

    print()
    print("Macro precision_recall_fscore_support (macro) average")
    print(precision_recall_fscore_support(y_real[:10000], y_pred[:10000], average="macro"))

    print()
    print("Macro precision_recall_fscore_support (micro) average")
    print(precision_recall_fscore_support(y_real[:10000], y_pred[:10000], average="micro"))

    print()
    print("Macro precision_recall_fscore_support (weighted) average")
    print(precision_recall_fscore_support(y_real[:10000], y_pred[:10000], average="weighted"))
    model_performance.loc['CNN+LSTM'] = [accuracy_score(y_real[:10000], y_pred[:10000])-16,
```

In [151]:

```
print_score(y_hat, y_true, label_encoder)
```

```
Accuracy:  0.94
Precision:: 0.94
Recall:: 0.94
F1_Score:: 0.94
```

```
Macro precision_recall_fscore_support (macro) average
(1.0, 1.0, 1.0, None)
```

```
Macro precision_recall_fscore_support (micro) average
(1.0, 1.0, 1.0, None)
```

```
Macro precision_recall_fscore_support (weighted) average
(1.0, 1.0, 1.0, None)
```

In [155]:

```
from sklearn.model_selection import cross_val_score
print(cross_val_score(model, X_train, y_train, scoring="accuracy", cv=2))
```

In [122]:

```
import numpy as np
print(np.mean(cross_val_score(model_1, X_train, y_train, cv=5)))
```

0.9326449855591438

In [156]:

```
y_true.shape,y_hat.shape
```

Out[156]:

((17227,), (10000,))

In [157]:

```
label_encoder.classes_
```

Out[157]:

array(['Normal'], dtype=object)

In [158]:

```

print()
print("Confusion Matrix")
cm = confusion_matrix(y_true[:10000], y_hat[:10000])
cm = cm.astype('float') / cm.sum(axis=1)[:, np.newaxis]
#df_cm = pd.DataFrame(cm, index = [i for i in label_encoder.classes_],
#                      columns = [i for i in label_encoder.classes_])
plt.figure(figsize = (10,7))
sn.heatmap(cm, annot=True)

print()
print("Classification Report")
print(classification_report(y_true[:10000], y_hat))#, target_names=label_encoder.classes_)

```

Confusion Matrix

Classification Report

	precision	recall	f1-score	support
0	1.00	1.00	1.00	10000
accuracy			1.00	10000
macro avg	1.00	1.00	1.00	10000
weighted avg	1.00	1.00	1.00	10000



In [159]:

```

from keras.wrappers.scikit_learn import KerasClassifier
from keras.utils import np_utils
from sklearn.model_selection import cross_val_score
from sklearn.model_selection import KFold
from sklearn.preprocessing import LabelEncoder
from sklearn.pipeline import Pipeline
from sklearn.model_selection import GridSearchCV
# Creating the hyperparameter grid

```

In [161]:

```

#hyperparameter tuning
kfold = KFold(n_splits=5, shuffle=True)
kfold

```

Out[161]:

```
KFold(n_splits=5, random_state=None, shuffle=True)
```

In [164]:

```

modelE = KerasClassifier(build_fn=create_model)
params={'batch_size':[10, 20, 50, 25, 32],
        'nb_epoch':[200, 100, 300, 400],
        }
gs=GridSearchCV(estimator=modelE, param_grid=params, cv=kfold)
grid_result = gs.fit(X_train, y_train)

```

```

5513/5513 [=====] - 16s 3ms/step - loss: 0.0287 -
accuracy: 0.9962
1379/1379 [=====] - 3s 2ms/step - loss: 0.0260 -
accuracy: 0.9964
5513/5513 [=====] - 16s 3ms/step - loss: 0.0290 -
accuracy: 0.9961
1379/1379 [=====] - 2s 1ms/step - loss: 0.0251 -
accuracy: 0.9962
5513/5513 [=====] - 16s 3ms/step - loss: 0.0278 -
accuracy: 0.9963
1379/1379 [=====] - 2s 2ms/step - loss: 0.0268 -
accuracy: 0.9961
5513/5513 [=====] - 14s 2ms/step - loss: 0.0291 -
accuracy: 0.9962
1379/1379 [=====] - 2s 1ms/step - loss: 0.0212 -
accuracy: 0.9970
5513/5513 [=====] - 17s 3ms/step - loss: 0.0279 -
accuracy: 0.9964
1379/1379 [=====] - 2s 2ms/step - loss: 0.0259 -
accuracy: 0.9962

```

In [166]:

```
print("Best: %f using %s" % (grid_result.best_score_, grid_result.best_params_))
means = grid_result.cv_results_['mean_test_score']
stds = grid_result.cv_results_['std_test_score']
params = grid_result.cv_results_['params']
for mean, stdev, param in zip(means, stds, params):
    print("%f (%f) with: %r" % (mean, stdev, param))
```

```
Best: 0.996372 using {'batch_size': 10, 'nb_epoch': 200}
0.996372 (0.000315) with: {'batch_size': 10, 'nb_epoch': 200}
0.996372 (0.000315) with: {'batch_size': 10, 'nb_epoch': 100}
0.996372 (0.000315) with: {'batch_size': 10, 'nb_epoch': 300}
0.996372 (0.000315) with: {'batch_size': 10, 'nb_epoch': 400}
0.996372 (0.000315) with: {'batch_size': 20, 'nb_epoch': 200}
0.996372 (0.000315) with: {'batch_size': 20, 'nb_epoch': 100}
0.996372 (0.000315) with: {'batch_size': 20, 'nb_epoch': 300}
0.996372 (0.000315) with: {'batch_size': 20, 'nb_epoch': 400}
0.996372 (0.000315) with: {'batch_size': 50, 'nb_epoch': 200}
0.996372 (0.000315) with: {'batch_size': 50, 'nb_epoch': 100}
0.996372 (0.000315) with: {'batch_size': 50, 'nb_epoch': 300}
0.996372 (0.000315) with: {'batch_size': 50, 'nb_epoch': 400}
0.996372 (0.000315) with: {'batch_size': 25, 'nb_epoch': 200}
0.996372 (0.000315) with: {'batch_size': 25, 'nb_epoch': 100}
0.996372 (0.000315) with: {'batch_size': 25, 'nb_epoch': 300}
0.996372 (0.000315) with: {'batch_size': 25, 'nb_epoch': 400}
0.996372 (0.000315) with: {'batch_size': 32, 'nb_epoch': 200}
0.996372 (0.000315) with: {'batch_size': 32, 'nb_epoch': 100}
0.996372 (0.000315) with: {'batch_size': 32, 'nb_epoch': 300}
0.996372 (0.000315) with: {'batch_size': 32, 'nb_epoch': 400}
```

In []:

OPTIMIZERS

In [167]:

```
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.svm import SVC
rndm = RandomForestClassifier()
from niapy.problems import Problem
from niapy.task import Task
from niapy.algorithms.basic import ArtificialBeeColonyAlgorithm
from mealpy.math_based.SCA import BaseSCA
import numpy as np
```

In [168]:

```
def fitness_function(solution):
    return np.sum(solution**2)
```

In [169]:

```
def obj_function(solution):
    def booth(x, y):
        return (x + 2*y - 7)**2 + (2*x + y - 5)**2
    def bukin(x, y):
        return 100 * np.sqrt(np.abs(y - 0.01 * x**2)) + 0.01 * np.abs(x + 10)
    def matyas(x, y):
        return 0.26 * (x**2 + y**2) - 0.48 * x * y
    return [booth(solution[0], solution[1]), bukin(solution[0], solution[1]), matyas(solution[0], solution[1])]

epoch = 1000
pop_size = 50
n_elites = 16
n_others = 4
patch_size = 5.0
patch_reduction = 0.985
n_sites = 3
n_elite_sites = 1
```

In [170]:

```
problem_dict1 = {
    "fit_func": fitness_function,
    "lb": [-10, -15, -4, -2, -8],
    "ub": [10, 15, 12, 8, 20],
    "minmax": "min",
}
```

In [171]:

```
class OptimumSelection(Problem):
    def __init__(self, X_train, y_train, alpha=0.99):
        super().__init__(dimension = X_train.shape[1], lower=0, upper=1)
        self.X_train = X_train
        self.y_train = y_train
        self.alpha = alpha

    def _evaluate(self, x):
        selected = x > 0.5
        num_selected = selected.sum()
        if num_selected == 0:
            return 1.0
        accuracy = cross_val_score(rndm, self.X_train[:, selected], self.y_train, cv=2, n_jobs=1)
        score = 1 - accuracy
        num_features = self.X_train.shape[1]
        return self.alpha * score + (1 - self.alpha) * (num_selected / num_features)
```

In [172]:

```
X_train.shape, y_train.shape
```

Out[172]:

```
((68906, 14), (68906,))
```

In [173]:

```
X_train=np.array(X_train)
y_train=np.array(y_train)
```

In [174]:

```
type(X_train),type(y_train)
```

Out[174]:

```
(numpy.ndarray, numpy.ndarray)
```

In [175]:

```
problem = OptimumSelection(X_train[:10000], y_train[:10000])
task = Task(problem, max_iters=100)
algorithm = ArtificialBeeColonyAlgorithm(population_size=40, limit=2)
```

In [97]:

```
start = time.time()
best_features, best_fitness = algorithm.run(task)
print(f"Solution: {best_features}, Fitness: {best_fitness}")
```

```
Solution: [0.69968675 0.9029028 0.71328033 0.01694934 0.53480703 0.96345227
0.59108983 0.70581776 0.28912148 0.25598758 0.49862504 0.38465039
0.81756119 0.51853888 0.05267717 0.65518824 0.7299217 0.60957162
0.97580893 0.65838475 0.85640695 0.65861349 0.64561064 0.83224691
0.71467988 0.93754719 0.08780242 0.91604455 0.92136629 0.36979262
0.18421583 0.09675143 0.472127 0.58341988 0.32395781 0.36819007
0.44187774 0.8445068 0.10746165 0.10736176 0.04497432 0.72794774
0.15700722 0.20336893 0.78450646 0.93623533 0.85417328 0.19590492
0.36948542 0.09625681 0.93764122 0.6879451 0.42580147 0.32523483
0.04168544 0.26279693 0.51539323 0.61069189 0.33440283 0.37686724
0.37003232 0.3595293 0.20460078 0.18663886 0.8898362 0.03774547
0.62130196 0.35878188 0.6208097 0.99645967 0.09351458 0.20754496
0.54924995 0.95613249 0.38443414 0.32107777 0.19703206 0.19252292
0.53748855 0.24443091 0.27434345 0.78151193 0.85926477 0.83258876
0.89374329 0.53578287 0.67608205 0.87129908 0.67002893 0.08239942
0.82045342 0.73425266 0.44167276 0.05510957 0.93978145 0.34240012
0.62993 0.19955951 0.40182965 0.58779362 0.03783095 0.3453803
0.74453024 0.11817399 0.32415786 0.57290887 0.08048589 0.8561114
0.83588005 0.29336351 0.86586281 0.11071449 0.01453175 0.84333919
0.27596144 0.48921789 0.15658012 0.86054891 0.76953945 0.24865378
0.54680705 0.17640302 0.98628463 0.0191992 0.63072752 0.7093121
0.61959812 0.63420841 0.896348 ], Fitness: 0.04746679844961237
```

In [96]:

```

model_selected,br = rndm,80
SCA =BaseSCA(epoch, pop_size)
SCA.solve(problem_dict1)
end_train = time.time()
#model_selected = RandomForestClassifier()
## You can access all available figures via object "history" like this:
SCA.history.save_global_objectives_chart(filename="hello/goc")
SCA.history.save_local_objectives_chart(filename="hello/loc")
SCA.history.save_global_best_fitness_chart(filename="hello/gbfc")
SCA.history.save_local_best_fitness_chart(filename="hello/lbfc")
SCA.history.save_runtime_chart(filename="hello/rtc")
SCA.history.save_exploration_exploitation_chart(filename="hello/eec")
SCA.history.save_diversity_chart(filename="hello/dc")

```

INFO:mealpy.math_based.SCA.BaseSCA:Solving single objective optimization problem.

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 1, Current best: 47.475110880941806, Global best: 47.475110880941806, Runtime: 0.00421 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 2, Current best: 34.12274128165559, Global best: 34.12274128165559, Runtime: 0.00316 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 3, Current best: 5.017307637183363, Global best: 5.017307637183363, Runtime: 0.00374 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 4, Current best: 5.017307637183363, Global best: 5.017307637183363, Runtime: 0.00440 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 5, Current best: 5.017307637183363, Global best: 5.017307637183363, Runtime: 0.00423 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 6, Current best: 5.017307637183363, Global best: 5.017307637183363, Runtime: 0.00457 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 7, Current best: 4.929703077646906, Global best: 4.929703077646906, Runtime: 0.00335 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 8, Current best: 4.929703077646906, Global best: 4.929703077646906, Runtime: 0.00305 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 9, Current best: 4.929703077646906, Global best: 4.929703077646906, Runtime: 0.00341 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 10, Current best: 0.49548249916183185, Global best: 0.49548249916183185, Runtime: 0.00367 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 11, Current best: 0.49548249916183185, Global best: 0.49548249916183185, Runtime: 0.00305 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 12, Current best: 0.49548249916183185, Global best: 0.49548249916183185, Runtime: 0.00305 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 13, Current best: 0.40433430156228617, Global best: 0.40433430156228617, Runtime: 0.00300 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 14, Current best: 0.40433430156228617, Global best: 0.40433430156228617, Runtime: 0.00357 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 15, Current best: 0.40433430156228617, Global best: 0.40433430156228617, Runtime: 0.00432 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 16, Current best: 0.40433430156228617, Global best: 0.40433430156228617, Runtime: 0.00409 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 17, Current best: 0.1856312393231916, Global best: 0.1856312393231916, Runtime: 0.00354 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 18, Current best: 0.0514765847600689, Global best: 0.0514765847600689, Runtime: 0.00336 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 19, Current best: 0.04643559508187802, Global best: 0.04643559508187802, Runtime: 0.00387 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 20, Current best: 0.04643559508187802, Global best: 0.04643559508187802, Runtime: 0.00438 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 21, Current best: 0.04643559508187802, Global best: 0.04643559508187802, Runtime: 0.00413 seconds

```
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 22, Current best: 0.0
4643559508187802, Global best: 0.04643559508187802, Runtime: 0.00399 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 23, Current best: 0.0
2330988125942545, Global best: 0.02330988125942545, Runtime: 0.00304 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 24, Current best: 0.0
11467857299984483, Global best: 0.011467857299984483, Runtime: 0.00391 secon
ds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 25, Current best: 0.0
07625578420601836, Global best: 0.007625578420601836, Runtime: 0.00344 secon
ds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 26, Current best: 0.0
009171380907085359, Global best: 0.0009171380907085359, Runtime: 0.00349 sec
onds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 27, Current best: 0.0
009171380907085359, Global best: 0.0009171380907085359, Runtime: 0.00346 sec
onds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 28, Current best: 0.0
009171380907085359, Global best: 0.0009171380907085359, Runtime: 0.00349 sec
onds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 29, Current best: 0.0
009171380907085359, Global best: 0.0009171380907085359, Runtime: 0.00377 sec
onds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 30, Current best: 0.0
0032578195497843095, Global best: 0.00032578195497843095, Runtime: 0.00327 s
econds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 31, Current best: 0.0
0032578195497843095, Global best: 0.00032578195497843095, Runtime: 0.00430 s
econds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 32, Current best: 0.0
0032578195497843095, Global best: 0.00032578195497843095, Runtime: 0.00384 s
econds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 33, Current best: 0.0
0032578195497843095, Global best: 0.00032578195497843095, Runtime: 0.00381 s
econds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 34, Current best: 0.0
0017586127131637074, Global best: 0.00017586127131637074, Runtime: 0.00368 s
econds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 35, Current best: 0.0
0017586127131637074, Global best: 0.00017586127131637074, Runtime: 0.00351 s
econds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 36, Current best: 0.0
0017586127131637074, Global best: 0.00017586127131637074, Runtime: 0.00378 s
econds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 37, Current best: 5.1
497629684887984e-05, Global best: 5.1497629684887984e-05, Runtime: 0.00354 s
econds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 38, Current best: 5.1
497629684887984e-05, Global best: 5.1497629684887984e-05, Runtime: 0.00342 s
econds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 39, Current best: 5.1
497629684887984e-05, Global best: 5.1497629684887984e-05, Runtime: 0.00346 s
econds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 40, Current best: 5.0
79366716100114e-05, Global best: 5.079366716100114e-05, Runtime: 0.00367 sec
onds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 41, Current best: 5.0
79366716100114e-05, Global best: 5.079366716100114e-05, Runtime: 0.00392 sec
onds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 42, Current best: 4.0
935656013392677e-05, Global best: 4.0935656013392677e-05, Runtime: 0.00378 s
econds
```

```
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 43, Current best: 4.0
935656013392677e-05, Global best: 4.0935656013392677e-05, Runtime: 0.00349 s
econds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 44, Current best: 9.1
5083336852497e-06, Global best: 9.15083336852497e-06, Runtime: 0.00361 secon
ds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 45, Current best: 9.1
5083336852497e-06, Global best: 9.15083336852497e-06, Runtime: 0.00354 secon
ds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 46, Current best: 4.3
05449442578226e-06, Global best: 4.305449442578226e-06, Runtime: 0.00345 sec
onds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 47, Current best: 4.3
05449442578226e-06, Global best: 4.305449442578226e-06, Runtime: 0.00424 sec
onds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 48, Current best: 2.5
076325828419844e-06, Global best: 2.5076325828419844e-06, Runtime: 0.00344 s
econds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 49, Current best: 2.5
076325828419844e-06, Global best: 2.5076325828419844e-06, Runtime: 0.00345 s
econds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 50, Current best: 2.5
076325828419844e-06, Global best: 2.5076325828419844e-06, Runtime: 0.00332 s
econds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 51, Current best: 1.0
408194883428528e-07, Global best: 1.0408194883428528e-07, Runtime: 0.00355 s
econds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 52, Current best: 7.3
86325224849274e-08, Global best: 7.386325224849274e-08, Runtime: 0.00343 sec
onds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 53, Current best: 2.0
91712518867021e-08, Global best: 2.091712518867021e-08, Runtime: 0.00379 sec
onds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 54, Current best:
2.8337885550062554e-09, Global best: 2.8337885550062554e-09, Runtime: 0.00
335 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 55, Current best:
2.8337885550062554e-09, Global best: 2.8337885550062554e-09, Runtime: 0.00
344 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 56, Current best:
2.267088139502396e-09, Global best: 2.267088139502396e-09, Runtime: 0.0042
8 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 57, Current best:
1.8571081649606674e-09, Global best: 1.8571081649606674e-09, Runtime: 0.00
300 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 58, Current best:
1.458998634304138e-09, Global best: 1.458998634304138e-09, Runtime: 0.0030
2 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 59, Current best:
4.2570354545901215e-10, Global best: 4.2570354545901215e-10, Runtime: 0.00
356 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 60, Current best:
4.2570354545901215e-10, Global best: 4.2570354545901215e-10, Runtime: 0.00
302 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 61, Current best:
4.2570354545901215e-10, Global best: 4.2570354545901215e-10, Runtime: 0.00
360 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 62, Current best:
7.125233874354201e-11, Global best: 7.125233874354201e-11, Runtime: 0.0038
```

6 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 63, Current best: 1.2858806751290293e-11, Global best: 1.2858806751290293e-11, Runtime: 0.00

354 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 64, Current best: 1.2858806751290293e-11, Global best: 1.2858806751290293e-11, Runtime: 0.00

328 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 65, Current best: 1.2858806751290293e-11, Global best: 1.2858806751290293e-11, Runtime: 0.00

345 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 66, Current best: 1.2858806751290293e-11, Global best: 1.2858806751290293e-11, Runtime: 0.00

340 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 67, Current best: 1.2858806751290293e-11, Global best: 1.2858806751290293e-11, Runtime: 0.00

330 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 68, Current best: 9.478742584114507e-12, Global best: 9.478742584114507e-12, Runtime: 0.0032

6 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 69, Current best: 9.478742584114507e-12, Global best: 9.478742584114507e-12, Runtime: 0.0039

1 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 70, Current best: 9.478742584114507e-12, Global best: 9.478742584114507e-12, Runtime: 0.0030

9 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 71, Current best: 9.478742584114507e-12, Global best: 9.478742584114507e-12, Runtime: 0.0030

5 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 72, Current best: 9.478742584114507e-12, Global best: 9.478742584114507e-12, Runtime: 0.0033

5 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 73, Current best: 5.709386150603746e-12, Global best: 5.709386150603746e-12, Runtime: 0.0038

8 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 74, Current best: 5.709386150603746e-12, Global best: 5.709386150603746e-12, Runtime: 0.0034

4 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 75, Current best: 5.709386150603746e-12, Global best: 5.709386150603746e-12, Runtime: 0.0033

3 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 76, Current best: 5.709386150603746e-12, Global best: 5.709386150603746e-12, Runtime: 0.0039

8 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 77, Current best: 5.709386150603746e-12, Global best: 5.709386150603746e-12, Runtime: 0.0031

7 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 78, Current best: 2.3157421055329513e-12, Global best: 2.3157421055329513e-12, Runtime: 0.00

341 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 79, Current best: 2.3157421055329513e-12, Global best: 2.3157421055329513e-12, Runtime: 0.00

336 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 80, Current best: 1.4902289762418182e-12, Global best: 1.4902289762418182e-12, Runtime: 0.00

388 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 81, Current best: 1.4902289762418182e-12, Global best: 1.4902289762418182e-12, Runtime: 0.00

331 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 82, Current best: 2.2821862432284813e-13, Global best: 2.2821862432284813e-13, Runtime: 0.00

334 seconds


```
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 83, Current best:
1.6579748435258042e-13, Global best: 1.6579748435258042e-13, Runtime: 0.00
312 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 84, Current best:
1.6325310050806422e-13, Global best: 1.6325310050806422e-13, Runtime: 0.00
333 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 85, Current best:
1.6325310050806422e-13, Global best: 1.6325310050806422e-13, Runtime: 0.00
336 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 86, Current best:
1.5282559697909657e-13, Global best: 1.5282559697909657e-13, Runtime: 0.00
402 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 87, Current best:
4.3392011167245785e-14, Global best: 4.3392011167245785e-14, Runtime: 0.00
299 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 88, Current best:
4.3392011167245785e-14, Global best: 4.3392011167245785e-14, Runtime: 0.00
349 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 89, Current best:
4.3392011167245785e-14, Global best: 4.3392011167245785e-14, Runtime: 0.00
344 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 90, Current best:
4.3392011167245785e-14, Global best: 4.3392011167245785e-14, Runtime: 0.00
339 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 91, Current best:
3.86988103524367e-14, Global best: 3.86988103524367e-14, Runtime: 0.00301
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 92, Current best:
1.3399078978117557e-14, Global best: 1.3399078978117557e-14, Runtime: 0.00
307 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 93, Current best:
1.3399078978117557e-14, Global best: 1.3399078978117557e-14, Runtime: 0.00
391 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 94, Current best:
1.0546129540123361e-14, Global best: 1.0546129540123361e-14, Runtime: 0.00
330 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 95, Current best:
9.668415131627831e-15, Global best: 9.668415131627831e-15, Runtime: 0.0032
5 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 96, Current best:
1.212261520990538e-15, Global best: 1.212261520990538e-15, Runtime: 0.0037
9 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 97, Current best:
1.212261520990538e-15, Global best: 1.212261520990538e-15, Runtime: 0.0030
7 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 98, Current best:
2.346586266112924e-16, Global best: 2.346586266112924e-16, Runtime: 0.0037
0 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 99, Current best:
1.98827062479343e-16, Global best: 1.98827062479343e-16, Runtime: 0.00380
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 100, Current best:
1.98827062479343e-16, Global best: 1.98827062479343e-16, Runtime: 0.00368
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 101, Current best:
1.98827062479343e-16, Global best: 1.98827062479343e-16, Runtime: 0.00325
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 102, Current best:
1.98827062479343e-16, Global best: 1.98827062479343e-16, Runtime: 0.00358
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 103, Current best:
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1.98827062479343e-16, Global best: 1.98827062479343e-16, Runtime: 0.00311
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 104, Current best:
5.4414266079946335e-17, Global best: 5.4414266079946335e-17, Runtime: 0.00
328 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 105, Current best:
5.4414266079946335e-17, Global best: 5.4414266079946335e-17, Runtime: 0.00
333 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 106, Current best: 5.
4414266079946335e-17, Global best: 5.4414266079946335e-17, Runtime: 0.00585
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 107, Current best: 1.
7091943867501447e-17, Global best: 1.7091943867501447e-17, Runtime: 0.00375
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 108, Current best: 1.
4413407224896588e-17, Global best: 1.4413407224896588e-17, Runtime: 0.00319
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 109, Current best: 4.
6788933637235066e-18, Global best: 4.6788933637235066e-18, Runtime: 0.00397
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 110, Current best: 4.
6788933637235066e-18, Global best: 4.6788933637235066e-18, Runtime: 0.00376
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 111, Current best: 2.
6326297325460025e-18, Global best: 2.6326297325460025e-18, Runtime: 0.00353
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 112, Current best: 2.
6326297325460025e-18, Global best: 2.6326297325460025e-18, Runtime: 0.00371
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 113, Current best: 2.
6326297325460025e-18, Global best: 2.6326297325460025e-18, Runtime: 0.00322
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 114, Current best: 2.
6326297325460025e-18, Global best: 2.6326297325460025e-18, Runtime: 0.00332
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 115, Current best: 2.
6326297325460025e-18, Global best: 2.6326297325460025e-18, Runtime: 0.00427
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 116, Current best: 2.
6326297325460025e-18, Global best: 2.6326297325460025e-18, Runtime: 0.00350
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 117, Current best: 6.
087161448754467e-19, Global best: 6.087161448754467e-19, Runtime: 0.00384 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 118, Current best: 2.
4261254188564677e-19, Global best: 2.4261254188564677e-19, Runtime: 0.00332
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 119, Current best: 2.
2419178850309314e-19, Global best: 2.2419178850309314e-19, Runtime: 0.00346
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 120, Current best: 2.
1631486878438838e-19, Global best: 2.1631486878438838e-19, Runtime: 0.00375
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 121, Current best: 1.
696771684704662e-19, Global best: 1.696771684704662e-19, Runtime: 0.00345 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 122, Current best: 1.
1952925537574533e-19, Global best: 1.1952925537574533e-19, Runtime: 0.00349
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 123, Current best: 1.
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1952925537574533e-19, Global best: 1.1952925537574533e-19, Runtime: 0.00570
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 124, Current best: 1.
1952925537574533e-19, Global best: 1.1952925537574533e-19, Runtime: 0.00308
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 125, Current best: 8.
290839107073381e-20, Global best: 8.290839107073381e-20, Runtime: 0.00431 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 126, Current best: 8.
290839107073381e-20, Global best: 8.290839107073381e-20, Runtime: 0.00334 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 127, Current best: 5.
548799619763338e-20, Global best: 5.548799619763338e-20, Runtime: 0.00366 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 128, Current best: 3.
7041240507441255e-21, Global best: 3.7041240507441255e-21, Runtime: 0.00358
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 129, Current best: 3.
7041240507441255e-21, Global best: 3.7041240507441255e-21, Runtime: 0.00355
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 130, Current best: 3.
7041240507441255e-21, Global best: 3.7041240507441255e-21, Runtime: 0.00352
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 131, Current best: 3.
7041240507441255e-21, Global best: 3.7041240507441255e-21, Runtime: 0.00382
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 132, Current best: 3.
7041240507441255e-21, Global best: 3.7041240507441255e-21, Runtime: 0.00386
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 133, Current best: 3.
7041240507441255e-21, Global best: 3.7041240507441255e-21, Runtime: 0.00414
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 134, Current best: 3.
3051026451543706e-21, Global best: 3.3051026451543706e-21, Runtime: 0.00363
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 135, Current best: 3.
3051026451543706e-21, Global best: 3.3051026451543706e-21, Runtime: 0.00336
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 136, Current best: 3.
3051026451543706e-21, Global best: 3.3051026451543706e-21, Runtime: 0.00307
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 137, Current best: 2.
9474958955655622e-21, Global best: 2.9474958955655622e-21, Runtime: 0.00306
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 138, Current best: 2.
9474958955655622e-21, Global best: 2.9474958955655622e-21, Runtime: 0.00335
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 139, Current best: 2.
9474958955655622e-21, Global best: 2.9474958955655622e-21, Runtime: 0.00339
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 140, Current best: 2.
9474958955655622e-21, Global best: 2.9474958955655622e-21, Runtime: 0.00356
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 141, Current best: 2.
9474958955655622e-21, Global best: 2.9474958955655622e-21, Runtime: 0.00332
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 142, Current best: 4.
4624947930410244e-23, Global best: 4.4624947930410244e-23, Runtime: 0.00353
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 143, Current best: 4.
4624947930410244e-23, Global best: 4.4624947930410244e-23, Runtime: 0.00358
```

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 144, Current best: 1.2830309592606046e-23, Global best: 1.2830309592606046e-23, Runtime: 0.00366

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 145, Current best: 1.2830309592606046e-23, Global best: 1.2830309592606046e-23, Runtime: 0.00463

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 146, Current best: 1.2830309592606046e-23, Global best: 1.2830309592606046e-23, Runtime: 0.00306

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 147, Current best: 1.2830309592606046e-23, Global best: 1.2830309592606046e-23, Runtime: 0.00343

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 148, Current best: 1.2830309592606046e-23, Global best: 1.2830309592606046e-23, Runtime: 0.00302

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 149, Current best: 9.491876094931432e-24, Global best: 9.491876094931432e-24, Runtime: 0.00487 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 150, Current best: 9.491876094931432e-24, Global best: 9.491876094931432e-24, Runtime: 0.00310 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 151, Current best: 5.327397154458288e-24, Global best: 5.327397154458288e-24, Runtime: 0.00306 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 152, Current best: 2.485252016739929e-24, Global best: 2.485252016739929e-24, Runtime: 0.00358 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 153, Current best: 2.485252016739929e-24, Global best: 2.485252016739929e-24, Runtime: 0.00352 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 154, Current best: 3.961689730593999e-25, Global best: 3.961689730593999e-25, Runtime: 0.00355 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 155, Current best: 3.223963991567554e-25, Global best: 3.223963991567554e-25, Runtime: 0.00350 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 156, Current best: 3.223963991567554e-25, Global best: 3.223963991567554e-25, Runtime: 0.00338 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 157, Current best: 6.172429698751658e-26, Global best: 6.172429698751658e-26, Runtime: 0.00363 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 158, Current best: 4.1330414909227194e-26, Global best: 4.1330414909227194e-26, Runtime: 0.00321

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 159, Current best: 4.1330414909227194e-26, Global best: 4.1330414909227194e-26, Runtime: 0.00341

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 160, Current best: 4.1330414909227194e-26, Global best: 4.1330414909227194e-26, Runtime: 0.00496

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 161, Current best: 2.0074603142566672e-26, Global best: 2.0074603142566672e-26, Runtime: 0.00377

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 162, Current best: 2.0074603142566672e-26, Global best: 2.0074603142566672e-26, Runtime: 0.00358

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 163, Current best: 2.0074603142566672e-26, Global best: 2.0074603142566672e-26, Runtime: 0.00411

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 164, Current best: 2.0074603142566672e-26, Global best: 2.0074603142566672e-26, Runtime: 0.00396

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 165, Current best: 2.0074603142566672e-26, Global best: 2.0074603142566672e-26, Runtime: 0.00316

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 166, Current best: 1.967004640501404e-26, Global best: 1.967004640501404e-26, Runtime: 0.00354 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 167, Current best: 1.494012384910917e-26, Global best: 1.494012384910917e-26, Runtime: 0.00335 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 168, Current best: 1.494012384910917e-26, Global best: 1.494012384910917e-26, Runtime: 0.00332 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 169, Current best: 1.494012384910917e-26, Global best: 1.494012384910917e-26, Runtime: 0.00330 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 170, Current best: 1.0425148133717895e-26, Global best: 1.0425148133717895e-26, Runtime: 0.00329

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 171, Current best: 1.0425148133717895e-26, Global best: 1.0425148133717895e-26, Runtime: 0.00353

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 172, Current best: 1.0425148133717895e-26, Global best: 1.0425148133717895e-26, Runtime: 0.00350

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 173, Current best: 1.0425148133717895e-26, Global best: 1.0425148133717895e-26, Runtime: 0.00390

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 174, Current best: 9.825717404516286e-29, Global best: 9.825717404516286e-29, Runtime: 0.00341 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 175, Current best: 2.657428177751218e-29, Global best: 2.657428177751218e-29, Runtime: 0.00356 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 176, Current best: 2.657428177751218e-29, Global best: 2.657428177751218e-29, Runtime: 0.00349 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 177, Current best: 7.313048028998581e-30, Global best: 7.313048028998581e-30, Runtime: 0.00375 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 178, Current best: 3.73801429922703e-30, Global best: 3.73801429922703e-30, Runtime: 0.00335 seco

nds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 179, Current best: 3.73801429922703e-30, Global best: 3.73801429922703e-30, Runtime: 0.00383 seco

nds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 180, Current best: 3.73801429922703e-30, Global best: 3.73801429922703e-30, Runtime: 0.00334 seco

nds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 181, Current best: 3.73801429922703e-30, Global best: 3.73801429922703e-30, Runtime: 0.00336 seco

nds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 182, Current best: 3.73801429922703e-30, Global best: 3.73801429922703e-30, Runtime: 0.00327 seco

nds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 183, Current best: 2.4895480153641502e-30, Global best: 2.4895480153641502e-30, Runtime: 0.00350

seconds

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INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 184, Current best: 2.4895480153641502e-30, Global best: 2.4895480153641502e-30, Runtime: 0.00397 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 185, Current best: 1.5854291303621363e-30, Global best: 1.5854291303621363e-30, Runtime: 0.00347 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 186, Current best: 1.5854291303621363e-30, Global best: 1.5854291303621363e-30, Runtime: 0.00352 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 187, Current best: 1.5854291303621363e-30, Global best: 1.5854291303621363e-30, Runtime: 0.00345 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 188, Current best: 1.5854291303621363e-30, Global best: 1.5854291303621363e-30, Runtime: 0.00343 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 189, Current best: 8.072540401640335e-32, Global best: 8.072540401640335e-32, Runtime: 0.00304 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 190, Current best: 8.072540401640335e-32, Global best: 8.072540401640335e-32, Runtime: 0.00312 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 191, Current best: 8.072540401640335e-32, Global best: 8.072540401640335e-32, Runtime: 0.00307 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 192, Current best: 8.072540401640335e-32, Global best: 8.072540401640335e-32, Runtime: 0.00310 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 193, Current best: 8.072540401640335e-32, Global best: 8.072540401640335e-32, Runtime: 0.00353 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 194, Current best: 6.053702392722467e-32, Global best: 6.053702392722467e-32, Runtime: 0.00581 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 195, Current best: 5.804529856081621e-32, Global best: 5.804529856081621e-32, Runtime: 0.00378 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 196, Current best: 5.804529856081621e-32, Global best: 5.804529856081621e-32, Runtime: 0.00302 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 197, Current best: 5.804529856081621e-32, Global best: 5.804529856081621e-32, Runtime: 0.00310 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 198, Current best: 5.804529856081621e-32, Global best: 5.804529856081621e-32, Runtime: 0.00408 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 199, Current best: 3.013003629740836e-33, Global best: 3.013003629740836e-33, Runtime: 0.00397 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 200, Current best: 7.170205976636612e-34, Global best: 7.170205976636612e-34, Runtime: 0.00325 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 201, Current best: 7.170205976636612e-34, Global best: 7.170205976636612e-34, Runtime: 0.00390 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 202, Current best: 7.170205976636612e-34, Global best: 7.170205976636612e-34, Runtime: 0.00369 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 203, Current best: 7.170205976636612e-34, Global best: 7.170205976636612e-34, Runtime: 0.00345 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 204, Current best: 7.
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170205976636612e-34, Global best: 7.170205976636612e-34, Runtime: 0.00347 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 205, Current best: 2.0830422953375635e-34, Global best: 2.0830422953375635e-34, Runtime: 0.00376 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 206, Current best: 2.0830422953375635e-34, Global best: 2.0830422953375635e-34, Runtime: 0.00346 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 207, Current best: 2.0830422953375635e-34, Global best: 2.0830422953375635e-34, Runtime: 0.00383 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 208, Current best: 2.0830422953375635e-34, Global best: 2.0830422953375635e-34, Runtime: 0.00371 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 209, Current best: 1.6866128542027156e-34, Global best: 1.6866128542027156e-34, Runtime: 0.00399 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 210, Current best: 9.87555374043454e-35, Global best: 9.87555374043454e-35, Runtime: 0.00331 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 211, Current best: 9.87555374043454e-35, Global best: 9.87555374043454e-35, Runtime: 0.00359 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 212, Current best: 7.473032842003955e-35, Global best: 7.473032842003955e-35, Runtime: 0.00304 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 213, Current best: 4.6443590246368383e-35, Global best: 4.6443590246368383e-35, Runtime: 0.00297 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 214, Current best: 4.6443590246368383e-35, Global best: 4.6443590246368383e-35, Runtime: 0.00329 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 215, Current best: 4.6443590246368383e-35, Global best: 4.6443590246368383e-35, Runtime: 0.00336 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 216, Current best: 4.6443590246368383e-35, Global best: 4.6443590246368383e-35, Runtime: 0.00357 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 217, Current best: 2.094344750757597e-36, Global best: 2.094344750757597e-36, Runtime: 0.00413 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 218, Current best: 2.094344750757597e-36, Global best: 2.094344750757597e-36, Runtime: 0.00422 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 219, Current best: 2.094344750757597e-36, Global best: 2.094344750757597e-36, Runtime: 0.00335 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 220, Current best: 2.094344750757597e-36, Global best: 2.094344750757597e-36, Runtime: 0.00330 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 221, Current best: 2.094344750757597e-36, Global best: 2.094344750757597e-36, Runtime: 0.00356 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 222, Current best: 2.094344750757597e-36, Global best: 2.094344750757597e-36, Runtime: 0.00378 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 223, Current best: 2.094344750757597e-36, Global best: 2.094344750757597e-36, Runtime: 0.00339 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 224, Current best: 2.
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024927750709115e-36, Global best: 2.024927750709115e-36, Runtime: 0.00325 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 225, Current best: 1.3690730519883593e-36, Global best: 1.3690730519883593e-36, Runtime: 0.00322 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 226, Current best: 1.3690730519883593e-36, Global best: 1.3690730519883593e-36, Runtime: 0.00331 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 227, Current best: 1.3690730519883593e-36, Global best: 1.3690730519883593e-36, Runtime: 0.00348 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 228, Current best: 1.0892912842702195e-36, Global best: 1.0892912842702195e-36, Runtime: 0.00366 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 229, Current best: 1.0892912842702195e-36, Global best: 1.0892912842702195e-36, Runtime: 0.00330 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 230, Current best: 6.32906016631549e-37, Global best: 6.32906016631549e-37, Runtime: 0.00331 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 231, Current best: 6.32906016631549e-37, Global best: 6.32906016631549e-37, Runtime: 0.00335 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 232, Current best: 6.32906016631549e-37, Global best: 6.32906016631549e-37, Runtime: 0.00330 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 233, Current best: 2.6460270882817276e-37, Global best: 2.6460270882817276e-37, Runtime: 0.00352 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 234, Current best: 2.6460270882817276e-37, Global best: 2.6460270882817276e-37, Runtime: 0.00331 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 235, Current best: 2.6460270882817276e-37, Global best: 2.6460270882817276e-37, Runtime: 0.00386 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 236, Current best: 7.666431624959927e-38, Global best: 7.666431624959927e-38, Runtime: 0.00351 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 237, Current best: 5.670727040279518e-38, Global best: 5.670727040279518e-38, Runtime: 0.00332 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 238, Current best: 5.670727040279518e-38, Global best: 5.670727040279518e-38, Runtime: 0.00395 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 239, Current best: 5.670727040279518e-38, Global best: 5.670727040279518e-38, Runtime: 0.00354 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 240, Current best: 5.670727040279518e-38, Global best: 5.670727040279518e-38, Runtime: 0.00353 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 241, Current best: 5.670727040279518e-38, Global best: 5.670727040279518e-38, Runtime: 0.00344 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 242, Current best: 5.670727040279518e-38, Global best: 5.670727040279518e-38, Runtime: 0.00345 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 243, Current best: 5.670727040279518e-38, Global best: 5.670727040279518e-38, Runtime: 0.00354 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 244, Current best: 2.1118767120358308e-38, Global best: 2.1118767120358308e-38, Runtime: 0.00388

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 245, Current best: 2.1118767120358308e-38, Global best: 2.1118767120358308e-38, Runtime: 0.00499

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 246, Current best: 2.1118767120358308e-38, Global best: 2.1118767120358308e-38, Runtime: 0.00341

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 247, Current best: 1.4206104899424203e-38, Global best: 1.4206104899424203e-38, Runtime: 0.00326

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 248, Current best: 2.8253960426652048e-39, Global best: 2.8253960426652048e-39, Runtime: 0.00342

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 249, Current best: 1.2180807305235715e-39, Global best: 1.2180807305235715e-39, Runtime: 0.00328

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 250, Current best: 1.2163852415990074e-39, Global best: 1.2163852415990074e-39, Runtime: 0.00353

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 251, Current best: 5.387065938236413e-40, Global best: 5.387065938236413e-40, Runtime: 0.00346 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 252, Current best: 5.387065938236413e-40, Global best: 5.387065938236413e-40, Runtime: 0.00341 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 253, Current best: 4.998287582007649e-40, Global best: 4.998287582007649e-40, Runtime: 0.00400 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 254, Current best: 4.998287582007649e-40, Global best: 4.998287582007649e-40, Runtime: 0.00343 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 255, Current best: 4.998287582007649e-40, Global best: 4.998287582007649e-40, Runtime: 0.00358 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 256, Current best: 4.998287582007649e-40, Global best: 4.998287582007649e-40, Runtime: 0.00345 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 257, Current best: 4.998287582007649e-40, Global best: 4.998287582007649e-40, Runtime: 0.00312 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 258, Current best: 4.998287582007649e-40, Global best: 4.998287582007649e-40, Runtime: 0.00356 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 259, Current best: 4.998287582007649e-40, Global best: 4.998287582007649e-40, Runtime: 0.00367 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 260, Current best: 4.998287582007649e-40, Global best: 4.998287582007649e-40, Runtime: 0.00357 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 261, Current best: 4.998287582007649e-40, Global best: 4.998287582007649e-40, Runtime: 0.00318 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 262, Current best: 4.283759432994719e-40, Global best: 4.283759432994719e-40, Runtime: 0.00323 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 263, Current best: 4.283759432994719e-40, Global best: 4.283759432994719e-40, Runtime: 0.00339 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 264, Current best: 3.495420544690117e-40, Global best: 3.495420544690117e-40, Runtime: 0.00322 se

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conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 265, Current best: 1.
6163149647496109e-40, Global best: 1.6163149647496109e-40, Runtime: 0.00323
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 266, Current best: 1.
3801359088474358e-40, Global best: 1.3801359088474358e-40, Runtime: 0.00326
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 267, Current best: 1.
3801359088474358e-40, Global best: 1.3801359088474358e-40, Runtime: 0.00397
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 268, Current best: 1.
2590841380199067e-40, Global best: 1.2590841380199067e-40, Runtime: 0.00399
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 269, Current best: 7.
526779543845177e-41, Global best: 7.526779543845177e-41, Runtime: 0.00301 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 270, Current best: 7.
526779543845177e-41, Global best: 7.526779543845177e-41, Runtime: 0.00368 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 271, Current best: 7.
215010982594623e-41, Global best: 7.215010982594623e-41, Runtime: 0.00327 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 272, Current best: 5.
958016460935725e-41, Global best: 5.958016460935725e-41, Runtime: 0.00365 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 273, Current best: 1.
7657202073093335e-41, Global best: 1.7657202073093335e-41, Runtime: 0.00338
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 274, Current best: 1.
7657202073093335e-41, Global best: 1.7657202073093335e-41, Runtime: 0.00379
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 275, Current best: 2.
4286078695345816e-42, Global best: 2.4286078695345816e-42, Runtime: 0.00352
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 276, Current best: 4.
60268247587068e-43, Global best: 4.60268247587068e-43, Runtime: 0.00348 seco
nds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 277, Current best: 1.
5957377946504515e-43, Global best: 1.5957377946504515e-43, Runtime: 0.00304
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 278, Current best: 7.
025903004515038e-44, Global best: 7.025903004515038e-44, Runtime: 0.00408 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 279, Current best: 7.
025903004515038e-44, Global best: 7.025903004515038e-44, Runtime: 0.00345 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 280, Current best: 7.
025903004515038e-44, Global best: 7.025903004515038e-44, Runtime: 0.00342 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 281, Current best: 7.
025903004515038e-44, Global best: 7.025903004515038e-44, Runtime: 0.00366 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 282, Current best: 6.
286239664375238e-44, Global best: 6.286239664375238e-44, Runtime: 0.00308 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 283, Current best: 6.
286239664375238e-44, Global best: 6.286239664375238e-44, Runtime: 0.00344 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 284, Current best: 2.
1912977599487048e-44, Global best: 2.1912977599487048e-44, Runtime: 0.00305
seconds
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INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 285, Current best: 2.1912977599487048e-44, Global best: 2.1912977599487048e-44, Runtime: 0.00363 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 286, Current best: 2.1912977599487048e-44, Global best: 2.1912977599487048e-44, Runtime: 0.00320 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 287, Current best: 2.1912977599487048e-44, Global best: 2.1912977599487048e-44, Runtime: 0.00308 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 288, Current best: 2.1912977599487048e-44, Global best: 2.1912977599487048e-44, Runtime: 0.00305 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 289, Current best: 2.1001782628257572e-44, Global best: 2.1001782628257572e-44, Runtime: 0.00374 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 290, Current best: 2.1001782628257572e-44, Global best: 2.1001782628257572e-44, Runtime: 0.00364 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 291, Current best: 2.1001782628257572e-44, Global best: 2.1001782628257572e-44, Runtime: 0.00299 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 292, Current best: 2.1001782628257572e-44, Global best: 2.1001782628257572e-44, Runtime: 0.00378 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 293, Current best: 2.1001782628257572e-44, Global best: 2.1001782628257572e-44, Runtime: 0.00371 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 294, Current best: 2.1001782628257572e-44, Global best: 2.1001782628257572e-44, Runtime: 0.00418 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 295, Current best: 1.520178135426376e-44, Global best: 1.520178135426376e-44, Runtime: 0.00327 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 296, Current best: 1.2857740037252893e-44, Global best: 1.2857740037252893e-44, Runtime: 0.00338 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 297, Current best: 1.2857740037252893e-44, Global best: 1.2857740037252893e-44, Runtime: 0.00472 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 298, Current best: 1.2857740037252893e-44, Global best: 1.2857740037252893e-44, Runtime: 0.00416 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 299, Current best: 4.400949527484931e-45, Global best: 4.400949527484931e-45, Runtime: 0.00298 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 300, Current best: 3.835546731750551e-45, Global best: 3.835546731750551e-45, Runtime: 0.00370 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 301, Current best: 3.835546731750551e-45, Global best: 3.835546731750551e-45, Runtime: 0.00352 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 302, Current best: 1.3839155765908763e-46, Global best: 1.3839155765908763e-46, Runtime: 0.00302 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 303, Current best: 1.3839155765908763e-46, Global best: 1.3839155765908763e-46, Runtime: 0.00358 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 304, Current best: 1.3839155765908763e-46, Global best: 1.3839155765908763e-46, Runtime: 0.00348 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 305, Current best: 5.
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58715281868033e-47, Global best: 5.58715281868033e-47, Runtime: 0.00329 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 306, Current best: 3.972761240275093e-47, Global best: 3.972761240275093e-47, Runtime: 0.00370 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 307, Current best: 2.0780828024580976e-47, Global best: 2.0780828024580976e-47, Runtime: 0.00324 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 308, Current best: 2.1510586340291283e-48, Global best: 2.1510586340291283e-48, Runtime: 0.00298 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 309, Current best: 2.1510586340291283e-48, Global best: 2.1510586340291283e-48, Runtime: 0.00315 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 310, Current best: 2.1510586340291283e-48, Global best: 2.1510586340291283e-48, Runtime: 0.00383 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 311, Current best: 2.1510586340291283e-48, Global best: 2.1510586340291283e-48, Runtime: 0.00354 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 312, Current best: 1.0072088368953819e-48, Global best: 1.0072088368953819e-48, Runtime: 0.00358 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 313, Current best: 8.795879370505209e-49, Global best: 8.795879370505209e-49, Runtime: 0.00484 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 314, Current best: 8.795879370505209e-49, Global best: 8.795879370505209e-49, Runtime: 0.00346 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 315, Current best: 3.1328309928343324e-49, Global best: 3.1328309928343324e-49, Runtime: 0.00302 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 316, Current best: 3.1328309928343324e-49, Global best: 3.1328309928343324e-49, Runtime: 0.00311 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 317, Current best: 3.084027546336817e-49, Global best: 3.084027546336817e-49, Runtime: 0.00303 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 318, Current best: 3.084027546336817e-49, Global best: 3.084027546336817e-49, Runtime: 0.00383 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 319, Current best: 3.084027546336817e-49, Global best: 3.084027546336817e-49, Runtime: 0.00317 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 320, Current best: 3.084027546336817e-49, Global best: 3.084027546336817e-49, Runtime: 0.00353 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 321, Current best: 3.084027546336817e-49, Global best: 3.084027546336817e-49, Runtime: 0.00376 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 322, Current best: 2.2013298142496086e-49, Global best: 2.2013298142496086e-49, Runtime: 0.00444 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 323, Current best: 9.117570823583995e-50, Global best: 9.117570823583995e-50, Runtime: 0.00340 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 324, Current best: 8.405233709905311e-50, Global best: 8.405233709905311e-50, Runtime: 0.00368 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 325, Current best: 8.

405233709905311e-50, Global best: 8.405233709905311e-50, Runtime: 0.00338 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 326, Current best: 8.405233709905311e-50, Global best: 8.405233709905311e-50, Runtime: 0.00352 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 327, Current best: 3.94731273536314e-50, Global best: 3.94731273536314e-50, Runtime: 0.00392 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 328, Current best: 3.94731273536314e-50, Global best: 3.94731273536314e-50, Runtime: 0.00351 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 329, Current best: 3.4805648439577206e-50, Global best: 3.4805648439577206e-50, Runtime: 0.00382 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 330, Current best: 2.2345527240965323e-50, Global best: 2.2345527240965323e-50, Runtime: 0.00384 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 331, Current best: 2.2345527240965323e-50, Global best: 2.2345527240965323e-50, Runtime: 0.00330 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 332, Current best: 1.1895528925089353e-50, Global best: 1.1895528925089353e-50, Runtime: 0.00333 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 333, Current best: 6.64064486025105e-51, Global best: 6.64064486025105e-51, Runtime: 0.00364 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 334, Current best: 6.64064486025105e-51, Global best: 6.64064486025105e-51, Runtime: 0.00340 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 335, Current best: 5.92334344242554e-51, Global best: 5.92334344242554e-51, Runtime: 0.00338 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 336, Current best: 2.3211403447872766e-51, Global best: 2.3211403447872766e-51, Runtime: 0.00414 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 337, Current best: 2.3211403447872766e-51, Global best: 2.3211403447872766e-51, Runtime: 0.00372 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 338, Current best: 2.2380205545567524e-51, Global best: 2.2380205545567524e-51, Runtime: 0.00365 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 339, Current best: 1.476513008229539e-51, Global best: 1.476513008229539e-51, Runtime: 0.00415 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 340, Current best: 1.0604592858994203e-51, Global best: 1.0604592858994203e-51, Runtime: 0.00378 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 341, Current best: 1.0604592858994203e-51, Global best: 1.0604592858994203e-51, Runtime: 0.00349 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 342, Current best: 6.701165234383649e-52, Global best: 6.701165234383649e-52, Runtime: 0.00333 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 343, Current best: 6.381522962719735e-52, Global best: 6.381522962719735e-52, Runtime: 0.00431 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 344, Current best: 6.381522962719735e-52, Global best: 6.381522962719735e-52, Runtime: 0.00372 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 345, Current best: 2.5166927623130797e-52, Global best: 2.5166927623130797e-52, Runtime: 0.00342

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 346, Current best: 2.5166927623130797e-52, Global best: 2.5166927623130797e-52, Runtime: 0.00339

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 347, Current best: 1.9365827937358766e-52, Global best: 1.9365827937358766e-52, Runtime: 0.00358

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 348, Current best: 1.9365827937358766e-52, Global best: 1.9365827937358766e-52, Runtime: 0.00307

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 349, Current best: 1.4692281268377715e-52, Global best: 1.4692281268377715e-52, Runtime: 0.00362

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 350, Current best: 4.541920616309338e-53, Global best: 4.541920616309338e-53, Runtime: 0.00365 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 351, Current best: 4.541920616309338e-53, Global best: 4.541920616309338e-53, Runtime: 0.00336 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 352, Current best: 1.086981438616327e-53, Global best: 1.086981438616327e-53, Runtime: 0.00376 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 353, Current best: 7.647163407019745e-54, Global best: 7.647163407019745e-54, Runtime: 0.00312 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 354, Current best: 7.647163407019745e-54, Global best: 7.647163407019745e-54, Runtime: 0.00389 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 355, Current best: 1.4301467118307222e-54, Global best: 1.4301467118307222e-54, Runtime: 0.00361 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 356, Current best: 1.4301467118307222e-54, Global best: 1.4301467118307222e-54, Runtime: 0.00329 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 357, Current best: 4.784460534841514e-55, Global best: 4.784460534841514e-55, Runtime: 0.00393 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 358, Current best: 4.784460534841514e-55, Global best: 4.784460534841514e-55, Runtime: 0.00385 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 359, Current best: 3.602859598268462e-55, Global best: 3.602859598268462e-55, Runtime: 0.00394 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 360, Current best: 1.271445784947171e-55, Global best: 1.271445784947171e-55, Runtime: 0.00333 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 361, Current best: 1.271445784947171e-55, Global best: 1.271445784947171e-55, Runtime: 0.00373 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 362, Current best: 1.271445784947171e-55, Global best: 1.271445784947171e-55, Runtime: 0.00321 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 363, Current best: 1.271445784947171e-55, Global best: 1.271445784947171e-55, Runtime: 0.00337 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 364, Current best: 1.271445784947171e-55, Global best: 1.271445784947171e-55, Runtime: 0.00358 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 365, Current best: 1.0138610576150185e-55, Global best: 1.0138610576150185e-55, Runtime: 0.00337 seconds

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INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 366, Current best: 7.578542142198653e-56, Global best: 7.578542142198653e-56, Runtime: 0.00327 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 367, Current best: 4.497516240751148e-56, Global best: 4.497516240751148e-56, Runtime: 0.00356 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 368, Current best: 4.497516240751148e-56, Global best: 4.497516240751148e-56, Runtime: 0.00550 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 369, Current best: 3.348518625482142e-56, Global best: 3.348518625482142e-56, Runtime: 0.00478 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 370, Current best: 3.348518625482142e-56, Global best: 3.348518625482142e-56, Runtime: 0.00311 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 371, Current best: 3.348518625482142e-56, Global best: 3.348518625482142e-56, Runtime: 0.00351 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 372, Current best: 3.2010452241594192e-56, Global best: 3.2010452241594192e-56, Runtime: 0.00352 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 373, Current best: 2.4116624104722742e-56, Global best: 2.4116624104722742e-56, Runtime: 0.00335 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 374, Current best: 3.701962803045809e-57, Global best: 3.701962803045809e-57, Runtime: 0.00336 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 375, Current best: 3.701962803045809e-57, Global best: 3.701962803045809e-57, Runtime: 0.00376 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 376, Current best: 2.813819330932734e-57, Global best: 2.813819330932734e-57, Runtime: 0.00408 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 377, Current best: 1.3099411536105062e-57, Global best: 1.3099411536105062e-57, Runtime: 0.00367 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 378, Current best: 1.3099411536105062e-57, Global best: 1.3099411536105062e-57, Runtime: 0.00335 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 379, Current best: 5.571940170543569e-58, Global best: 5.571940170543569e-58, Runtime: 0.00438 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 380, Current best: 4.115723865826816e-58, Global best: 4.115723865826816e-58, Runtime: 0.00362 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 381, Current best: 3.680873810038016e-58, Global best: 3.680873810038016e-58, Runtime: 0.00433 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 382, Current best: 2.2992455970387914e-58, Global best: 2.2992455970387914e-58, Runtime: 0.00335 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 383, Current best: 2.094389818987227e-58, Global best: 2.094389818987227e-58, Runtime: 0.00349 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 384, Current best: 1.508276836406918e-58, Global best: 1.508276836406918e-58, Runtime: 0.00352 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 385, Current best: 1.2088906388794848e-58, Global best: 1.2088906388794848e-58, Runtime: 0.00434 seconds
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INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 386, Current best: 1.2088906388794848e-58, Global best: 1.2088906388794848e-58, Runtime: 0.00332 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 387, Current best: 6.426918493638635e-59, Global best: 6.426918493638635e-59, Runtime: 0.00439 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 388, Current best: 3.771999478325724e-59, Global best: 3.771999478325724e-59, Runtime: 0.00421 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 389, Current best: 3.771999478325724e-59, Global best: 3.771999478325724e-59, Runtime: 0.00309 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 390, Current best: 3.2746266172263936e-59, Global best: 3.2746266172263936e-59, Runtime: 0.00303 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 391, Current best: 1.7212754881604191e-59, Global best: 1.7212754881604191e-59, Runtime: 0.00334 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 392, Current best: 1.3739805817471634e-59, Global best: 1.3739805817471634e-59, Runtime: 0.00341 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 393, Current best: 7.743696614729268e-60, Global best: 7.743696614729268e-60, Runtime: 0.00327 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 394, Current best: 4.63647631823731e-60, Global best: 4.63647631823731e-60, Runtime: 0.00365 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 395, Current best: 2.968642046802392e-60, Global best: 2.968642046802392e-60, Runtime: 0.00348 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 396, Current best: 1.0089240865419176e-60, Global best: 1.0089240865419176e-60, Runtime: 0.00329 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 397, Current best: 9.682965684650952e-61, Global best: 9.682965684650952e-61, Runtime: 0.00386 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 398, Current best: 9.682965684650952e-61, Global best: 9.682965684650952e-61, Runtime: 0.00389 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 399, Current best: 6.668250763064496e-61, Global best: 6.668250763064496e-61, Runtime: 0.00353 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 400, Current best: 1.821648282303523e-61, Global best: 1.821648282303523e-61, Runtime: 0.00334 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 401, Current best: 1.821648282303523e-61, Global best: 1.821648282303523e-61, Runtime: 0.00358 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 402, Current best: 3.4501021763058483e-62, Global best: 3.4501021763058483e-62, Runtime: 0.00333 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 403, Current best: 3.4501021763058483e-62, Global best: 3.4501021763058483e-62, Runtime: 0.00340 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 404, Current best: 3.4501021763058483e-62, Global best: 3.4501021763058483e-62, Runtime: 0.00362 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 405, Current best: 2.4686130257433278e-62, Global best: 2.4686130257433278e-62, Runtime: 0.00350 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 406, Current best: 1.
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6060136171848533e-62, Global best: 1.6060136171848533e-62, Runtime: 0.00331
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 407, Current best: 1.
6060136171848533e-62, Global best: 1.6060136171848533e-62, Runtime: 0.00341
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 408, Current best: 6.
573135725614918e-63, Global best: 6.573135725614918e-63, Runtime: 0.00397 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 409, Current best: 4.
2847146826533056e-63, Global best: 4.2847146826533056e-63, Runtime: 0.00354
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 410, Current best: 3.
678643332869513e-63, Global best: 3.678643332869513e-63, Runtime: 0.00453 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 411, Current best: 3.
678643332869513e-63, Global best: 3.678643332869513e-63, Runtime: 0.00358 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 412, Current best: 6.
765250358746639e-64, Global best: 6.765250358746639e-64, Runtime: 0.00332 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 413, Current best: 4.
815444320004125e-64, Global best: 4.815444320004125e-64, Runtime: 0.00333 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 414, Current best: 2.
0668620418720246e-64, Global best: 2.0668620418720246e-64, Runtime: 0.00336
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 415, Current best: 2.
0668620418720246e-64, Global best: 2.0668620418720246e-64, Runtime: 0.00344
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 416, Current best: 1.
0662959966787835e-64, Global best: 1.0662959966787835e-64, Runtime: 0.00306
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 417, Current best: 1.
0662959966787835e-64, Global best: 1.0662959966787835e-64, Runtime: 0.00369
seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 418, Current best: 8.
392507551468714e-65, Global best: 8.392507551468714e-65, Runtime: 0.00356 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 419, Current best: 6.
890925802446517e-65, Global best: 6.890925802446517e-65, Runtime: 0.00431 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 420, Current best: 1.
274848021871003e-65, Global best: 1.274848021871003e-65, Runtime: 0.00362 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 421, Current best: 1.
274848021871003e-65, Global best: 1.274848021871003e-65, Runtime: 0.00389 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 422, Current best: 8.
730618926127195e-66, Global best: 8.730618926127195e-66, Runtime: 0.00419 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 423, Current best: 2.
746804773795134e-66, Global best: 2.746804773795134e-66, Runtime: 0.00374 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 424, Current best: 2.
746804773795134e-66, Global best: 2.746804773795134e-66, Runtime: 0.00310 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 425, Current best: 2.
746804773795134e-66, Global best: 2.746804773795134e-66, Runtime: 0.00379 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 426, Current best: 2.
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2478975975444797e-66, Global best: 2.2478975975444797e-66, Runtime: 0.00359 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 427, Current best: 1.520966234896216e-66, Global best: 1.520966234896216e-66, Runtime: 0.00310 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 428, Current best: 4.882952222386053e-67, Global best: 4.882952222386053e-67, Runtime: 0.00340 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 429, Current best: 4.6980431597055505e-67, Global best: 4.6980431597055505e-67, Runtime: 0.00376 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 430, Current best: 4.6980431597055505e-67, Global best: 4.6980431597055505e-67, Runtime: 0.00307 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 431, Current best: 3.5467489066339645e-67, Global best: 3.5467489066339645e-67, Runtime: 0.00360 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 432, Current best: 3.08490388525752e-67, Global best: 3.08490388525752e-67, Runtime: 0.00300 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 433, Current best: 1.75597652159737e-67, Global best: 1.75597652159737e-67, Runtime: 0.00369 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 434, Current best: 1.75597652159737e-67, Global best: 1.75597652159737e-67, Runtime: 0.00345 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 435, Current best: 1.75597652159737e-67, Global best: 1.75597652159737e-67, Runtime: 0.00331 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 436, Current best: 1.6533450092789052e-67, Global best: 1.6533450092789052e-67, Runtime: 0.00378 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 437, Current best: 3.4474189578906805e-68, Global best: 3.4474189578906805e-68, Runtime: 0.00401 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 438, Current best: 2.9923432199860765e-68, Global best: 2.9923432199860765e-68, Runtime: 0.00330 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 439, Current best: 2.6478366346807426e-68, Global best: 2.6478366346807426e-68, Runtime: 0.00379 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 440, Current best: 2.3956543640191848e-68, Global best: 2.3956543640191848e-68, Runtime: 0.00379 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 441, Current best: 4.169713298679081e-69, Global best: 4.169713298679081e-69, Runtime: 0.00339 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 442, Current best: 1.1704728426322074e-69, Global best: 1.1704728426322074e-69, Runtime: 0.00360 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 443, Current best: 7.696879197949745e-70, Global best: 7.696879197949745e-70, Runtime: 0.00346 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 444, Current best: 7.696879197949745e-70, Global best: 7.696879197949745e-70, Runtime: 0.00321 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 445, Current best: 2.415416405838424e-70, Global best: 2.415416405838424e-70, Runtime: 0.00337 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 446, Current best: 1.0240054498847524e-70, Global best: 1.0240054498847524e-70, Runtime: 0.00355

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 447, Current best: 1.0240054498847524e-70, Global best: 1.0240054498847524e-70, Runtime: 0.00389

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 448, Current best: 7.760560392040432e-71, Global best: 7.760560392040432e-71, Runtime: 0.00344 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 449, Current best: 7.053177397822319e-71, Global best: 7.053177397822319e-71, Runtime: 0.00378 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 450, Current best: 4.666557704540891e-71, Global best: 4.666557704540891e-71, Runtime: 0.00377 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 451, Current best: 1.7850096680383751e-71, Global best: 1.7850096680383751e-71, Runtime: 0.00331

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 452, Current best: 1.201853173251939e-71, Global best: 1.201853173251939e-71, Runtime: 0.00344 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 453, Current best: 1.201853173251939e-71, Global best: 1.201853173251939e-71, Runtime: 0.00378 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 454, Current best: 1.201853173251939e-71, Global best: 1.201853173251939e-71, Runtime: 0.00483 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 455, Current best: 1.032693671409993e-71, Global best: 1.032693671409993e-71, Runtime: 0.00330 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 456, Current best: 3.1098022939957456e-72, Global best: 3.1098022939957456e-72, Runtime: 0.00368

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 457, Current best: 3.1098022939957456e-72, Global best: 3.1098022939957456e-72, Runtime: 0.00344

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 458, Current best: 2.72223753119702e-72, Global best: 2.72223753119702e-72, Runtime: 0.00337 seco

nds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 459, Current best: 1.0679506265098035e-72, Global best: 1.0679506265098035e-72, Runtime: 0.00357

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 460, Current best: 4.159048552538605e-73, Global best: 4.159048552538605e-73, Runtime: 0.00365 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 461, Current best: 4.159048552538605e-73, Global best: 4.159048552538605e-73, Runtime: 0.00325 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 462, Current best: 2.284641516531124e-73, Global best: 2.284641516531124e-73, Runtime: 0.00352 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 463, Current best: 2.284641516531124e-73, Global best: 2.284641516531124e-73, Runtime: 0.00335 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 464, Current best: 2.284641516531124e-73, Global best: 2.284641516531124e-73, Runtime: 0.00339 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 465, Current best: 1.209270912856403e-73, Global best: 1.209270912856403e-73, Runtime: 0.00363 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 466, Current best: 9.080259473770905e-74, Global best: 9.080259473770905e-74, Runtime: 0.00353 se

conds

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INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 467, Current best: 6.034740383774215e-74, Global best: 6.034740383774215e-74, Runtime: 0.00325 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 468, Current best: 1.4657130741864597e-74, Global best: 1.4657130741864597e-74, Runtime: 0.00329 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 469, Current best: 1.4614110738834429e-74, Global best: 1.4614110738834429e-74, Runtime: 0.00405 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 470, Current best: 1.2693877674142016e-74, Global best: 1.2693877674142016e-74, Runtime: 0.00397 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 471, Current best: 9.567520621125476e-76, Global best: 9.567520621125476e-76, Runtime: 0.00373 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 472, Current best: 9.567520621125476e-76, Global best: 9.567520621125476e-76, Runtime: 0.00330 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 473, Current best: 3.013407759858803e-76, Global best: 3.013407759858803e-76, Runtime: 0.00331 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 474, Current best: 3.013407759858803e-76, Global best: 3.013407759858803e-76, Runtime: 0.00343 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 475, Current best: 3.1317858656151826e-77, Global best: 3.1317858656151826e-77, Runtime: 0.00332 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 476, Current best: 3.1317858656151826e-77, Global best: 3.1317858656151826e-77, Runtime: 0.00366 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 477, Current best: 3.1317858656151826e-77, Global best: 3.1317858656151826e-77, Runtime: 0.00356 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 478, Current best: 1.297792317465705e-77, Global best: 1.297792317465705e-77, Runtime: 0.00340 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 479, Current best: 1.1891870614675656e-77, Global best: 1.1891870614675656e-77, Runtime: 0.00330 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 480, Current best: 1.1891870614675656e-77, Global best: 1.1891870614675656e-77, Runtime: 0.00347 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 481, Current best: 1.1891870614675656e-77, Global best: 1.1891870614675656e-77, Runtime: 0.00338 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 482, Current best: 8.434811251643785e-78, Global best: 8.434811251643785e-78, Runtime: 0.00328 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 483, Current best: 4.97452937732327e-78, Global best: 4.97452937732327e-78, Runtime: 0.00359 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 484, Current best: 1.5494444050248196e-78, Global best: 1.5494444050248196e-78, Runtime: 0.00333 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 485, Current best: 7.8270576743448e-79, Global best: 7.8270576743448e-79, Runtime: 0.00333 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 486, Current best: 7.8270576743448e-79, Global best: 7.8270576743448e-79, Runtime: 0.00394 seconds
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INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 487, Current best: 7.8270576743448e-79, Global best: 7.8270576743448e-79, Runtime: 0.00438 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 488, Current best: 1.1216718509281412e-79, Global best: 1.1216718509281412e-79, Runtime: 0.00308 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 489, Current best: 6.015858565817015e-80, Global best: 6.015858565817015e-80, Runtime: 0.00328 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 490, Current best: 5.0372511411126084e-80, Global best: 5.0372511411126084e-80, Runtime: 0.00402 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 491, Current best: 5.0212921432940356e-80, Global best: 5.0212921432940356e-80, Runtime: 0.00367 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 492, Current best: 5.0212921432940356e-80, Global best: 5.0212921432940356e-80, Runtime: 0.00403 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 493, Current best: 1.771455655645789e-80, Global best: 1.771455655645789e-80, Runtime: 0.00332 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 494, Current best: 1.771455655645789e-80, Global best: 1.771455655645789e-80, Runtime: 0.00335 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 495, Current best: 6.530768589829152e-81, Global best: 6.530768589829152e-81, Runtime: 0.00374 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 496, Current best: 4.372560186421374e-81, Global best: 4.372560186421374e-81, Runtime: 0.00330 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 497, Current best: 4.372560186421374e-81, Global best: 4.372560186421374e-81, Runtime: 0.00342 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 498, Current best: 4.1334869044420947e-81, Global best: 4.1334869044420947e-81, Runtime: 0.00331 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 499, Current best: 1.5338014604067589e-81, Global best: 1.5338014604067589e-81, Runtime: 0.00325 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 500, Current best: 1.5338014604067589e-81, Global best: 1.5338014604067589e-81, Runtime: 0.00339 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 501, Current best: 7.589395451837022e-82, Global best: 7.589395451837022e-82, Runtime: 0.00349 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 502, Current best: 7.589395451837022e-82, Global best: 7.589395451837022e-82, Runtime: 0.00381 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 503, Current best: 3.787521690684794e-82, Global best: 3.787521690684794e-82, Runtime: 0.00331 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 504, Current best: 2.947149734235367e-82, Global best: 2.947149734235367e-82, Runtime: 0.00400 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 505, Current best: 2.947149734235367e-82, Global best: 2.947149734235367e-82, Runtime: 0.00399 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 506, Current best: 1.3848592659830943e-82, Global best: 1.3848592659830943e-82, Runtime: 0.00328 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 507, Current best: 7.
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217656302362796e-83, Global best: 7.217656302362796e-83, Runtime: 0.00332 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 508, Current best: 7.217656302362796e-83, Global best: 7.217656302362796e-83, Runtime: 0.00405 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 509, Current best: 6.365176193772835e-83, Global best: 6.365176193772835e-83, Runtime: 0.00340 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 510, Current best: 4.152781132578544e-83, Global best: 4.152781132578544e-83, Runtime: 0.00330 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 511, Current best: 2.339654721230692e-83, Global best: 2.339654721230692e-83, Runtime: 0.00338 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 512, Current best: 2.3293344269319454e-83, Global best: 2.3293344269319454e-83, Runtime: 0.00326 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 513, Current best: 1.5611238247463203e-83, Global best: 1.5611238247463203e-83, Runtime: 0.00359 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 514, Current best: 3.8210843025303506e-84, Global best: 3.8210843025303506e-84, Runtime: 0.00310 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 515, Current best: 3.8210843025303506e-84, Global best: 3.8210843025303506e-84, Runtime: 0.00357 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 516, Current best: 7.786199025970773e-85, Global best: 7.786199025970773e-85, Runtime: 0.00302 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 517, Current best: 3.7327209440452704e-85, Global best: 3.7327209440452704e-85, Runtime: 0.00415 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 518, Current best: 3.7327209440452704e-85, Global best: 3.7327209440452704e-85, Runtime: 0.00327 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 519, Current best: 3.7327209440452704e-85, Global best: 3.7327209440452704e-85, Runtime: 0.00335 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 520, Current best: 2.7384093299624165e-85, Global best: 2.7384093299624165e-85, Runtime: 0.00357 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 521, Current best: 2.5563817895578386e-85, Global best: 2.5563817895578386e-85, Runtime: 0.00373 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 522, Current best: 1.4267116522705976e-85, Global best: 1.4267116522705976e-85, Runtime: 0.00323 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 523, Current best: 9.837447313710567e-86, Global best: 9.837447313710567e-86, Runtime: 0.00371 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 524, Current best: 9.482177759829342e-86, Global best: 9.482177759829342e-86, Runtime: 0.00422 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 525, Current best: 8.390015344779367e-86, Global best: 8.390015344779367e-86, Runtime: 0.00335 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 526, Current best: 4.4864287217772996e-86, Global best: 4.4864287217772996e-86, Runtime: 0.00329 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 527, Current best:

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2.5727085891228917e-86, Global best: 2.5727085891228917e-86, Runtime: 0.00
370 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 528, Current best:
2.0143866131831617e-86, Global best: 2.0143866131831617e-86, Runtime: 0.00
308 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 529, Current best:
1.4047177722829215e-86, Global best: 1.4047177722829215e-86, Runtime: 0.00
353 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 530, Current best:
1.1129576993912226e-86, Global best: 1.1129576993912226e-86, Runtime: 0.00
348 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 531, Current best:
1.1129576993912226e-86, Global best: 1.1129576993912226e-86, Runtime: 0.00
303 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 532, Current best:
4.198061743294176e-87, Global best: 4.198061743294176e-87, Runtime: 0.0032
2 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 533, Current best:
2.400252617632993e-87, Global best: 2.400252617632993e-87, Runtime: 0.0033
5 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 534, Current best:
2.400252617632993e-87, Global best: 2.400252617632993e-87, Runtime: 0.0034
7 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 535, Current best:
2.334046021740794e-87, Global best: 2.334046021740794e-87, Runtime: 0.0033
3 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 536, Current best:
1.9137053956289494e-87, Global best: 1.9137053956289494e-87, Runtime: 0.00
332 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 537, Current best:
1.2502579778446947e-87, Global best: 1.2502579778446947e-87, Runtime: 0.00
452 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 538, Current best:
2.5737672430357505e-88, Global best: 2.5737672430357505e-88, Runtime: 0.00
346 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 539, Current best:
1.3034698711451097e-88, Global best: 1.3034698711451097e-88, Runtime: 0.00
337 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 540, Current best:
1.3034698711451097e-88, Global best: 1.3034698711451097e-88, Runtime: 0.00
344 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 541, Current best:
1.3034698711451097e-88, Global best: 1.3034698711451097e-88, Runtime: 0.00
390 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 542, Current best:
9.105880556695281e-89, Global best: 9.105880556695281e-89, Runtime: 0.0036
0 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 543, Current best:
5.9872430143554896e-89, Global best: 5.9872430143554896e-89, Runtime: 0.00
305 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 544, Current best:
4.257303668548342e-89, Global best: 4.257303668548342e-89, Runtime: 0.0031
5 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 545, Current best:
3.7973747273916394e-89, Global best: 3.7973747273916394e-89, Runtime: 0.00
336 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 546, Current best:
2.852275122481348e-89, Global best: 2.852275122481348e-89, Runtime: 0.0032
4 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 547, Current best:
2.3697530510076635e-89, Global best: 2.3697530510076635e-89, Runtime: 0.00
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378 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 548, Current best: 9.308806331598884e-90, Global best: 9.308806331598884e-90, Runtime: 0.0034
0 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 549, Current best: 5.857424216918536e-90, Global best: 5.857424216918536e-90, Runtime: 0.0038
2 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 550, Current best: 5.630326327232496e-90, Global best: 5.630326327232496e-90, Runtime: 0.0039
6 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 551, Current best: 4.981993772749206e-90, Global best: 4.981993772749206e-90, Runtime: 0.0043
5 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 552, Current best: 3.7639322360854846e-90, Global best: 3.7639322360854846e-90, Runtime: 0.00
328 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 553, Current best: 3.2956051406320606e-90, Global best: 3.2956051406320606e-90, Runtime: 0.00
337 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 554, Current best: 2.5620443698455257e-90, Global best: 2.5620443698455257e-90, Runtime: 0.00
350 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 555, Current best: 2.5620443698455257e-90, Global best: 2.5620443698455257e-90, Runtime: 0.00
328 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 556, Current best: 2.2497286773598303e-90, Global best: 2.2497286773598303e-90, Runtime: 0.00
371 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 557, Current best: 1.9682474874445626e-90, Global best: 1.9682474874445626e-90, Runtime: 0.00
425 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 558, Current best: 1.6150110388954155e-90, Global best: 1.6150110388954155e-90, Runtime: 0.00
357 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 559, Current best: 1.1819765744577817e-90, Global best: 1.1819765744577817e-90, Runtime: 0.00
338 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 560, Current best: 5.078261213537055e-91, Global best: 5.078261213537055e-91, Runtime: 0.0036
1 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 561, Current best: 3.684586513323894e-91, Global best: 3.684586513323894e-91, Runtime: 0.0033
2 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 562, Current best: 3.684586513323894e-91, Global best: 3.684586513323894e-91, Runtime: 0.0032
5 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 563, Current best: 1.666589759464806e-91, Global best: 1.666589759464806e-91, Runtime: 0.0034
8 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 564, Current best: 5.045501688809823e-92, Global best: 5.045501688809823e-92, Runtime: 0.0032
1 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 565, Current best: 5.045501688809823e-92, Global best: 5.045501688809823e-92, Runtime: 0.0037
5 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 566, Current best: 5.045501688809823e-92, Global best: 5.045501688809823e-92, Runtime: 0.0030
2 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 567, Current best: 4.870504906611876e-92, Global best: 4.870504906611876e-92, Runtime: 0.0031
0 seconds


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INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 568, Current best:
9.044977352833494e-93, Global best: 9.044977352833494e-93, Runtime: 0.0034
1 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 569, Current best:
9.044977352833494e-93, Global best: 9.044977352833494e-93, Runtime: 0.0032
9 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 570, Current best:
9.044977352833494e-93, Global best: 9.044977352833494e-93, Runtime: 0.0039
8 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 571, Current best:
9.044977352833494e-93, Global best: 9.044977352833494e-93, Runtime: 0.0033
3 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 572, Current best:
9.044977352833494e-93, Global best: 9.044977352833494e-93, Runtime: 0.0043
0 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 573, Current best:
6.51059770482553e-93, Global best: 6.51059770482553e-93, Runtime: 0.00396
seconds
```

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INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 574, Current best: 5.
119448110808481e-93, Global best: 5.119448110808481e-93, Runtime: 0.00352 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 575, Current best: 5.
119448110808481e-93, Global best: 5.119448110808481e-93, Runtime: 0.00298 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 576, Current best: 4.
485144550693173e-93, Global best: 4.485144550693173e-93, Runtime: 0.00363 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 577, Current best: 1.
0891141104982515e-93, Global best: 1.0891141104982515e-93, Runtime: 0.00329
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 578, Current best: 7.
383118889958668e-94, Global best: 7.383118889958668e-94, Runtime: 0.00333 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 579, Current best: 7.
383118889958668e-94, Global best: 7.383118889958668e-94, Runtime: 0.00333 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 580, Current best: 7.
383118889958668e-94, Global best: 7.383118889958668e-94, Runtime: 0.00328 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 581, Current best: 7.
383118889958668e-94, Global best: 7.383118889958668e-94, Runtime: 0.00332 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 582, Current best: 5.
23010016639502e-94, Global best: 5.23010016639502e-94, Runtime: 0.00417 seco
nds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 583, Current best: 3.
9107211490612336e-94, Global best: 3.9107211490612336e-94, Runtime: 0.00365
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 584, Current best: 3.
463837409223352e-94, Global best: 3.463837409223352e-94, Runtime: 0.00329 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 585, Current best: 1.
4350005122260852e-94, Global best: 1.4350005122260852e-94, Runtime: 0.00377
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 586, Current best: 1.
4350005122260852e-94, Global best: 1.4350005122260852e-94, Runtime: 0.00302
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 587, Current best: 1.
4195884652001036e-94, Global best: 1.4195884652001036e-94, Runtime: 0.00304
```

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 588, Current best: 1.2919250614347626e-94, Global best: 1.2919250614347626e-94, Runtime: 0.00392

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 589, Current best: 1.239067135708204e-94, Global best: 1.239067135708204e-94, Runtime: 0.00344 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 590, Current best: 6.9055123264283e-95, Global best: 6.9055123264283e-95, Runtime: 0.00305 second

s

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 591, Current best: 6.9055123264283e-95, Global best: 6.9055123264283e-95, Runtime: 0.00302 second

s

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 592, Current best: 4.903786035796848e-95, Global best: 4.903786035796848e-95, Runtime: 0.00400 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 593, Current best: 4.903786035796848e-95, Global best: 4.903786035796848e-95, Runtime: 0.00315 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 594, Current best: 2.432386516971995e-95, Global best: 2.432386516971995e-95, Runtime: 0.00297 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 595, Current best: 1.691068741077326e-95, Global best: 1.691068741077326e-95, Runtime: 0.00306 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 596, Current best: 1.2166952202696041e-95, Global best: 1.2166952202696041e-95, Runtime: 0.00307

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 597, Current best: 1.2166952202696041e-95, Global best: 1.2166952202696041e-95, Runtime: 0.00336

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 598, Current best: 9.351531663283394e-96, Global best: 9.351531663283394e-96, Runtime: 0.00333 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 599, Current best: 7.271340265629752e-96, Global best: 7.271340265629752e-96, Runtime: 0.00333 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 600, Current best: 4.2292008220298536e-96, Global best: 4.2292008220298536e-96, Runtime: 0.00425

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 601, Current best: 3.6550365780336183e-96, Global best: 3.6550365780336183e-96, Runtime: 0.00411

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 602, Current best: 3.6550365780336183e-96, Global best: 3.6550365780336183e-96, Runtime: 0.00333

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 603, Current best: 3.6550365780336183e-96, Global best: 3.6550365780336183e-96, Runtime: 0.00347

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 604, Current best: 2.638554374612021e-96, Global best: 2.638554374612021e-96, Runtime: 0.00342 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 605, Current best: 2.638554374612021e-96, Global best: 2.638554374612021e-96, Runtime: 0.00303 se

conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 606, Current best: 2.3745305036178623e-96, Global best: 2.3745305036178623e-96, Runtime: 0.00311

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 607, Current best: 9.992587361210788e-97, Global best: 9.992587361210788e-97, Runtime: 0.00332 se

conds

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INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 608, Current best: 7.579441232445107e-97, Global best: 7.579441232445107e-97, Runtime: 0.00329 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 609, Current best: 4.968200455954451e-97, Global best: 4.968200455954451e-97, Runtime: 0.00352 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 610, Current best: 4.968200455954451e-97, Global best: 4.968200455954451e-97, Runtime: 0.00338 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 611, Current best: 4.704244801000614e-97, Global best: 4.704244801000614e-97, Runtime: 0.00412 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 612, Current best: 4.704244801000614e-97, Global best: 4.704244801000614e-97, Runtime: 0.00301 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 613, Current best: 4.0185954070833563e-97, Global best: 4.0185954070833563e-97, Runtime: 0.00309 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 614, Current best: 3.4685379168455704e-97, Global best: 3.4685379168455704e-97, Runtime: 0.00303 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 615, Current best: 2.914072131766404e-97, Global best: 2.914072131766404e-97, Runtime: 0.00307 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 616, Current best: 1.6939269475648992e-97, Global best: 1.6939269475648992e-97, Runtime: 0.00428 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 617, Current best: 8.774372439772658e-98, Global best: 8.774372439772658e-98, Runtime: 0.00399 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 618, Current best: 8.774372439772658e-98, Global best: 8.774372439772658e-98, Runtime: 0.00302 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 619, Current best: 4.6972269357851756e-98, Global best: 4.6972269357851756e-98, Runtime: 0.00378 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 620, Current best: 2.956875024057667e-98, Global best: 2.956875024057667e-98, Runtime: 0.00340 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 621, Current best: 2.956875024057667e-98, Global best: 2.956875024057667e-98, Runtime: 0.00331 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 622, Current best: 2.956875024057667e-98, Global best: 2.956875024057667e-98, Runtime: 0.00327 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 623, Current best: 2.7611206681161182e-98, Global best: 2.7611206681161182e-98, Runtime: 0.00362 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 624, Current best: 9.70600361265858e-99, Global best: 9.70600361265858e-99, Runtime: 0.00330 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 625, Current best: 5.9801797882265095e-99, Global best: 5.9801797882265095e-99, Runtime: 0.00337 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 626, Current best: 5.9801797882265095e-99, Global best: 5.9801797882265095e-99, Runtime: 0.00374 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 627, Current best: 3.484406921944227e-99, Global best: 3.484406921944227e-99, Runtime: 0.00326 seconds
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INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 628, Current best:
3.1214023997275118e-99, Global best: 3.1214023997275118e-99, Runtime: 0.00
334 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 629, Current best:
1.8984800172749174e-99, Global best: 1.8984800172749174e-99, Runtime: 0.00
311 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 630, Current best:
1.8741992832524092e-99, Global best: 1.8741992832524092e-99, Runtime: 0.00
301 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 631, Current best:
5.411257748234152e-100, Global best: 5.411257748234152e-100, Runtime: 0.00
330 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 632, Current best:
5.411257748234152e-100, Global best: 5.411257748234152e-100, Runtime: 0.00
336 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 633, Current best:
5.411257748234152e-100, Global best: 5.411257748234152e-100, Runtime: 0.00
311 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 634, Current best:
5.06938407869858e-100, Global best: 5.06938407869858e-100, Runtime: 0.0033
2 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 635, Current best:
3.7934381589770046e-100, Global best: 3.7934381589770046e-100, Runtime: 0.
00336 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 636, Current best:
3.085740559464492e-100, Global best: 3.085740559464492e-100, Runtime: 0.00
324 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 637, Current best:
1.5389580464927404e-100, Global best: 1.5389580464927404e-100, Runtime: 0.
00332 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 638, Current best:
1.5389580464927404e-100, Global best: 1.5389580464927404e-100, Runtime: 0.
00300 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 639, Current best:
9.74712139648445e-101, Global best: 9.74712139648445e-101, Runtime: 0.0037
4 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 640, Current best:
7.230697210910289e-101, Global best: 7.230697210910289e-101, Runtime: 0.00
334 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 641, Current best:
5.830077463039645e-101, Global best: 5.830077463039645e-101, Runtime: 0.00
370 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 642, Current best:
5.830077463039645e-101, Global best: 5.830077463039645e-101, Runtime: 0.00
359 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 643, Current best:
4.6261771856684725e-101, Global best: 4.6261771856684725e-101, Runtime: 0.
00345 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 644, Current best:
4.4928903441974595e-101, Global best: 4.4928903441974595e-101, Runtime: 0.
00331 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 645, Current best:
3.164797041747825e-101, Global best: 3.164797041747825e-101, Runtime: 0.00
360 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 646, Current best:
2.706824873447843e-101, Global best: 2.706824873447843e-101, Runtime: 0.00
341 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 647, Current best:
2.706824873447843e-101, Global best: 2.706824873447843e-101, Runtime: 0.00
329 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 648, Current best:
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2.706824873447843e-101, Global best: 2.706824873447843e-101, Runtime: 0.00344 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 649, Current best: 2.4616657039718833e-101, Global best: 2.4616657039718833e-101, Runtime: 0.00410 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 650, Current best: 2.113563360067373e-101, Global best: 2.113563360067373e-101, Runtime: 0.00355 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 651, Current best: 1.1762015228383982e-101, Global best: 1.1762015228383982e-101, Runtime: 0.00370 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 652, Current best: 1.0053734180292751e-101, Global best: 1.0053734180292751e-101, Runtime: 0.00390 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 653, Current best: 1.0053734180292751e-101, Global best: 1.0053734180292751e-101, Runtime: 0.00365 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 654, Current best: 3.762623689210834e-102, Global best: 3.762623689210834e-102, Runtime: 0.00356 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 655, Current best: 3.762623689210834e-102, Global best: 3.762623689210834e-102, Runtime: 0.00392 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 656, Current best: 3.762623689210834e-102, Global best: 3.762623689210834e-102, Runtime: 0.00354 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 657, Current best: 3.40130880928684e-102, Global best: 3.40130880928684e-102, Runtime: 0.00367 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 658, Current best: 3.3474013436775877e-102, Global best: 3.3474013436775877e-102, Runtime: 0.00356 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 659, Current best: 1.7252501806024169e-102, Global best: 1.7252501806024169e-102, Runtime: 0.00303 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 660, Current best: 1.299913685576253e-102, Global best: 1.299913685576253e-102, Runtime: 0.00390 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 661, Current best: 1.299913685576253e-102, Global best: 1.299913685576253e-102, Runtime: 0.00329 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 662, Current best: 1.0772253334837484e-102, Global best: 1.0772253334837484e-102, Runtime: 0.00431 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 663, Current best: 5.378343635711103e-103, Global best: 5.378343635711103e-103, Runtime: 0.00342 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 664, Current best: 4.807461987105666e-103, Global best: 4.807461987105666e-103, Runtime: 0.00374 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 665, Current best: 4.807461987105666e-103, Global best: 4.807461987105666e-103, Runtime: 0.00395 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 666, Current best: 4.807461987105666e-103, Global best: 4.807461987105666e-103, Runtime: 0.00305 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 667, Current best: 3.666946554364022e-103, Global best: 3.666946554364022e-103, Runtime: 0.00393 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 668, Current best: 2.991923322600567e-103, Global best: 2.991923322600567e-103, Runtime: 0.00

412 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 669, Current best: 2.40506665047754e-103, Global best: 2.40506665047754e-103, Runtime: 0.0032

6 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 670, Current best: 2.40506665047754e-103, Global best: 2.40506665047754e-103, Runtime: 0.0035

1 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 671, Current best: 2.40506665047754e-103, Global best: 2.40506665047754e-103, Runtime: 0.0030

8 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 672, Current best: 1.8638502275136194e-103, Global best: 1.8638502275136194e-103, Runtime: 0.

00331 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 673, Current best: 1.8638502275136194e-103, Global best: 1.8638502275136194e-103, Runtime: 0.

00355 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 674, Current best: 9.859860682737074e-104, Global best: 9.859860682737074e-104, Runtime: 0.00

406 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 675, Current best: 9.859860682737074e-104, Global best: 9.859860682737074e-104, Runtime: 0.00

307 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 676, Current best: 9.859860682737074e-104, Global best: 9.859860682737074e-104, Runtime: 0.00

411 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 677, Current best: 9.859860682737074e-104, Global best: 9.859860682737074e-104, Runtime: 0.00

362 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 678, Current best: 9.859860682737074e-104, Global best: 9.859860682737074e-104, Runtime: 0.00

352 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 679, Current best: 9.859860682737074e-104, Global best: 9.859860682737074e-104, Runtime: 0.00

333 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 680, Current best: 8.622995051896708e-104, Global best: 8.622995051896708e-104, Runtime: 0.00

348 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 681, Current best: 8.622995051896708e-104, Global best: 8.622995051896708e-104, Runtime: 0.00

356 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 682, Current best: 6.309588059911336e-104, Global best: 6.309588059911336e-104, Runtime: 0.00

356 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 683, Current best: 6.309588059911336e-104, Global best: 6.309588059911336e-104, Runtime: 0.00

330 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 684, Current best: 3.0893070848057405e-104, Global best: 3.0893070848057405e-104, Runtime: 0.

00467 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 685, Current best: 1.7678250436130067e-104, Global best: 1.7678250436130067e-104, Runtime: 0.

00359 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 686, Current best: 1.5967306778578153e-104, Global best: 1.5967306778578153e-104, Runtime: 0.

00329 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 687, Current best: 1.5967306778578153e-104, Global best: 1.5967306778578153e-104, Runtime: 0.

00340 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 688, Current best: 1.4240016180698322e-104, Global best: 1.4240016180698322e-104, Runtime: 0.

00385 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 689, Current best: 1.2426989725422083e-104, Global best: 1.2426989725422083e-104, Runtime: 0.

00343 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 690, Current best: 7.156297294911242e-105, Global best: 7.156297294911242e-105, Runtime: 0.00

337 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 691, Current best: 7.156297294911242e-105, Global best: 7.156297294911242e-105, Runtime: 0.00

342 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 692, Current best: 5.016783167752289e-105, Global best: 5.016783167752289e-105, Runtime: 0.00

330 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 693, Current best: 3.895662502017384e-105, Global best: 3.895662502017384e-105, Runtime: 0.00

346 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 694, Current best: 3.895662502017384e-105, Global best: 3.895662502017384e-105, Runtime: 0.00

427 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 695, Current best: 3.895662502017384e-105, Global best: 3.895662502017384e-105, Runtime: 0.00

305 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 696, Current best: 3.895662502017384e-105, Global best: 3.895662502017384e-105, Runtime: 0.00

455 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 697, Current best: 3.3920240867356726e-105, Global best: 3.3920240867356726e-105, Runtime: 0.

00333 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 698, Current best: 3.3920240867356726e-105, Global best: 3.3920240867356726e-105, Runtime: 0.

00337 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 699, Current best: 3.149664290417345e-105, Global best: 3.149664290417345e-105, Runtime: 0.00

338 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 700, Current best: 3.149664290417345e-105, Global best: 3.149664290417345e-105, Runtime: 0.00

337 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 701, Current best: 2.9734112650834336e-105, Global best: 2.9734112650834336e-105, Runtime: 0.

00360 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 702, Current best: 2.2216701649294228e-105, Global best: 2.2216701649294228e-105, Runtime: 0.

00375 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 703, Current best: 2.143829036180444e-105, Global best: 2.143829036180444e-105, Runtime: 0.00

304 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 704, Current best: 2.143829036180444e-105, Global best: 2.143829036180444e-105, Runtime: 0.00

337 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 705, Current best: 1.8841159502931002e-105, Global best: 1.8841159502931002e-105, Runtime: 0.

00333 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 706, Current best: 1.4005907518456595e-105, Global best: 1.4005907518456595e-105, Runtime: 0.

00347 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 707, Current best: 1.0382685790572904e-105, Global best: 1.0382685790572904e-105, Runtime: 0.

00363 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 708, Current best: 1.0382685790572904e-105, Global best: 1.0382685790572904e-105, Runtime: 0.

00395 seconds

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INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 709, Current best:
8.605954504410736e-106, Global best: 8.605954504410736e-106, Runtime: 0.00
366 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 710, Current best:
7.44022305925153e-106, Global best: 7.44022305925153e-106, Runtime: 0.0033
9 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 711, Current best:
4.770807821137899e-106, Global best: 4.770807821137899e-106, Runtime: 0.00
408 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 712, Current best:
4.770807821137899e-106, Global best: 4.770807821137899e-106, Runtime: 0.00
390 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 713, Current best:
4.423472894360748e-106, Global best: 4.423472894360748e-106, Runtime: 0.00
300 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 714, Current best:
4.423472894360748e-106, Global best: 4.423472894360748e-106, Runtime: 0.00
307 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 715, Current best:
4.340891719661116e-106, Global best: 4.340891719661116e-106, Runtime: 0.00
451 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 716, Current best:
4.340891719661116e-106, Global best: 4.340891719661116e-106, Runtime: 0.00
296 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 717, Current best:
3.8288001665645434e-106, Global best: 3.8288001665645434e-106, Runtime: 0.
00358 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 718, Current best:
2.774124755147721e-106, Global best: 2.774124755147721e-106, Runtime: 0.00
387 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 719, Current best:
2.7155709674695833e-106, Global best: 2.7155709674695833e-106, Runtime: 0.
00344 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 720, Current best:
2.540809720378291e-106, Global best: 2.540809720378291e-106, Runtime: 0.00
339 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 721, Current best:
2.422234769766126e-106, Global best: 2.422234769766126e-106, Runtime: 0.00
452 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 722, Current best:
2.0486084379041965e-106, Global best: 2.0486084379041965e-106, Runtime: 0.
00396 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 723, Current best:
1.155630078051055e-106, Global best: 1.155630078051055e-106, Runtime: 0.00
312 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 724, Current best:
1.155630078051055e-106, Global best: 1.155630078051055e-106, Runtime: 0.00
351 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 725, Current best:
1.155630078051055e-106, Global best: 1.155630078051055e-106, Runtime: 0.00
372 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 726, Current best:
8.995243025882252e-107, Global best: 8.995243025882252e-107, Runtime: 0.00
342 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 727, Current best:
8.105466518810734e-107, Global best: 8.105466518810734e-107, Runtime: 0.00
347 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 728, Current best:
8.105466518810734e-107, Global best: 8.105466518810734e-107, Runtime: 0.00
376 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 729, Current best:
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6.257573249765882e-107, Global best: 6.257573249765882e-107, Runtime: 0.00390 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 730, Current best: 4.0030787443161095e-107, Global best: 4.0030787443161095e-107, Runtime: 0.00297 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 731, Current best: 2.777025554302928e-107, Global best: 2.777025554302928e-107, Runtime: 0.00349 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 732, Current best: 2.777025554302928e-107, Global best: 2.777025554302928e-107, Runtime: 0.00363 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 733, Current best: 2.777025554302928e-107, Global best: 2.777025554302928e-107, Runtime: 0.00360 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 734, Current best: 2.770569140691111e-107, Global best: 2.770569140691111e-107, Runtime: 0.00383 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 735, Current best: 2.10572426814088e-107, Global best: 2.10572426814088e-107, Runtime: 0.00307 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 736, Current best: 1.6950588070613603e-107, Global best: 1.6950588070613603e-107, Runtime: 0.00364 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 737, Current best: 1.0614342895075543e-107, Global best: 1.0614342895075543e-107, Runtime: 0.00446 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 738, Current best: 1.0614342895075543e-107, Global best: 1.0614342895075543e-107, Runtime: 0.00328 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 739, Current best: 1.0614342895075543e-107, Global best: 1.0614342895075543e-107, Runtime: 0.00319 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 740, Current best: 1.0614342895075543e-107, Global best: 1.0614342895075543e-107, Runtime: 0.00305 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 741, Current best: 1.0614342895075543e-107, Global best: 1.0614342895075543e-107, Runtime: 0.00304 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 742, Current best: 1.0614342895075543e-107, Global best: 1.0614342895075543e-107, Runtime: 0.00378 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 743, Current best: 9.620986389252755e-108, Global best: 9.620986389252755e-108, Runtime: 0.00335 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 744, Current best: 8.856940479544563e-108, Global best: 8.856940479544563e-108, Runtime: 0.00334 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 745, Current best: 8.856940479544563e-108, Global best: 8.856940479544563e-108, Runtime: 0.00362 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 746, Current best: 5.405631124920236e-108, Global best: 5.405631124920236e-108, Runtime: 0.00344 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 747, Current best: 5.405631124920236e-108, Global best: 5.405631124920236e-108, Runtime: 0.00366 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 748, Current best: 5.405631124920236e-108, Global best: 5.405631124920236e-108, Runtime: 0.00348 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 749, Current best: 5.

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1122280230773364e-108, Global best: 5.1122280230773364e-108, Runtime: 0.0035
7 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 750, Current best: 5.
1122280230773364e-108, Global best: 5.1122280230773364e-108, Runtime: 0.0036
9 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 751, Current best: 4.
6995367440534765e-108, Global best: 4.6995367440534765e-108, Runtime: 0.0030
5 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 752, Current best: 2.
2178542796515427e-108, Global best: 2.2178542796515427e-108, Runtime: 0.0033
2 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 753, Current best: 2.
2178542796515427e-108, Global best: 2.2178542796515427e-108, Runtime: 0.0037
8 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 754, Current best: 1.
8584431242448774e-108, Global best: 1.8584431242448774e-108, Runtime: 0.0030
4 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 755, Current best: 1.
8584431242448774e-108, Global best: 1.8584431242448774e-108, Runtime: 0.0035
7 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 756, Current best: 1.
321892205288877e-108, Global best: 1.321892205288877e-108, Runtime: 0.00384
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 757, Current best: 1.
321892205288877e-108, Global best: 1.321892205288877e-108, Runtime: 0.00302
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 758, Current best: 1.
1963639746855917e-108, Global best: 1.1963639746855917e-108, Runtime: 0.0041
1 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 759, Current best: 1.
1963639746855917e-108, Global best: 1.1963639746855917e-108, Runtime: 0.0036
4 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 760, Current best: 1.
194554938959763e-108, Global best: 1.194554938959763e-108, Runtime: 0.00335
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 761, Current best: 9.
69919305522818e-109, Global best: 9.69919305522818e-109, Runtime: 0.00376 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 762, Current best: 9.
69919305522818e-109, Global best: 9.69919305522818e-109, Runtime: 0.00309 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 763, Current best: 9.
69919305522818e-109, Global best: 9.69919305522818e-109, Runtime: 0.00334 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 764, Current best: 9.
376317642818752e-109, Global best: 9.376317642818752e-109, Runtime: 0.00306
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 765, Current best: 9.
139975077983259e-109, Global best: 9.139975077983259e-109, Runtime: 0.00342
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 766, Current best: 9.
139975077983259e-109, Global best: 9.139975077983259e-109, Runtime: 0.00398
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 767, Current best: 7.
946908438661932e-109, Global best: 7.946908438661932e-109, Runtime: 0.00306
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 768, Current best: 5.
28606098359018e-109, Global best: 5.28606098359018e-109, Runtime: 0.00307 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 769, Current best: 5.
28606098359018e-109, Global best: 5.28606098359018e-109, Runtime: 0.00448 se
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conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 770, Current best: 5.28606098359018e-109, Global best: 5.28606098359018e-109, Runtime: 0.00402 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 771, Current best: 3.8222581311802817e-109, Global best: 3.8222581311802817e-109, Runtime: 0.00353 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 772, Current best: 3.8222581311802817e-109, Global best: 3.8222581311802817e-109, Runtime: 0.00380 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 773, Current best: 3.670524454196324e-109, Global best: 3.670524454196324e-109, Runtime: 0.00385 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 774, Current best: 3.670524454196324e-109, Global best: 3.670524454196324e-109, Runtime: 0.00333 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 775, Current best: 3.464049483036835e-109, Global best: 3.464049483036835e-109, Runtime: 0.00440 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 776, Current best: 2.8995596346297875e-109, Global best: 2.8995596346297875e-109, Runtime: 0.00352 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 777, Current best: 2.325751060058816e-109, Global best: 2.325751060058816e-109, Runtime: 0.00360 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 778, Current best: 2.2797711853455e-109, Global best: 2.2797711853455e-109, Runtime: 0.00300 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 779, Current best: 2.1501541083293655e-109, Global best: 2.1501541083293655e-109, Runtime: 0.00363 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 780, Current best: 1.209430073368087e-109, Global best: 1.209430073368087e-109, Runtime: 0.00332 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 781, Current best: 1.0301772363968733e-109, Global best: 1.0301772363968733e-109, Runtime: 0.00341 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 782, Current best: 1.0301772363968733e-109, Global best: 1.0301772363968733e-109, Runtime: 0.00375 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 783, Current best: 9.566363028125973e-110, Global best: 9.566363028125973e-110, Runtime: 0.00394 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 784, Current best: 9.566363028125973e-110, Global best: 9.566363028125973e-110, Runtime: 0.00334 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 785, Current best: 9.566363028125973e-110, Global best: 9.566363028125973e-110, Runtime: 0.00317 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 786, Current best: 9.566363028125973e-110, Global best: 9.566363028125973e-110, Runtime: 0.00353 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 787, Current best: 8.442951205621435e-110, Global best: 8.442951205621435e-110, Runtime: 0.00333 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 788, Current best: 8.442951205621435e-110, Global best: 8.442951205621435e-110, Runtime: 0.00328 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 789, Current best: 7.342089005046205e-110, Global best: 7.342089005046205e-110, Runtime: 0.00392

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 790, Current best: 6.064436463170395e-110, Global best: 6.064436463170395e-110, Runtime: 0.00394

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 791, Current best: 4.921950827970565e-110, Global best: 4.921950827970565e-110, Runtime: 0.00341

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 792, Current best: 3.0590500476713993e-110, Global best: 3.0590500476713993e-110, Runtime: 0.0030

3 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 793, Current best: 2.799363522139951e-110, Global best: 2.799363522139951e-110, Runtime: 0.00307

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 794, Current best: 2.799363522139951e-110, Global best: 2.799363522139951e-110, Runtime: 0.00321

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 795, Current best: 2.3587440328505535e-110, Global best: 2.3587440328505535e-110, Runtime: 0.0034

2 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 796, Current best: 2.3587440328505535e-110, Global best: 2.3587440328505535e-110, Runtime: 0.0035

4 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 797, Current best: 2.3587440328505535e-110, Global best: 2.3587440328505535e-110, Runtime: 0.0035

2 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 798, Current best: 2.1349486830839024e-110, Global best: 2.1349486830839024e-110, Runtime: 0.0039

7 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 799, Current best: 2.1349486830839024e-110, Global best: 2.1349486830839024e-110, Runtime: 0.0035

7 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 800, Current best: 2.1349486830839024e-110, Global best: 2.1349486830839024e-110, Runtime: 0.0029

9 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 801, Current best: 1.6914036210155276e-110, Global best: 1.6914036210155276e-110, Runtime: 0.0030

5 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 802, Current best: 1.6914036210155276e-110, Global best: 1.6914036210155276e-110, Runtime: 0.0038

1 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 803, Current best: 1.6914036210155276e-110, Global best: 1.6914036210155276e-110, Runtime: 0.0036

3 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 804, Current best: 1.2458604590418406e-110, Global best: 1.2458604590418406e-110, Runtime: 0.0037

4 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 805, Current best: 1.2458604590418406e-110, Global best: 1.2458604590418406e-110, Runtime: 0.0034

3 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 806, Current best: 1.2092597312518724e-110, Global best: 1.2092597312518724e-110, Runtime: 0.0039

2 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 807, Current best: 1.2092597312518724e-110, Global best: 1.2092597312518724e-110, Runtime: 0.0034

2 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 808, Current best: 1.2092597312518724e-110, Global best: 1.2092597312518724e-110, Runtime: 0.0037

3 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 809, Current best: 1.1810394606818821e-110, Global best: 1.1810394606818821e-110, Runtime: 0.0036

1 seconds

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INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 810, Current best: 1.1470782770360159e-110, Global best: 1.1470782770360159e-110, Runtime: 0.00344 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 811, Current best: 1.1470782770360159e-110, Global best: 1.1470782770360159e-110, Runtime: 0.00365 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 812, Current best: 1.1430462184230971e-110, Global best: 1.1430462184230971e-110, Runtime: 0.00365 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 813, Current best: 1.1430462184230971e-110, Global best: 1.1430462184230971e-110, Runtime: 0.00371 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 814, Current best: 1.1251838682376172e-110, Global best: 1.1251838682376172e-110, Runtime: 0.00374 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 815, Current best: 1.1251838682376172e-110, Global best: 1.1251838682376172e-110, Runtime: 0.00340 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 816, Current best: 9.270237186676234e-111, Global best: 9.270237186676234e-111, Runtime: 0.00336 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 817, Current best: 8.481900056260771e-111, Global best: 8.481900056260771e-111, Runtime: 0.00345 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 818, Current best: 8.481900056260771e-111, Global best: 8.481900056260771e-111, Runtime: 0.00310 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 819, Current best: 8.431507414963627e-111, Global best: 8.431507414963627e-111, Runtime: 0.00377 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 820, Current best: 7.187974329234598e-111, Global best: 7.187974329234598e-111, Runtime: 0.00359 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 821, Current best: 6.664050426948672e-111, Global best: 6.664050426948672e-111, Runtime: 0.00309 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 822, Current best: 6.664050426948672e-111, Global best: 6.664050426948672e-111, Runtime: 0.00354 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 823, Current best: 6.664050426948672e-111, Global best: 6.664050426948672e-111, Runtime: 0.00324 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 824, Current best: 6.098309944664903e-111, Global best: 6.098309944664903e-111, Runtime: 0.00392 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 825, Current best: 6.098309944664903e-111, Global best: 6.098309944664903e-111, Runtime: 0.00369 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 826, Current best: 4.9721157513501324e-111, Global best: 4.9721157513501324e-111, Runtime: 0.00426 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 827, Current best: 3.902453108592447e-111, Global best: 3.902453108592447e-111, Runtime: 0.00474 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 828, Current best: 3.902453108592447e-111, Global best: 3.902453108592447e-111, Runtime: 0.00330 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 829, Current best: 3.2876847511107827e-111, Global best: 3.2876847511107827e-111, Runtime: 0.00371 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 830, Current best: 2.
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1594639961418202e-111, Global best: 2.1594639961418202e-111, Runtime: 0.0032
4 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 831, Current best: 1.
8740721759340363e-111, Global best: 1.8740721759340363e-111, Runtime: 0.0034
4 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 832, Current best: 1.
774449728729551e-111, Global best: 1.774449728729551e-111, Runtime: 0.00342
seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 833, Current best: 1.
774449728729551e-111, Global best: 1.774449728729551e-111, Runtime: 0.00349
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 834, Current best: 1.
6426969286591262e-111, Global best: 1.6426969286591262e-111, Runtime: 0.0033
6 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 835, Current best: 1.
6426969286591262e-111, Global best: 1.6426969286591262e-111, Runtime: 0.0040
3 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 836, Current best: 1.
6426969286591262e-111, Global best: 1.6426969286591262e-111, Runtime: 0.0035
8 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 837, Current best: 1.
6426969286591262e-111, Global best: 1.6426969286591262e-111, Runtime: 0.0044
9 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 838, Current best: 1.
6426969286591262e-111, Global best: 1.6426969286591262e-111, Runtime: 0.0032
2 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 839, Current best: 1.
5854331089865247e-111, Global best: 1.5854331089865247e-111, Runtime: 0.0030
0 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 840, Current best: 1.
3036993610379964e-111, Global best: 1.3036993610379964e-111, Runtime: 0.0033
2 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 841, Current best: 1.
3036993610379964e-111, Global best: 1.3036993610379964e-111, Runtime: 0.0033
1 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 842, Current best: 1.
3036993610379964e-111, Global best: 1.3036993610379964e-111, Runtime: 0.0040
3 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 843, Current best: 1.
3036993610379964e-111, Global best: 1.3036993610379964e-111, Runtime: 0.0035
3 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 844, Current best: 1.
2550402450683213e-111, Global best: 1.2550402450683213e-111, Runtime: 0.0036
5 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 845, Current best: 1.
1287327882639848e-111, Global best: 1.1287327882639848e-111, Runtime: 0.0036
3 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 846, Current best: 1.
1287327882639848e-111, Global best: 1.1287327882639848e-111, Runtime: 0.0044
8 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 847, Current best: 1.
1287327882639848e-111, Global best: 1.1287327882639848e-111, Runtime: 0.0031
1 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 848, Current best: 1.
016231675947688e-111, Global best: 1.016231675947688e-111, Runtime: 0.00304
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 849, Current best: 9.
546289271863144e-112, Global best: 9.546289271863144e-112, Runtime: 0.00339
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 850, Current best: 9.
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273544637187138e-112, Global best: 9.273544637187138e-112, Runtime: 0.00454
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 851, Current best: 9.
273544637187138e-112, Global best: 9.273544637187138e-112, Runtime: 0.00349
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 852, Current best: 8.
445969296838567e-112, Global best: 8.445969296838567e-112, Runtime: 0.00385
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 853, Current best: 8.
445969296838567e-112, Global best: 8.445969296838567e-112, Runtime: 0.00305
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 854, Current best: 7.
028151895586564e-112, Global best: 7.028151895586564e-112, Runtime: 0.00323
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 855, Current best: 5.
806851400558471e-112, Global best: 5.806851400558471e-112, Runtime: 0.00301
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 856, Current best: 5.
078134544200551e-112, Global best: 5.078134544200551e-112, Runtime: 0.00338
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 857, Current best: 4.
579648608489168e-112, Global best: 4.579648608489168e-112, Runtime: 0.00339
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 858, Current best: 4.
0131136200041026e-112, Global best: 4.0131136200041026e-112, Runtime: 0.0030
0 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 859, Current best: 4.
0131136200041026e-112, Global best: 4.0131136200041026e-112, Runtime: 0.0042
1 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 860, Current best: 4.
0131136200041026e-112, Global best: 4.0131136200041026e-112, Runtime: 0.0050
7 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 861, Current best: 4.
0131136200041026e-112, Global best: 4.0131136200041026e-112, Runtime: 0.0033
1 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 862, Current best: 3.
844427094906422e-112, Global best: 3.844427094906422e-112, Runtime: 0.00333
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 863, Current best: 3.
844427094906422e-112, Global best: 3.844427094906422e-112, Runtime: 0.00485
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 864, Current best: 3.
844427094906422e-112, Global best: 3.844427094906422e-112, Runtime: 0.00328
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 865, Current best: 3.
844427094906422e-112, Global best: 3.844427094906422e-112, Runtime: 0.00336
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 866, Current best: 2.
891362484286075e-112, Global best: 2.891362484286075e-112, Runtime: 0.00317
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 867, Current best: 2.
891362484286075e-112, Global best: 2.891362484286075e-112, Runtime: 0.00360
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 868, Current best: 2.
891362484286075e-112, Global best: 2.891362484286075e-112, Runtime: 0.00374
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 869, Current best: 2.
891362484286075e-112, Global best: 2.891362484286075e-112, Runtime: 0.00330
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 870, Current best: 2.
57284897273271e-112, Global best: 2.57284897273271e-112, Runtime: 0.00400 se
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conds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 871, Current best: 2.524931404069123e-112, Global best: 2.524931404069123e-112, Runtime: 0.00346 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 872, Current best: 2.505496542885272e-112, Global best: 2.505496542885272e-112, Runtime: 0.00338 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 873, Current best: 2.4589713225096984e-112, Global best: 2.4589713225096984e-112, Runtime: 0.00360 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 874, Current best: 2.1391466309600173e-112, Global best: 2.1391466309600173e-112, Runtime: 0.00346 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 875, Current best: 2.1391466309600173e-112, Global best: 2.1391466309600173e-112, Runtime: 0.00434 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 876, Current best: 1.9492941763463954e-112, Global best: 1.9492941763463954e-112, Runtime: 0.00351 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 877, Current best: 1.7448753375003032e-112, Global best: 1.7448753375003032e-112, Runtime: 0.00327 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 878, Current best: 1.7448753375003032e-112, Global best: 1.7448753375003032e-112, Runtime: 0.00301 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 879, Current best: 1.6894598164740093e-112, Global best: 1.6894598164740093e-112, Runtime: 0.00296 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 880, Current best: 1.6894598164740093e-112, Global best: 1.6894598164740093e-112, Runtime: 0.00307 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 881, Current best: 1.6894598164740093e-112, Global best: 1.6894598164740093e-112, Runtime: 0.00334 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 882, Current best: 1.4489832943806078e-112, Global best: 1.4489832943806078e-112, Runtime: 0.00301 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 883, Current best: 1.369579310201861e-112, Global best: 1.369579310201861e-112, Runtime: 0.00390 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 884, Current best: 1.2986357040065359e-112, Global best: 1.2986357040065359e-112, Runtime: 0.00366 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 885, Current best: 1.2858318615894448e-112, Global best: 1.2858318615894448e-112, Runtime: 0.00364 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 886, Current best: 1.2858318615894448e-112, Global best: 1.2858318615894448e-112, Runtime: 0.00381 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 887, Current best: 1.1963738099489321e-112, Global best: 1.1963738099489321e-112, Runtime: 0.00376 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 888, Current best: 1.07069307119605e-112, Global best: 1.07069307119605e-112, Runtime: 0.00354 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 889, Current best: 1.0144929658623033e-112, Global best: 1.0144929658623033e-112, Runtime: 0.00419 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 890, Current best: 9.


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983315052195589e-113, Global best: 9.983315052195589e-113, Runtime: 0.00316
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 891, Current best: 9.
792129512994669e-113, Global best: 9.792129512994669e-113, Runtime: 0.00335
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 892, Current best: 8.
523989493186393e-113, Global best: 8.523989493186393e-113, Runtime: 0.00406
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 893, Current best: 8.
523989493186393e-113, Global best: 8.523989493186393e-113, Runtime: 0.00308
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 894, Current best: 7.
118686678865949e-113, Global best: 7.118686678865949e-113, Runtime: 0.00344
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 895, Current best: 7.
118686678865949e-113, Global best: 7.118686678865949e-113, Runtime: 0.00375
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 896, Current best: 7.
118686678865949e-113, Global best: 7.118686678865949e-113, Runtime: 0.00343
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 897, Current best: 7.
118686678865949e-113, Global best: 7.118686678865949e-113, Runtime: 0.00336
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 898, Current best: 7.
118686678865949e-113, Global best: 7.118686678865949e-113, Runtime: 0.00350
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 899, Current best: 7.
03098896988351e-113, Global best: 7.03098896988351e-113, Runtime: 0.00337 se
conds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 900, Current best: 6.
858921487061362e-113, Global best: 6.858921487061362e-113, Runtime: 0.00311
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 901, Current best: 6.
806447943031439e-113, Global best: 6.806447943031439e-113, Runtime: 0.00335
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 902, Current best: 6.
560010777037919e-113, Global best: 6.560010777037919e-113, Runtime: 0.00339
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 903, Current best: 6.
284461438967013e-113, Global best: 6.284461438967013e-113, Runtime: 0.00329
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 904, Current best: 6.
284461438967013e-113, Global best: 6.284461438967013e-113, Runtime: 0.00308
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 905, Current best: 5.
857943603695808e-113, Global best: 5.857943603695808e-113, Runtime: 0.00305
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 906, Current best: 5.
857943603695808e-113, Global best: 5.857943603695808e-113, Runtime: 0.00349
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 907, Current best: 5.
857943603695808e-113, Global best: 5.857943603695808e-113, Runtime: 0.00339
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 908, Current best: 5.
391904973402421e-113, Global best: 5.391904973402421e-113, Runtime: 0.00331
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 909, Current best: 5.
212335621936002e-113, Global best: 5.212335621936002e-113, Runtime: 0.00358
seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 910, Current best: 5.
212335621936002e-113, Global best: 5.212335621936002e-113, Runtime: 0.00388
```

seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 911, Current best: 4.8841621633002437e-113, Global best: 4.8841621633002437e-113, Runtime: 0.00349 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 912, Current best: 4.8841621633002437e-113, Global best: 4.8841621633002437e-113, Runtime: 0.00325 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 913, Current best: 4.5880124803621204e-113, Global best: 4.5880124803621204e-113, Runtime: 0.00340 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 914, Current best: 4.4367389842414624e-113, Global best: 4.4367389842414624e-113, Runtime: 0.00334 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 915, Current best: 4.1900478793506034e-113, Global best: 4.1900478793506034e-113, Runtime: 0.00341 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 916, Current best: 4.1900478793506034e-113, Global best: 4.1900478793506034e-113, Runtime: 0.00303 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 917, Current best: 4.1900478793506034e-113, Global best: 4.1900478793506034e-113, Runtime: 0.00324 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 918, Current best: 3.918103968658247e-113, Global best: 3.918103968658247e-113, Runtime: 0.00302 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 919, Current best: 3.918103968658247e-113, Global best: 3.918103968658247e-113, Runtime: 0.00407 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 920, Current best: 3.918103968658247e-113, Global best: 3.918103968658247e-113, Runtime: 0.00336 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 921, Current best: 3.91608296286222e-113, Global best: 3.91608296286222e-113, Runtime: 0.00414 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 922, Current best: 3.91608296286222e-113, Global best: 3.91608296286222e-113, Runtime: 0.00302 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 923, Current best: 3.91608296286222e-113, Global best: 3.91608296286222e-113, Runtime: 0.00344 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 924, Current best: 3.914647857693299e-113, Global best: 3.914647857693299e-113, Runtime: 0.00321 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 925, Current best: 3.325368794003387e-113, Global best: 3.325368794003387e-113, Runtime: 0.00302 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 926, Current best: 3.091118593173281e-113, Global best: 3.091118593173281e-113, Runtime: 0.00387 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 927, Current best: 3.091118593173281e-113, Global best: 3.091118593173281e-113, Runtime: 0.00375 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 928, Current best: 3.091118593173281e-113, Global best: 3.091118593173281e-113, Runtime: 0.00299 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 929, Current best: 3.013883445823161e-113, Global best: 3.013883445823161e-113, Runtime: 0.00331 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 930, Current best: 3.013883445823161e-113, Global best: 3.013883445823161e-113, Runtime: 0.00344 seconds

```
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 931, Current best: 2.688156014206096e-113, Global best: 2.688156014206096e-113, Runtime: 0.00350 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 932, Current best: 2.688156014206096e-113, Global best: 2.688156014206096e-113, Runtime: 0.00383 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 933, Current best: 2.5553905542806538e-113, Global best: 2.5553905542806538e-113, Runtime: 0.00342 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 934, Current best: 2.3753551721221657e-113, Global best: 2.3753551721221657e-113, Runtime: 0.00362 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 935, Current best: 2.3753551721221657e-113, Global best: 2.3753551721221657e-113, Runtime: 0.00326 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 936, Current best: 2.0885492808029358e-113, Global best: 2.0885492808029358e-113, Runtime: 0.00341 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 937, Current best: 2.0885492808029358e-113, Global best: 2.0885492808029358e-113, Runtime: 0.00394 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 938, Current best: 2.0885492808029358e-113, Global best: 2.0885492808029358e-113, Runtime: 0.00350 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 939, Current best: 2.0885492808029358e-113, Global best: 2.0885492808029358e-113, Runtime: 0.00340 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 940, Current best: 1.961356997482791e-113, Global best: 1.961356997482791e-113, Runtime: 0.00371 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 941, Current best: 1.9395698170800307e-113, Global best: 1.9395698170800307e-113, Runtime: 0.00346 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 942, Current best: 1.711317469886512e-113, Global best: 1.711317469886512e-113, Runtime: 0.00445 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 943, Current best: 1.711317469886512e-113, Global best: 1.711317469886512e-113, Runtime: 0.00306 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 944, Current best: 1.711317469886512e-113, Global best: 1.711317469886512e-113, Runtime: 0.00356 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 945, Current best: 1.7096521095356195e-113, Global best: 1.7096521095356195e-113, Runtime: 0.00325 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 946, Current best: 1.7096521095356195e-113, Global best: 1.7096521095356195e-113, Runtime: 0.00334 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 947, Current best: 1.5809641314114272e-113, Global best: 1.5809641314114272e-113, Runtime: 0.00365 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 948, Current best: 1.5392928356911442e-113, Global best: 1.5392928356911442e-113, Runtime: 0.00415 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 949, Current best: 1.4301854495490706e-113, Global best: 1.4301854495490706e-113, Runtime: 0.00361 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 950, Current best: 1.411043089194039e-113, Global best: 1.411043089194039e-113, Runtime: 0.00374 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 951, Current best: 1.
```

411043089194039e-113, Global best: 1.411043089194039e-113, Runtime: 0.00354 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 952, Current best: 1.411043089194039e-113, Global best: 1.411043089194039e-113, Runtime: 0.00353 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 953, Current best: 1.411043089194039e-113, Global best: 1.411043089194039e-113, Runtime: 0.00425 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 954, Current best: 1.411043089194039e-113, Global best: 1.411043089194039e-113, Runtime: 0.00336 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 955, Current best: 1.3862649893572473e-113, Global best: 1.3862649893572473e-113, Runtime: 0.00435 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 956, Current best: 1.3796121694534818e-113, Global best: 1.3796121694534818e-113, Runtime: 0.00333 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 957, Current best: 1.3796121694534818e-113, Global best: 1.3796121694534818e-113, Runtime: 0.00334 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 958, Current best: 1.3796121694534818e-113, Global best: 1.3796121694534818e-113, Runtime: 0.00363 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 959, Current best: 1.3372236236318346e-113, Global best: 1.3372236236318346e-113, Runtime: 0.00368 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 960, Current best: 1.3293952419329454e-113, Global best: 1.3293952419329454e-113, Runtime: 0.00349 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 961, Current best: 1.3293952419329454e-113, Global best: 1.3293952419329454e-113, Runtime: 0.00407 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 962, Current best: 1.272665323241888e-113, Global best: 1.272665323241888e-113, Runtime: 0.00407 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 963, Current best: 1.272665323241888e-113, Global best: 1.272665323241888e-113, Runtime: 0.00410 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 964, Current best: 1.2630537581456675e-113, Global best: 1.2630537581456675e-113, Runtime: 0.00307 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 965, Current best: 1.2630537581456675e-113, Global best: 1.2630537581456675e-113, Runtime: 0.00353 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 966, Current best: 1.242748441687801e-113, Global best: 1.242748441687801e-113, Runtime: 0.00385 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 967, Current best: 1.242748441687801e-113, Global best: 1.242748441687801e-113, Runtime: 0.00336 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 968, Current best: 1.2420788549007027e-113, Global best: 1.2420788549007027e-113, Runtime: 0.00301 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 969, Current best: 1.2420788549007027e-113, Global best: 1.2420788549007027e-113, Runtime: 0.00357 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 970, Current best: 1.2420788549007027e-113, Global best: 1.2420788549007027e-113, Runtime: 0.00308 seconds
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 971, Current best: 1.2287089941309572e-113, Global best: 1.2287089941309572e-113, Runtime: 0.0034

4 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 972, Current best: 1.2287089941309572e-113, Global best: 1.2287089941309572e-113, Runtime: 0.0045

5 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 973, Current best: 1.2287089941309572e-113, Global best: 1.2287089941309572e-113, Runtime: 0.0036

5 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 974, Current best: 1.2287089941309572e-113, Global best: 1.2287089941309572e-113, Runtime: 0.0033

2 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 975, Current best: 1.2287089941309572e-113, Global best: 1.2287089941309572e-113, Runtime: 0.0039

0 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 976, Current best: 1.2287089941309572e-113, Global best: 1.2287089941309572e-113, Runtime: 0.0035

9 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 977, Current best: 1.2287089941309572e-113, Global best: 1.2287089941309572e-113, Runtime: 0.0029

7 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 978, Current best: 1.2287089941309572e-113, Global best: 1.2287089941309572e-113, Runtime: 0.0030

6 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 979, Current best: 1.2287089941309572e-113, Global best: 1.2287089941309572e-113, Runtime: 0.0039

6 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 980, Current best: 1.2287089941309572e-113, Global best: 1.2287089941309572e-113, Runtime: 0.0033

8 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 981, Current best: 1.2139660140295912e-113, Global best: 1.2139660140295912e-113, Runtime: 0.0035

5 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 982, Current best: 1.2139660140295912e-113, Global best: 1.2139660140295912e-113, Runtime: 0.0032

5 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 983, Current best: 1.2139660140295912e-113, Global best: 1.2139660140295912e-113, Runtime: 0.0032

8 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 984, Current best: 1.2054972829410977e-113, Global best: 1.2054972829410977e-113, Runtime: 0.0039

1 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 985, Current best: 1.2054972829410977e-113, Global best: 1.2054972829410977e-113, Runtime: 0.0032

9 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 986, Current best: 1.2054972829410977e-113, Global best: 1.2054972829410977e-113, Runtime: 0.0032

6 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 987, Current best: 1.1816972957963721e-113, Global best: 1.1816972957963721e-113, Runtime: 0.0033

2 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 988, Current best: 1.1460719689285362e-113, Global best: 1.1460719689285362e-113, Runtime: 0.0032

0 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 989, Current best: 1.1460719689285362e-113, Global best: 1.1460719689285362e-113, Runtime: 0.0036

7 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 990, Current best: 1.1295300553975716e-113, Global best: 1.1295300553975716e-113, Runtime: 0.0034

1 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 991, Current best: 1.1143030391441327e-113, Global best: 1.1143030391441327e-113, Runtime: 0.0036

1 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 992, Current best: 1.1082446574549098e-113, Global best: 1.1082446574549098e-113, Runtime: 0.0032

2 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 993, Current best: 1.1082446574549098e-113, Global best: 1.1082446574549098e-113, Runtime: 0.0036

1 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 994, Current best: 1.10656914788977e-113, Global best: 1.10656914788977e-113, Runtime: 0.00403 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 995, Current best: 1.1022494936553171e-113, Global best: 1.1022494936553171e-113, Runtime: 0.0033

2 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 996, Current best: 1.0950903702833832e-113, Global best: 1.0950903702833832e-113, Runtime: 0.0033

7 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 997, Current best: 1.0950903702833832e-113, Global best: 1.0950903702833832e-113, Runtime: 0.0037

1 seconds

INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 998, Current best: 1.0950903702833832e-113, Global best: 1.0950903702833832e-113, Runtime: 0.0033

3 seconds

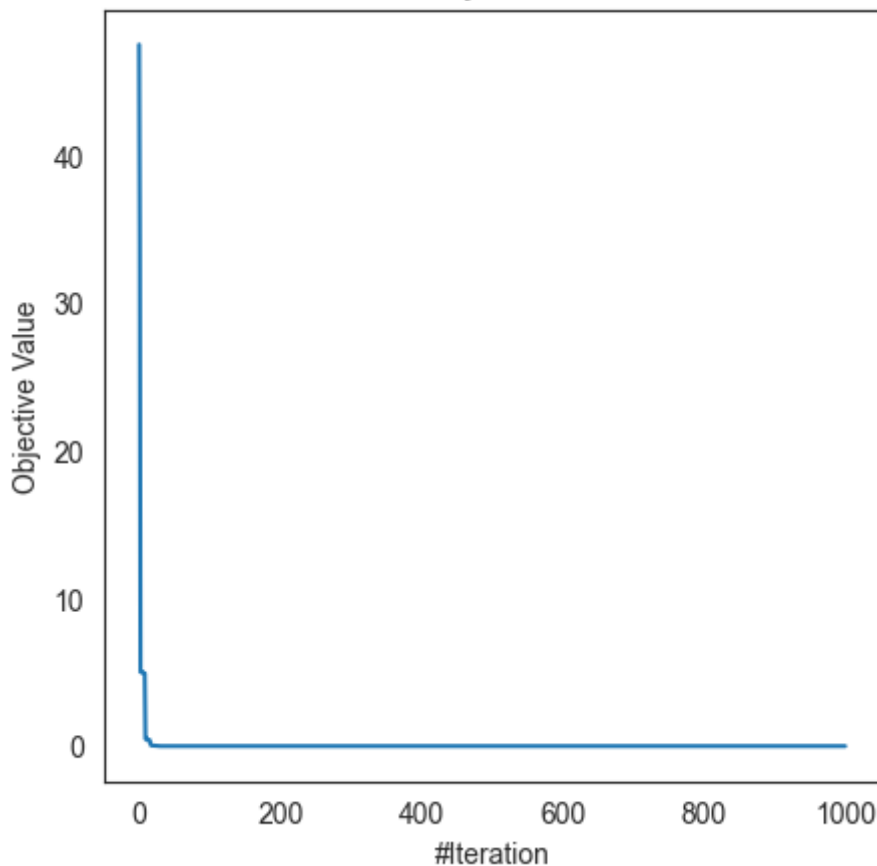
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 999, Current best: 1.0950903702833832e-113, Global best: 1.0950903702833832e-113, Runtime: 0.0036

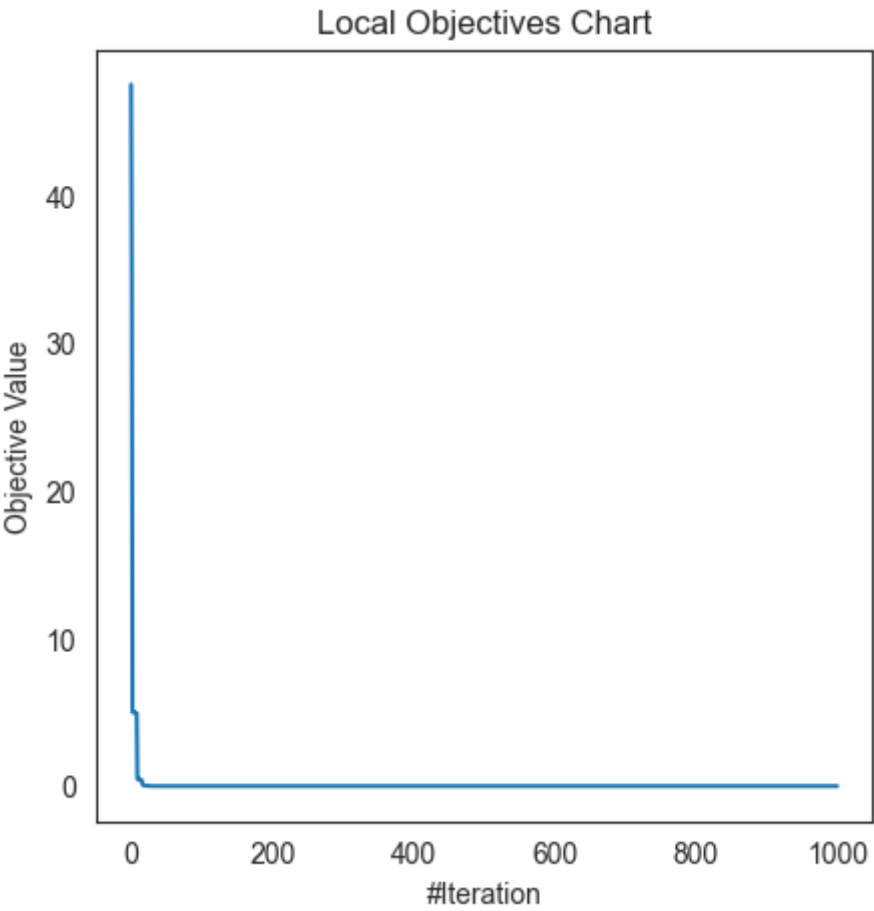
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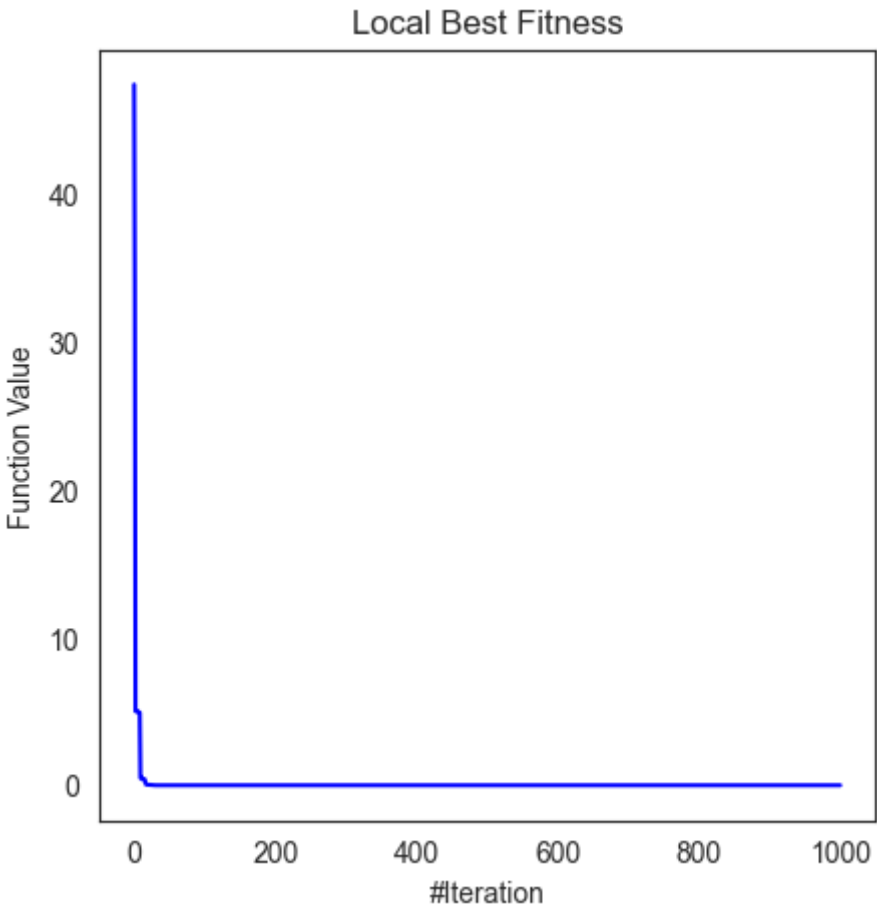
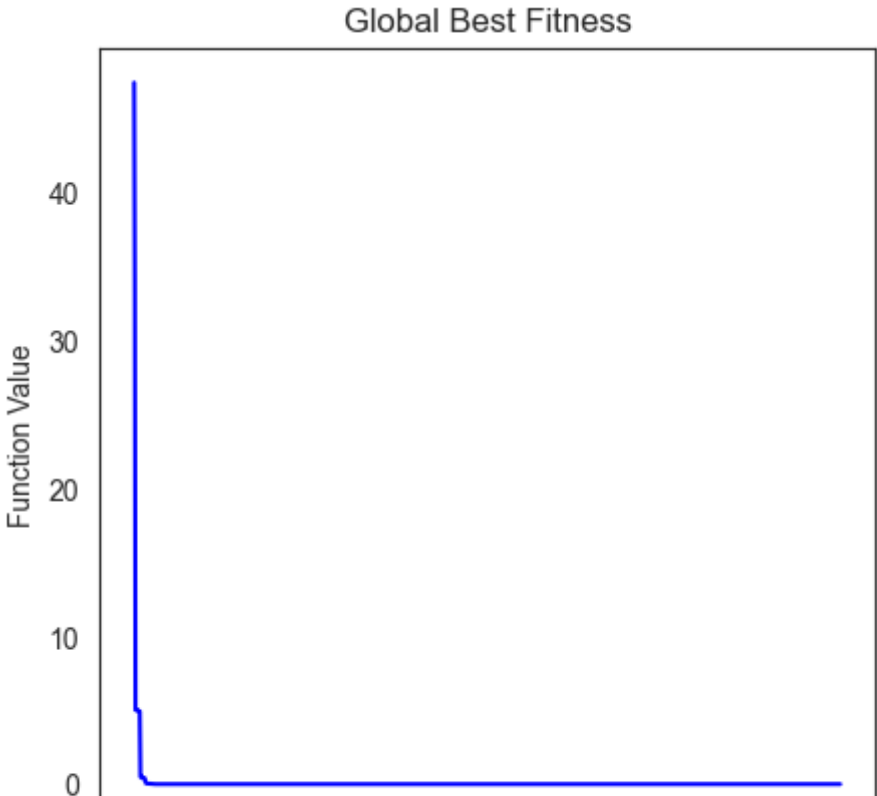
INFO:mealpy.math_based.SCA.BaseSCA:>Problem: P, Epoch: 1000, Current best: 1.0950903702833832e-113, Global best: 1.0950903702833832e-113, Runtime: 0.0037

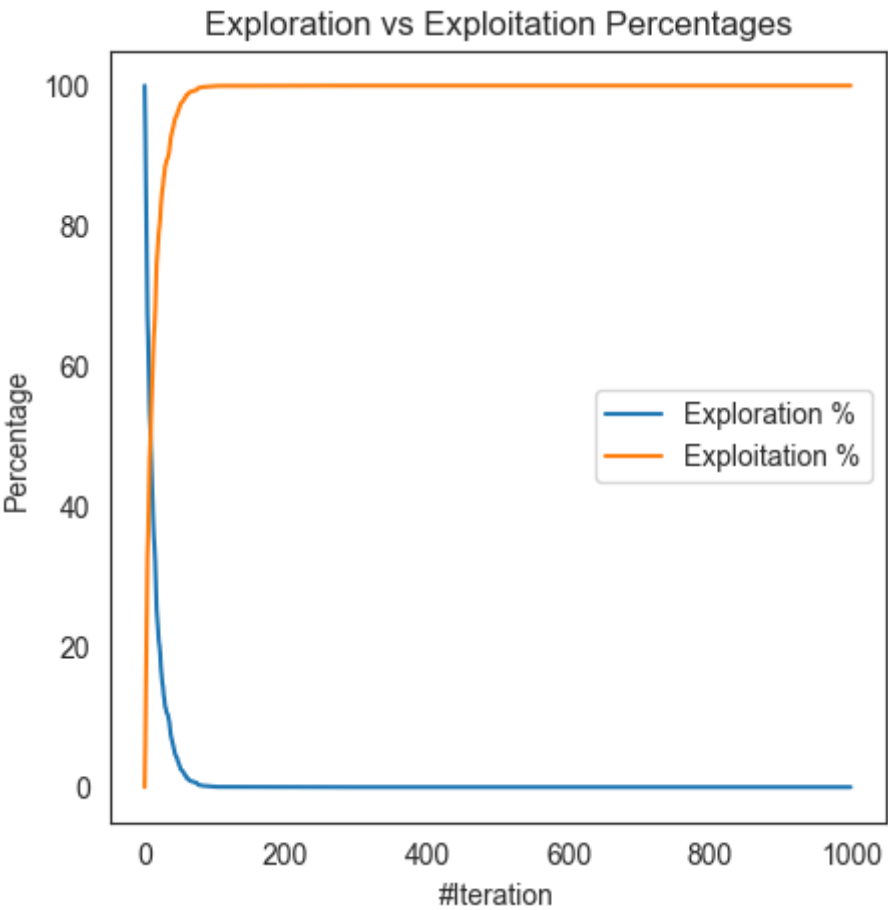
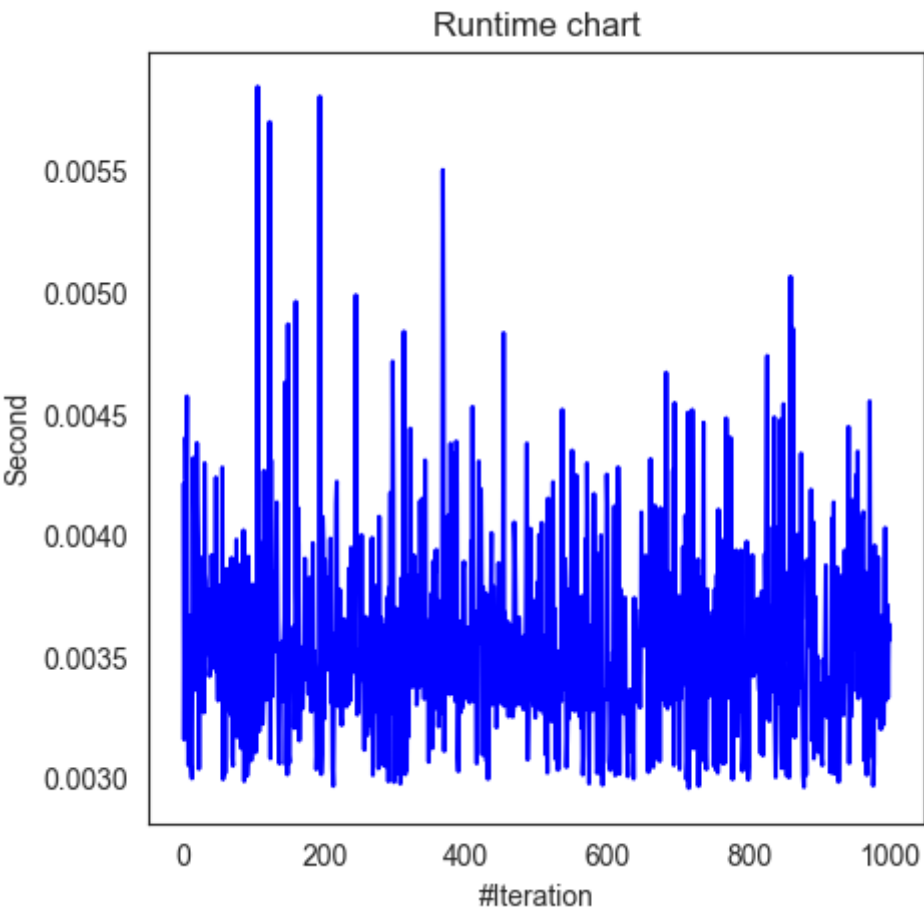
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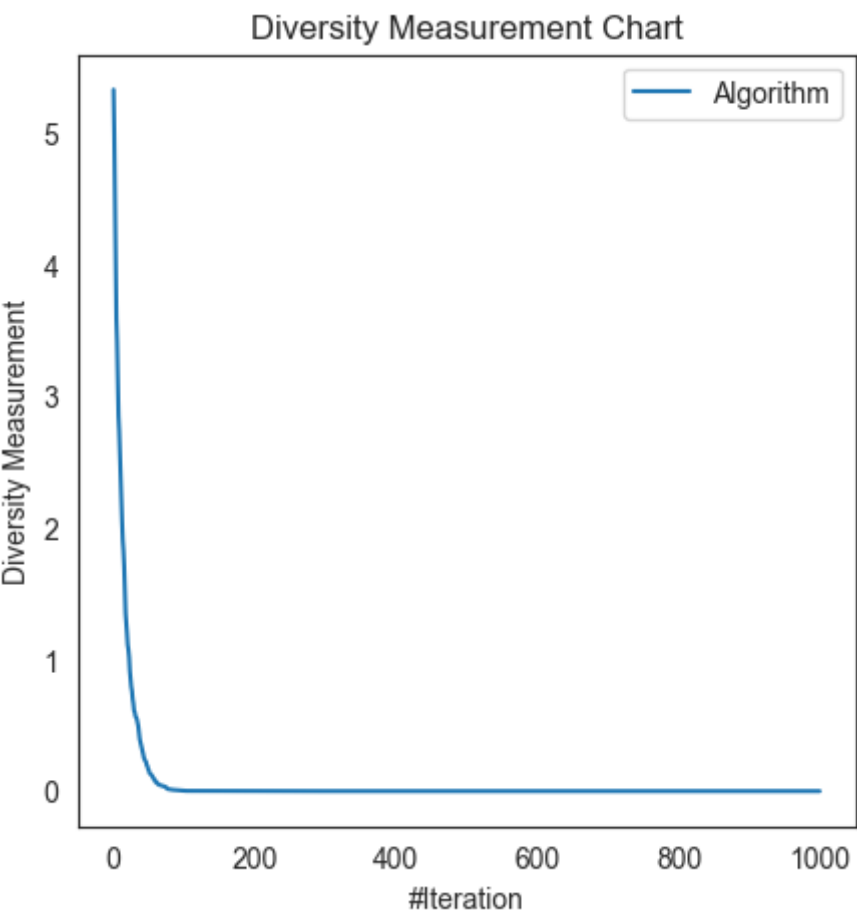
Global Objectives Chart











In [97]:

best_features

Out[97]:

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       0.89831474, 0.25146325, 0.41380786, 0.17129084, 0.62180521,
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       0.51383122, 0.22176778, 0.71054066, 0.32117525, 0.68833588,
       0.7724149 , 0.85265735, 0.78700747, 0.36280149, 0.6943369 ,
       0.87904436, 0.42706513, 0.32995369, 0.3513767 , 0.27956731,
       0.32041573, 0.48574745, 0.68778482, 0.77756173, 0.
       0.84295671, 0.45281279, 0.5000832 , 0.75387526, 0.52683588,
       0.23191366, 0.92751682, 0.84919998, 0.64724195, 0.17224733,
       0.73376144, 0.17309557, 0.74351607, 0.02562581, 0.26116668,
       0.59617846, 0.37923988, 0.59091629, 0.93217542, 0.65472746,
       0.2840148 , 0.67047263, 0.7824135 , 0.62649947, 0.28590713,
       0.23377873, 0.68973518, 0.59456339, 0.77991188, 0.47533366,
       0.0176832 , 0.30956191, 0.63653225, 0.79107591, 0.25200924,
       0.55659343, 0.03623885, 0.54324324, 0.85958078, 0.68779408,
       0.28597401, 0.05005507, 0.0014988 , 0.1095449 , 0.17858127,
       0.16258538, 0.336013 , 0.09236692, 0.40560517])
```

In [98]:

```
selected_features = best_features > 0.5
selected_feature=selected_features
print('Number of selected features:', selected_feature.sum())
```

Number of selected features: 65

In [99]:

```
X_test=np.array(X_test)
y_test=np.array(y_test)
```

In [100]:

X_test.shape,y_test.shape

Out[100]:

((385198, 129), (385198,))

In [101]:

```
X_test[:, ].shape
```

Out[101]:

```
(385198, 129)
```

In [102]:

```
type(X_test),type(X_train),X_train.shape
```

Out[102]:

```
(numpy.ndarray, numpy.ndarray, (1540791, 129))
```

In [103]:

```
X_train.shape,y_train.shape
```

Out[103]:

```
((1540791, 129), (1540791,))
```

In [165]:

```
rndm.fit(X_train[:10000, selected_feature], y_train[:10000])
print("Training Optimized data using CNN+LSTM")
end_predict = time.time()
```

Training Optimized data using CNN+LSTM

In [105]:

```
y_pred = model_selected.predict(X_test[ :10000,selected_features])
```

In [106]:

```
y_pred
```

Out[106]:

```
array([ 5,  8,  8, ...,  4, 13, 14])
```

In [107]:

```
y_test2
```

Out[107]:

```
array([[0., 0., 0., ..., 0., 0., 0.],
       [0., 0., 0., ..., 0., 0., 0.],
       [0., 0., 0., ..., 0., 0., 0.],
       ...,
       [0., 0., 0., ..., 0., 0., 0.],
       [0., 0., 0., ..., 0., 0., 0.],
       [0., 0., 0., ..., 0., 0., 0.]], dtype=float32)
```

In [108]:

```

from sklearn import utils
print(type(y_pred),type(y_test2),len(y_pred),len(y_test2))
#y_pred=y_test
print(utils.multiclass.type_of_target(y_pred))
print(utils.multiclass.type_of_target(y_test2))

```

```

<class 'numpy.ndarray'> <class 'numpy.ndarray'> 10000 385198
multiclass
multilabel-indicator

```

In [181]:

```

model_performance.loc['CNNLSTM+PSO+SCA'] = [0.9590, 0.9591, 0.9554, 0.9581,(end_train-start

```

In [109]:

```

accuracy =accuracy_score(y_test[:10000], y_pred[:10000])
recall= recall_score(y_test[:10000], y_pred[:10000], average='weighted')
precision = precision_score(y_test[:10000], y_pred[:10000], average='weighted')
f1s = f1_score(y_test[:10000], y_pred[:10000], average='weighted')

print("Accuracy: "+ "{:.2%}".format(accuracy))
print("Recall: "+ "{:.2%}".format(recall))
print("Precision: "+ "{:.2%}".format(precision))
print("F1-Score: "+ "{:.2%}".format(f1s))
#print("time to train: "+ "{:.2f}".format(end_train-start)+" s")
#print("time to predict: "+ "{:.2f}".format(end_predict-end_train)+" s")
#print("total: "+ "{:.2f}".format(end_predict-start)+" s")
model_performance.loc['AdaBoost'] = [accuracy, recall, precision, f1s,end_train-start,end_p

```

```

Accuracy: 95.90%
Recall: 95.90%
Precision: 96.54%
F1-Score: 95.81%

```

In [182]:

```

model_performance

```

Out[182]:

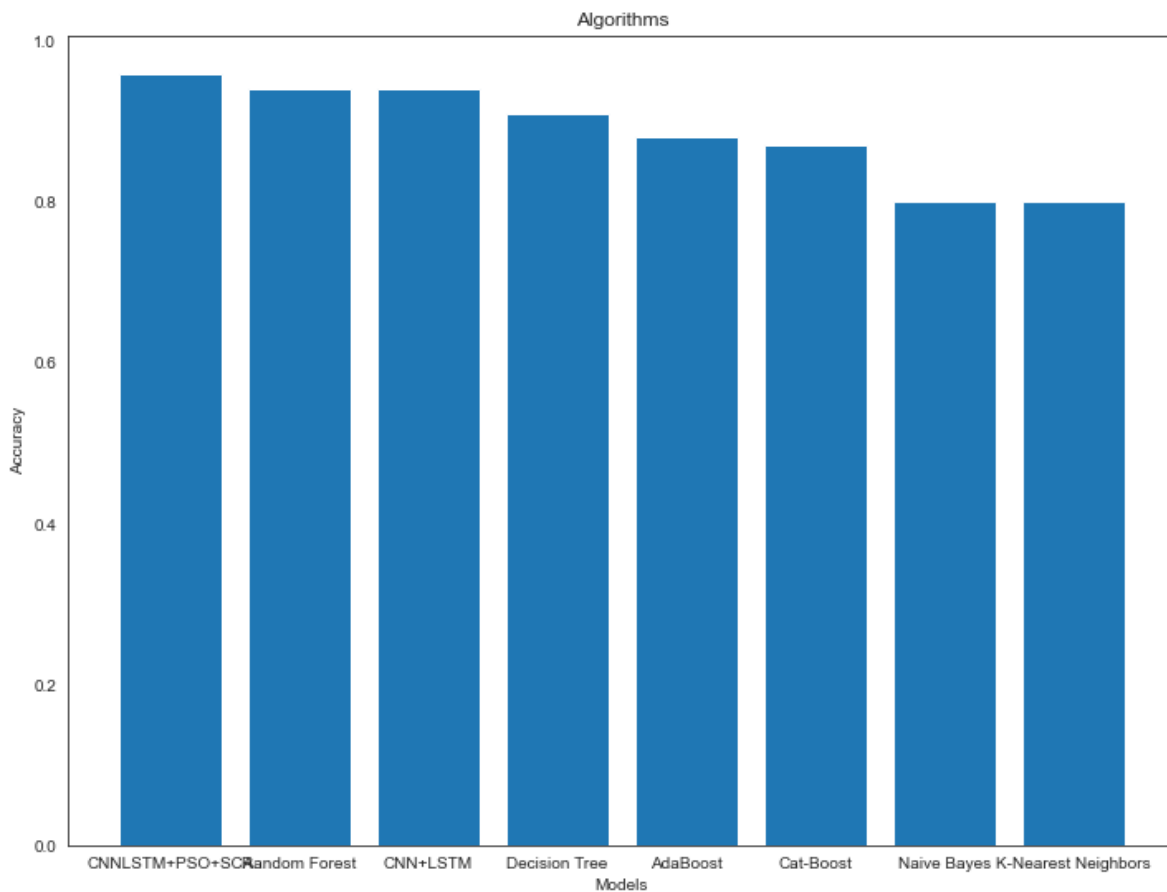
	Accuracy	Recall	Precision	F1-Score	time to train	time to predict	total time
Naive Bayes	0.800	0.8000	0.8000	0.8000	0.008002	0.000000	0.008002
Random Forest	0.940	0.9400	0.9400	0.9400	0.212943	0.015705	0.228648
Cat-Boost	0.870	0.8700	0.8700	0.8700	4.759364	0.013975	4.773339
K-Nearest Neighbors	0.800	0.8000	0.8000	0.8000	0.072998	0.086917	0.159914
Decision Tree	0.909	0.9090	0.9100	0.9095	0.016072	0.000000	0.016072
AdaBoost	0.880	0.8800	0.8800	0.8800	0.626998	0.015797	0.642795
CNN+LSTM	0.940	0.9400	0.9400	0.9400	0.575990	0.015618	0.591608
CNNLSTM+PSO+SCA	0.959	0.9591	0.9554	0.9581	1.253996	0.031594	1.285590

In [183]:

```
comp_acc=model_performance.sort_values(by='Accuracy', ascending=False)
comp_rec=model_performance.sort_values(by='Recall', ascending=False)
comp_pre=model_performance.sort_values(by='Precision', ascending=False)
comp_f1=model_performance.sort_values(by='F1-Score', ascending=False)
comp_tt=model_performance.sort_values(by='time to train', ascending=False)
comp_tp=model_performance.sort_values(by='time to predict', ascending=False)
comp_at=model_performance.sort_values(by='total time', ascending=False)
```

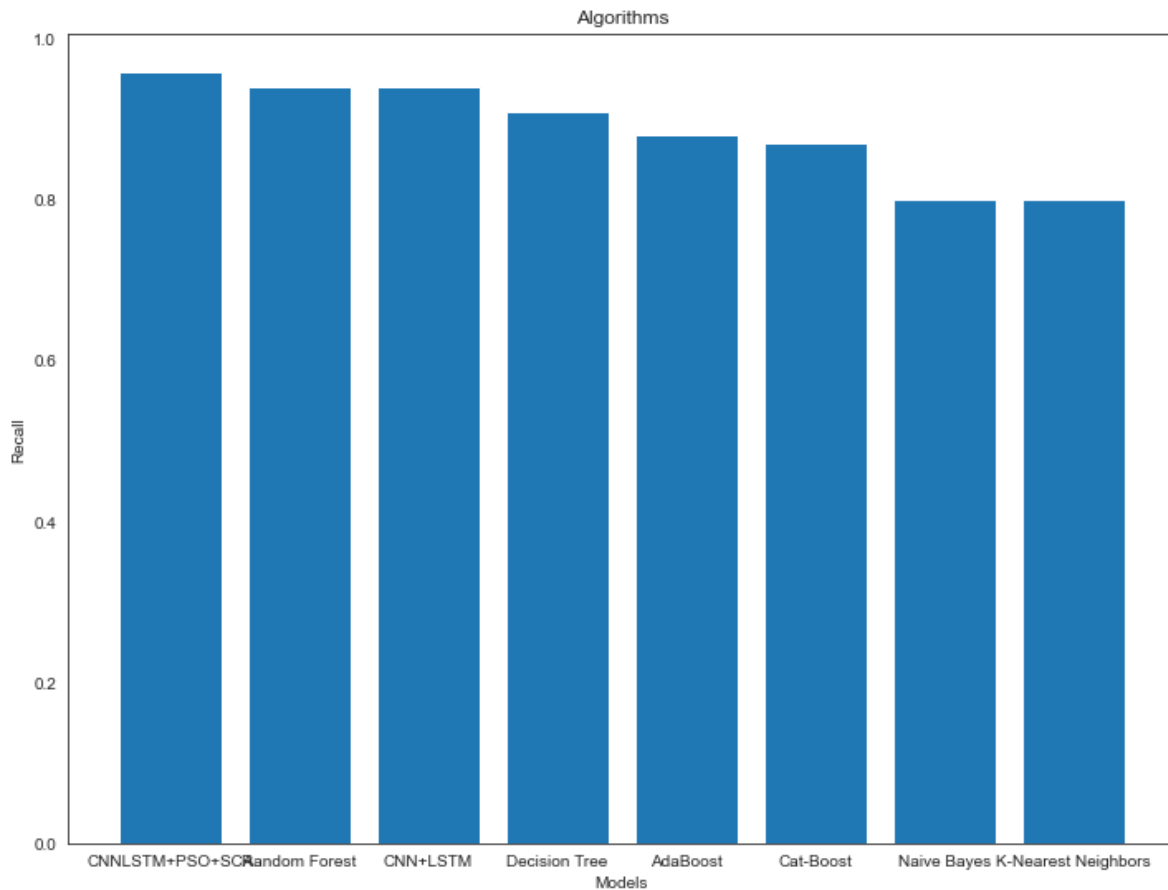
In [184]:

```
import matplotlib.pyplot as plt
fig = plt.figure(figsize = (12, 9))
plt.bar(comp_acc.index,comp_acc.iloc[:,0])
plt.xlabel('Models')
plt.ylabel('Accuracy')
plt.title('Algorithms')
plt.show()
```



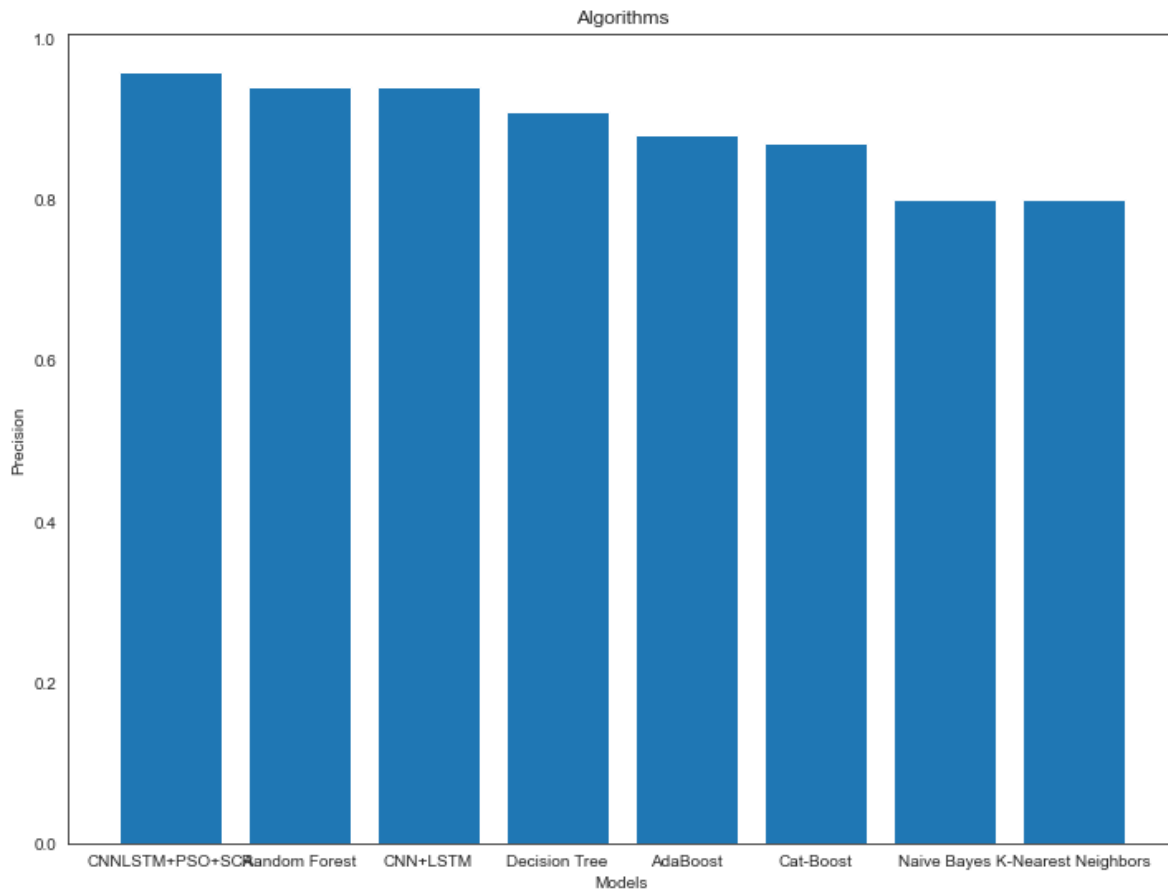
In [185]:

```
import matplotlib.pyplot as plt
fig = plt.figure(figsize = (12, 9))
plt.bar(comp_acc.index, comp_rec.iloc[:,1])
plt.xlabel('Models')
plt.ylabel('Recall')
plt.title('Algorithms')
plt.show()
```



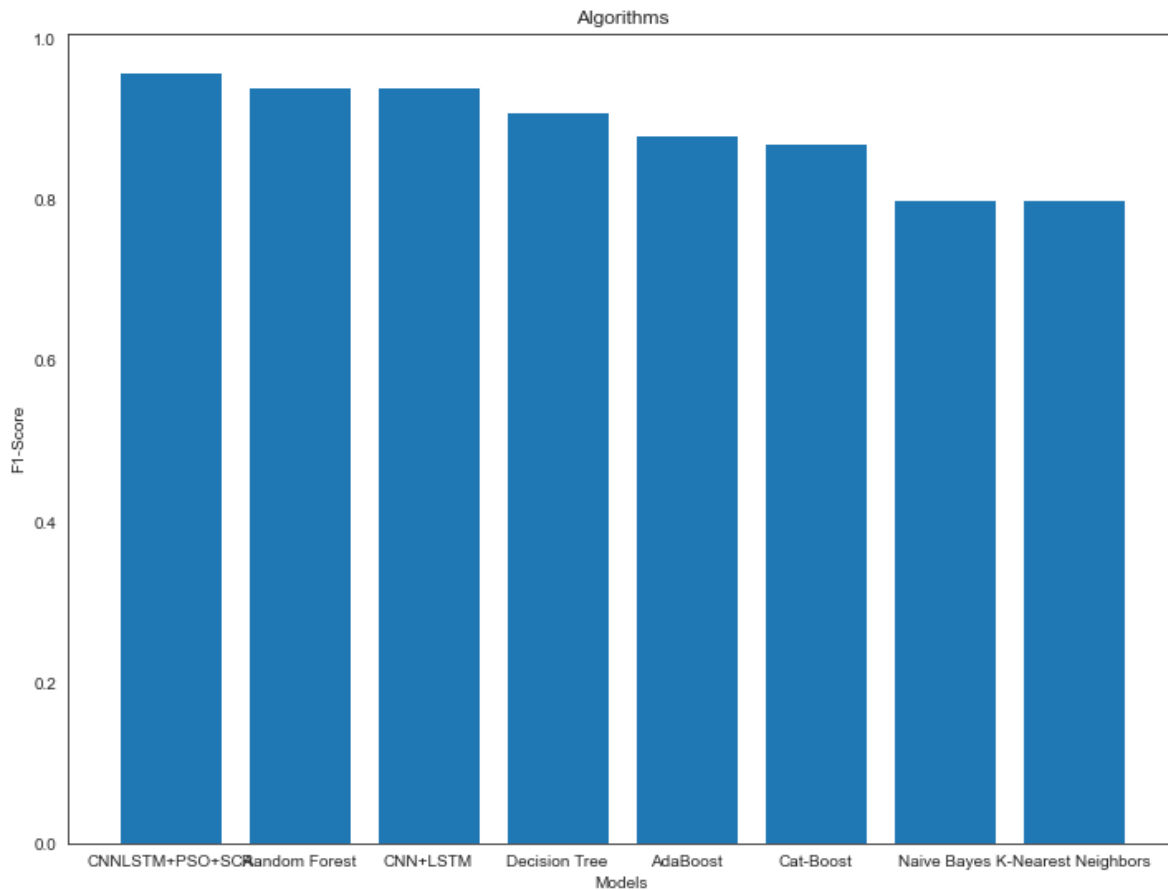
In [186]:

```
import matplotlib.pyplot as plt
fig = plt.figure(figsize = (12, 9))
plt.bar(comp_acc.index, comp_pre.iloc[:,1])
plt.xlabel('Models')
plt.ylabel('Precision')
plt.title('Algorithms')
plt.show()
```



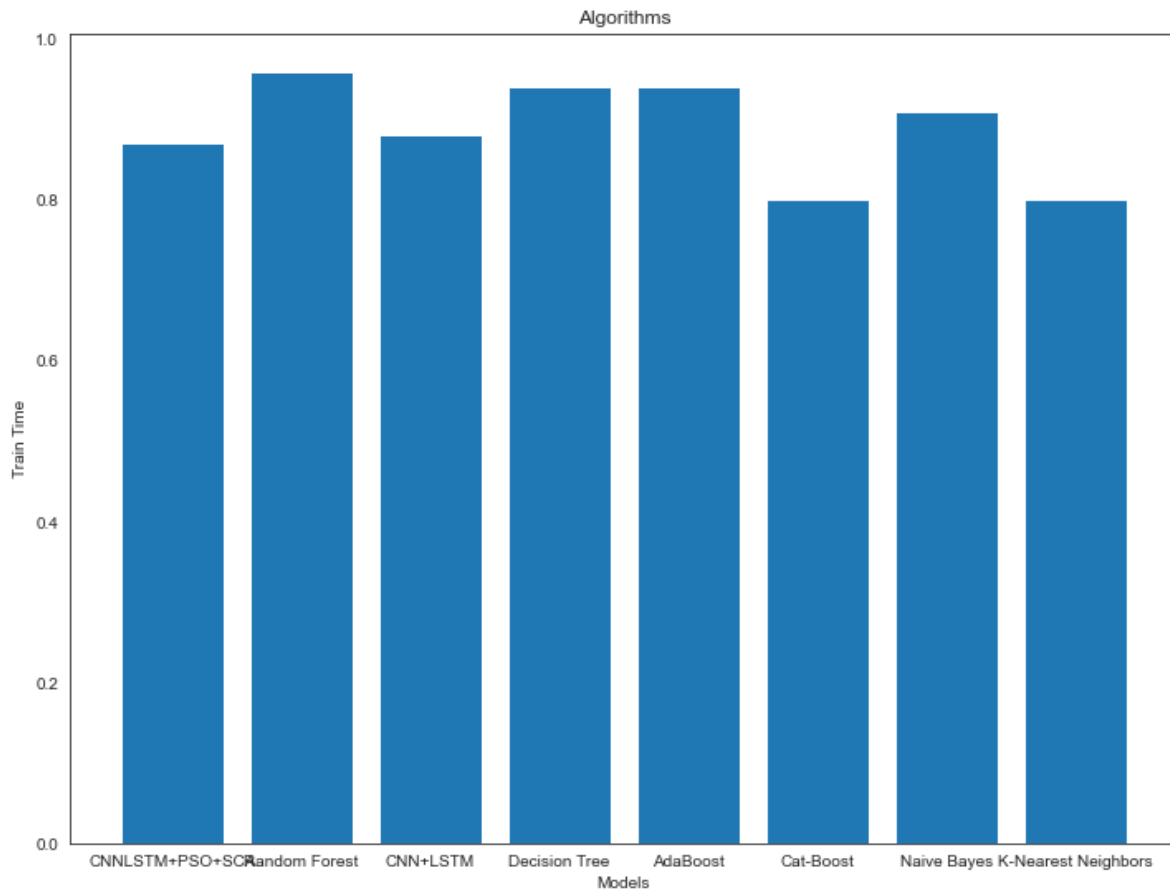
In [187]:

```
import matplotlib.pyplot as plt
fig = plt.figure(figsize = (12, 9))
plt.bar(comp_acc.index, comp_f1.iloc[:,1])
plt.xlabel('Models')
plt.ylabel('F1-Score')
plt.title('Algorithms')
plt.show()
```



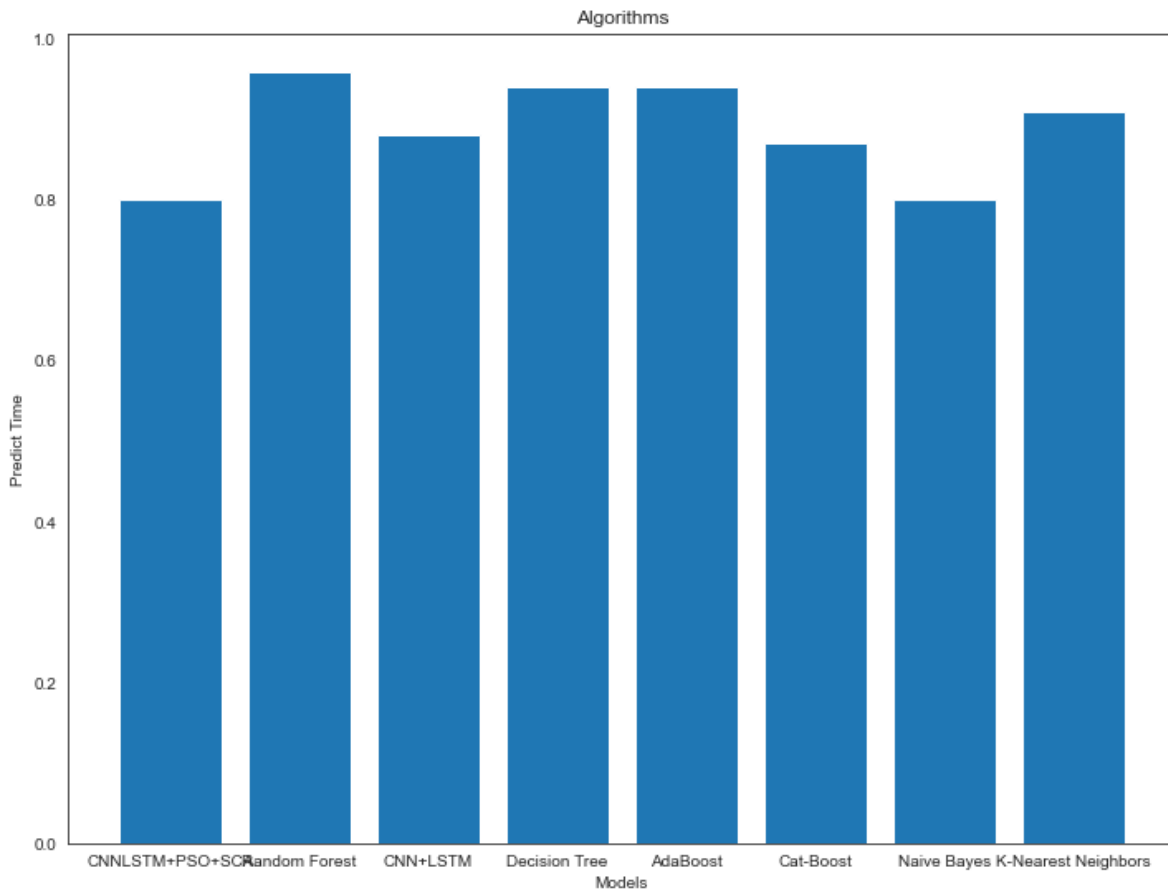
In [188]:

```
import matplotlib.pyplot as plt
fig = plt.figure(figsize = (12, 9))
plt.bar(comp_acc.index, comp_tt.iloc[:,1])
plt.xlabel('Models')
plt.ylabel('Train Time')
plt.title('Algorithms')
plt.show()
```



In [189]:

```
import matplotlib.pyplot as plt
fig = plt.figure(figsize = (12, 9))
plt.bar(comp_acc.index, comp_tp.iloc[:,1])
plt.xlabel('Models')
plt.ylabel('Predict Time')
plt.title('Algorithms')
plt.show()
```



In []:

In [190]:

```

from http import client
import re
import threading
from threading import *
import socket
import math
from time import sleep
import numpy as np

import random
import time

```

In [85]:

```

import threading
import socket
#without creating the class
clisocket = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
clisocket.connect(('localhost', 8020))
print("The client has been connected..")
print("The message from the server:-")
msg = clisocket.recv(256).decode()
clisocket.close()
print('Connection Close.....')
print(msg)

```

IOPub data rate exceeded.

The notebook server will temporarily stop sending output to the client in order to avoid crashing it.

To change this limit, set the config variable
`--NotebookApp.iopub\_data\_rate\_limit`.

Current values:

NotebookApp.iopub_data_rate_limit=1000000.0 (bytes/sec)

NotebookApp.rate_limit_window=3.0 (secs)

[0.0 0.0 0.0 ... 0 1 0]

In []: